Introduction to Probability

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Abstract

Lecture note of Introduction to Probability.

Contents

Chapter 1

Combinatorial Analysis

Lecture 1

24 Feb.

Definition 1.0.1 (Gerolamo Cardano (1501-1576) Basic Probability Model).

- Sample space: set of all possible outcomes.
- Event: $E \subseteq S$, the set of outcomes we are interested in.
- Probability: $\mathbb{P}(E) \in [0, 1]$.

Remark 1.0.1. In (finite) uniform model,

$$\mathbb{P}(E) = \frac{|E|}{|S|}.$$

Example 1.0.1. Rolling a (fair) die, then what is the probability of getting a six?

Proof. $S = \{1, 2, 3, 4, 5, 6\}, E = \{6\}$, then $\mathbb{P}(E) = \frac{|E|}{|S|} = \frac{1}{6}$. ⊛

Example 1.0.2. If we roll a fair die, then what is the probability of rolling a prime?

Proof. $S = \{1, 2, 3, 4, 5, 6\}, E = \{2, 3, 5\}$, then $\mathbb{P}(E) = \frac{|E|}{|S|} = \frac{3}{6} = \frac{1}{2}$. ⊛

Example 1.0.3. Standard deck of 52 cards. Draw a random card, then what is the probability of getting an ace?

Answer. $\frac{1}{13}$. ⊛

Example 1.0.4. Roll two fair dice, what is the probability of the sum being 7?

Proof.

$$S = \{(1, 1), (1, 2), \ldots, (1, 6), (2, 1), (2, 2), \ldots, (6, 6)\},$$

and

$$E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\},$$

so $\mathbb{P}(E) = \frac{|E|}{|S|} = \frac{6}{36} = \frac{1}{6}$. ⊛

1.1 Counting Rule

Theorem 1.1.1 (Counting Rule). If a set S is a disjoint union

$$S = S_1 \uplus S_2 \uplus \ldots \uplus S_n,$$

i.e. $S_i \cap S_j = \varnothing$ for all $i \neq j$, then

$$|S| = \sum_{i=1}^{n} |S_i|.$$

Example 1.1.1. Roll two fair dice. What is the probability of having at least one odd number?

Proof. $E = \{\text{at least one odd roll}\}$, then $E = E_1 \cup E_2$ where $E_1 = \{\text{first die is odd}\}$ and $E_2 = \{\text{second die is odd}\}$. However, $E_1 \cap E_2 \neq \varnothing$. Thus, instead, we define $E_1' = \{\text{first die is odd}\}$ and $E_2' = \{\text{first die is even and second die is odd}\}$, then we know

$$E = E_1' \uplus E_2',$$

so we have $|E| = |E_1'| + |E_2'|$, and $S = \{(x, y) : x, y \in [6]\}$, and we know

$$E_1' = \{(x, y) \mid x_1 \in \{1, 3, 5\}, y \in [6]\}$$

and

$$E_2' = \{(x, y) \mid x \in \{2, 4, 6\}, y \in \{1, 3, 5\}\},$$

which gives $|E_1'| = 18$ and $|E_2'| = 9$, and thus

$$\mathbb{P}(E) = \frac{|E|}{|S|} = \frac{18 + 9}{36} = \frac{3}{4}.$$

⊛

Theorem 1.1.2 (Product Rule). If a set S is the Cartesian product of sets $S_1, S_2, \ldots, S_n$, i.e.

$$S = S_1 \times S_2 \times \cdots \times S_n = \{(a_1, a_2, \ldots, a_n) : \forall i \in [n], a_i \in S_i\},$$

then $|S| = |S_1| \times |S_2| \times \ldots |S_n| = \prod_{i=1}^{n} |S_i|$.

Remark 1.1.1. Informally, if a big chain can be broken into a sequence of smaller choice, then the total number of options is the product of the number of options for each small choice.

Example 1.1.2. Roll two fair dice. What is the probability that the sum is odd?

Proof. $S = \{(x, y) \mid x, y \in [6]\}$. Also, we know

$$E = \{\text{sum is odd}\} = \{\text{exactly one die is odd and the other is even}\} = E_1 \uplus E_2,$$

where $E_1 = \{\text{first is odd and second is even}\}$ and $E_2 = \{\text{first is even and second is odd}\}$. Note that

$$E_1 = \{1, 3, 5\} \times \{2, 4, 6\} \text{ and } E_2 = \{2, 4, 6\} \times \{1, 3, 5\},$$

so $|E_1| = |E_2| = 9$. By the sum rule, we know $|E| = |E_1| + |E_2| = 9 + 9 = 18$, and so

$$\mathbb{P}(E) = \frac{|E|}{|S|} = \frac{18}{36} = \frac{1}{2}.$$

⊛

Theorem 1.1.3 (Advanced Product Rule). If we are making a series of n choices, and for the i-th choice, we always have k_i options available, then the total number of options is

$$k_1 \times k_2 \times \cdots \times k_n = \prod_{i=1}^{n} k_i.$$

Example 1.1.3. Roll two fair dice. What is the probability that the sum is odd?

Proof. We know $S = \{(x, y) : x, y \in [6]\}$, and $E = \{(x, y) \in S : 2 \nmid x + y\}$.

- First question: Which roll is odd?
- Second question: What is the first roll? How many options?
- Third question: What is the second roll? How many options?

For the first question, we have 2 options, the first and the second. For the second question, we know there are 3 options since we need the first die to be even or odd, and similarly we know the second roll also has 3 options. Hence, $|E| = 2 \times 3 \times 3 = 18$, and thus $\mathbb{P}(E) = \frac{1}{2}$ since $|S| = 36$. ⊛

1.1.1 Permutations

Claim 1.1.1. There are

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 1$$

ways to order n distinct elements.

Proof. Use the advanced product rule. For the first option, we have n choices, and the second has $n-1$ options, and so on. ⊛

1.1.2 Combinations

Question. How many subsets of size r are there of an n -element set?

Definition 1.1.1. The binomial coefficient $\binom{n}{r}$, "n choose r" counts the number of r-element subsets of an n-element set.

Claim 1.1.2. $\forall 0 \leq r \leq n$, we have

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

Proof. We count the number of ways of ordering all n items in two different ways. (Double counting)

- First method: Direct permutation, which has $n!$ ways.
- Second method:
  - Step 1: Choose which elements will be in the front r, which has $\binom{n}{r}$ elements.
  - Step 2: Order these r elements, which has $r!$ methods.
  - Step 3: Order the remaining $n - r$ elements, which has $(n-r)!$ methods.

  Thus, by advanced product rule, we know

$$n! = \binom{n}{r} r!(n-r)! \Rightarrow \binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

⊛

Observation. For all $0 \le r \le n$,
$$\binom{n}{r} = \binom{n}{n-r}.$$

Proof.
$$\binom{n}{n-r} = \frac{n!}{(n-r)!(n-(n-r))!} = \frac{n!}{(n-r)!r!} = \binom{n}{r}.$$
∎

Observation. Choosing a subset of r elements is equivalent to choose the $n-r$ elements that don't go in the subset. In fact, it means $\binom{n}{r} = \binom{n}{n-r}$.

Proposition 1.1.1 (Pascal's identity). $\forall 1 \le r \le n$,
$$\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}.$$

Proof.
$$\binom{n}{r} + \binom{n}{r+1} = \frac{n!}{r!(n-r)!} + \frac{n!}{(r+1)!(n-(r+1))!} = \frac{(n+1)!}{r!(n+1-r)!}.$$
∎

Lecture 2

26 Feb.

Another proof for Pascal's identity. Since we know
$$\binom{n}{r} = \#\text{ of subsets of size } r \text{ from a set of size } n\text{, say } [n] = \{1, 2, \ldots, n\}.$$
Let $S = \binom{[n]}{r}$ be the set of these subsets.

Notation. For any set X,
$$\binom{X}{r} = \{Y \subseteq X, |Y| = r\},$$
so $\left|\binom{X}{r}\right| = \binom{|X|}{r}$.

Let
$$S_1 = \left\{y \in \binom{[n]}{r} : n \notin Y\right\} \text{ and } S_2 = \left\{y \in \binom{[n]}{r} : n \in Y\right\},$$
then $S = S_1 \cup S_2$, and thus $|S| = |S_1| + |S_2|$. Now since $|S_1| = \binom{n-1}{r}$ and $|S_2| = \binom{n-1}{r-1}$, so we know
$$\binom{n}{r} = |S| = |S_1| + |S_2| = \binom{n-1}{r} + \binom{n-1}{r-1}.$$
∎

Remark 1.1.2. Pascal's identity gives a recursive formula for computing $\binom{n}{r}$, which is much simpler than $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

Theorem 1.1.4 (Binomial Theorem)**.** For any integer $n > 0$ and any $x, y \in \mathbb{R}$,

$$(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^r y^{n-r}.$$

Proof. Note that

$$(x+y)^n = \underbrace{(x+y)(x+y)\dots(x+y)}_{n \text{ times}}.$$

Each monomial comes from choosing one term from each factor(x or y), taken the from $x^r y^{n-r}$, where r is the number of factors from which we choose x, $0 \le r \le n$. Note that the coefficient of $x^r y^{n-r}$ is $\binom{n}{r}$ for all r, so

$$(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^r y^{n-r}.$$

■

Corollary 1.1.1. Total number of subsets of an n-element set is 2^n.

Proof. Apply the product rule, for each of the elements, ask if it is in the subset or not. We know there are 2 options per question, so the product rule gives 2^n subsets. ■

Another proof. Let S be the set of all subsets. Then for $0 \le r \le n$, let $S_r \subseteq S$ be the subsets of those subsets of size r. Then,

$$S = S_0 \uplus S_1 \uplus \dots \uplus S_n,$$

and by sum rule we know

$$|S| = \sum_{r=0}^{n} |S_r| = \sum_{r=0}^{n} \binom{n}{r} = \sum_{r=0}^{n} \binom{n}{r} 1^r 1^{n-r} (1+1)^n = 2^n.$$

■

Corollary 1.1.2. Let X be an n-element set, $n \ge 1$, then the number of even-sized subsets of X is equal to the number of odd-sized subsets of X.

An approach for odd n and even r.

$$\# \text{ of even-sized subsets} = \sum_{\substack{r \text{ even} \\ 0 \le r \le n}} \binom{n}{r} = \sum_{\substack{r \text{ even} \\ 0 \le r \le n}} \binom{n}{n-r}.$$

Now if n is odd and r even, then $n - r$ is odd, so

$$\sum_{\substack{r \text{ even} \\ 0 \le r \le n}} \binom{n}{n-r} = \sum_{\substack{r \text{ odd} \\ 0 \le r \le n}} \binom{n}{r}.$$

■

Proof. Want to show

$$\sum_{r \text{ even}} \binom{n}{r} - \sum_{r \text{ odd}} \binom{n}{r} = 0,$$

i.e.

$$\sum_{r=0}^{n} \binom{n}{r} (-1)^r = 0.$$

Note that

$$\sum_{r=0}^{n} \binom{n}{r} (-1)^r = \sum_{r=0}^{n} \binom{n}{r} (-1)^r 1^{n-r} = (1 + (-1))^n = 0.$$

■

Example 1.1.4. In poker, we are dealt a hand of five cards from a standard deck. What is the probability of getting a full house (three-of-a-kind and a pair)?

Proof. The sample space is the subsets of 5 cards from deck. If D means the deck of cards, then $S = \binom{D}{5}$ and $|S| = \binom{52}{5}$. Also, we know $E = \{\text{full houses}\} = \{\text{triple} + \text{pair}\}$, so we can first choose the triple then choose the pair, and for triple and pair, we have to choose suits and values. Thus, we have $13 \times \binom{4}{3}$ options for the triple and $12 \times \binom{4}{2}$ options for the pair. Thus, number of full houses is $13 \times \binom{4}{3} \times 12 \times \binom{4}{2}$, and thus

$$\mathbb{P}(\text{full houses}) = \frac{13 \times \binom{4}{3} \times 12 \times \binom{4}{2}}{\binom{52}{5}}.$$

⊛

1.1.3 Choosing with repetition

As previously seen.

$$\begin{aligned}\binom{n}{r} &= \#\text{ of subsets of size } r \text{ of a set of size } n \\ &= \#\text{ of ways of choosing } r \text{ items out of } n \text{ without order and without repetition.}\end{aligned}$$

Question. What if repetition is allowed? How many ways can we choose r items out of n, with repitition but without order?

Let x_i be the number of times the i-th element is chosen, then we want to count the number of $(x_1, x_2, \ldots, x_n)$ pairs satisfying

$$x_i \geq 0, \quad x_i \in \mathbb{Z}, \quad \sum_{i=1}^{n} x_i = r.$$

We can use a method call **stars and bars drawing** to count the number of such pairs. We represent our choices with stars, and use bars to separate the different elements. For example, $(1, 0, 2, 0, 0)$, which is a possible pair, and it corresponds to

$$*||**||,$$

and every possible pair corresponds to a diagram like this, and it is a bijection, while there are r stars and $n-1$ bars in each diagram, so there are $\binom{n+r-1}{n-1}$ possible pairs.

Example 1.1.5. There is a probability course with 77 students. Professor chooses 60 students to pass. How many options does the professor have.

Answer. $\binom{77}{60}$. ⊛

Example 1.1.6. What if the professor instead needs to assign grades to the students. Professor decides to distributes 4500 points between the 77 students. How many ways are there of doing this?

Proof. Let x_i be the number of points to student i, then $x_i \geq 0$ and $\sum_{i=1}^{77} x_i = 4500$. Thus, the number of solutions is $\binom{4576}{4500}$. ⊛

Example 1.1.7. What if every student should receive at least 10 points?

Proof. Now we have a restriction of $x_i \geq 10$ for all i, so we can define $y_i = x_i - 10$ for all i, and thus we want $y_i \geq 0$ for all i and

$$\sum_{i=1}^{77} y_i = \sum_{i=1}^{77} (x_i - 10) = 3730,$$

which means the number of ways of distribution is $\binom{3806}{3730}$.

$\circledast$

Chapter 2

Axiom of Probability

Suppose we roll a fair die 100 times and are interested in the sum of the rolls.

> **Question.** What is the sample space?

We have to define the definition of sample space first.

> **As previously seen.** The sample space is a set S of all possible outcomes, and sometimes denoted by Ω, and the events is a subset $E \subseteq S$ that we are interested in.

Lecture 3

3 March.

Definition 2.0.1. Given events $E_1, E_2, \cdots \subseteq S$,

- $\bigcup_{i=1}^{\infty} E_i = \{x \in S : \text{there exists } i \text{ with } x \in E_i\}$.
- $\bigcap_{i=1}^{\infty} E_i = \{x \in S : x \in E_i \text{ for all } i\}$.
- $E_1, E_2, \ldots$ are mutually exclusive (or pairwise disjoint) if $E_i \cap E_j = \varnothing$ for all $i \neq j$.
- Given $E \subseteq S$, the complement $E^C = \overline{E} = \{x \in S : x \notin E\}$.
- Given $E, F \subseteq S$, $F \setminus E = \{x \in F : x \notin E\} = F \cap E^C$.
- $E \Delta F = (E \setminus F) \cup (F \setminus E)$.

Proposition 2.0.1 (De Morgan's law)**.** Given sets $E_1, E_2, \cdots \subseteq S$,

$$\left(\bigcup_{i=1}^{\infty} E_i\right)^C = \bigcap_{i=1}^{\infty} E_i^C \text{ and } \left(\bigcap_{i=1}^{\infty} E_i\right)^C = \bigcup_{i=1}^{\infty} E_i^C.$$

Proof.

$$\begin{aligned}\left(\bigcup_{i=1}^{\infty} E_i\right)^C &= \left\{x \in S : x \notin \bigcup_{i=1}^{\infty} E_i\right\} = \{x \in S : \nexists i \text{ s.t. } x \in E_i\} \\ &= \{x \in S : \forall i, x \notin E_i\} = \{x \in S : \forall i, x \in E_i^C\} = \bigcap_{i=1}^{\infty} E_i^C.\end{aligned}$$

$$\left(\bigcap_{i=1}^{\infty} E_i\right)^C = \left(\bigcap_{i=1}^{\infty} (E_i^C)^C\right)^C = \left(\left(\bigcup_{i=1}^{\infty} E_i^C\right)^C\right)^C = \left(\bigcup_{i=1}^{\infty} E_i^C\right).$$

■

Definition 2.0.2. Given a sample space S, a probability measure is a function $\mathbb{P} : 2^S \to \mathbb{R}$ satisfying

(i) for every event E, we have $0 \leq \mathbb{P}(E) \leq 1$.

(ii) $\mathbb{P}(S) = 1$.

(iii) (countable additivity) for every sequence of mutually exclusive events $E_1, E_2, \ldots$

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(E_i).$$

Definition 2.0.3. Given a sample space S, and a probability measure $\mathbb{P} : 2^S \to \mathbb{R}$, we call $(S, \mathbb{P})$ a probability space.

Proposition 2.0.2. $\mathbb{P}(\varnothing) = 0$.

Proof. Consider the following sequence of events $E_1 = S$, $E_i = \varnothing$ for all $i \geq 2$. Observe that those are mutually exclusive:

$$\forall i \neq j, E_i \cap E_j = \varnothing$$

since one of E_i, E_j must be empty. Thus,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(E_i),$$

and since

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) = \mathbb{P}(S) = 1,$$

we know

$$1 = \sum_{i=1}^{\infty} \mathbb{P}(E_i) = \mathbb{P}(S) + \sum_{i=2}^{\infty} \mathbb{P}(\varnothing) = 1 + \sum_{i=2}^{\infty} \mathbb{P}(\varnothing).$$

This gives $\mathbb{P}(\varnothing) = 0$. ∎

Proposition 2.0.3. If $E_1, E_2, \ldots, E_n$ is a finite sequence of mutually exclusive events, then

$$\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} \mathbb{P}(E_i).$$

Proof. Consider the sequence

$$E_1, E_2, \ldots, E_n, \varnothing, \varnothing, \ldots,$$

then we have

$$\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(E_i) = \sum_{i=1}^{n} \mathbb{P}(E_i).$$

∎

Proposition 2.0.4. For any event E,

$$\mathbb{P}\left(E^C\right) = 1 - \mathbb{P}(E).$$

Proof. Consider the finite sequence

$$E_1 = E \text{ and } E_2 = E^C,$$

then

$$\sum_{i=1}^{2} \mathbb{P}(E_i) = \mathbb{P}\left(\bigcup_{i=1}^{2} E_i\right) = \mathbb{P}(S) = 1,$$

which gives

$$\mathbb{P}(E_1) = 1 - \mathbb{P}\left(E_1^C\right).$$

■

Proposition 2.0.5 (Monotonity). If $E \subseteq F$ are events, then

$$\mathbb{P}(E) \leq \mathbb{P}(F).$$

Proof. Note that $F = E \cup (F \setminus E)$, so

$$\mathbb{P}(F) = \mathbb{P}(E) + \mathbb{P}(F \setminus E) \geq \mathbb{P}(E).$$

■

Example 2.0.1. We have an unfair six-sided die, which rolls 1 with probability 0.1, a 2 with probability 0.2, a 3 with probability 0.25, a 4 with probability 0.15, and a 5 with probability 0.17. Then,

(a) What is the probability of rolling a 6?

(b) What is the probability of rolling a prime?

Proof. To define the probability over a countable sample space $S = \{1, 2, 3, 4, 5, 6\}$, it is enough to define the probability of individual outcomes $\{x\} \subseteq S$. Since for any $E \subseteq S$, we know $E = \bigcup_{x \in E} \{x\}$ and thus $\mathbb{P}(E) = \sum_{x \in E} \mathbb{P}(\{x\})$. Hence, we know

$$\mathbb{P}(\{6\}) = \mathbb{P}\left(\{1, 2, 3, 4, 5\}^C\right) = 1 - \sum_{i=1}^{5} \mathbb{P}(\{i\}) = 0.13.$$

Also,

$$\mathbb{P}(\text{rolling a prime}) = 0.62.$$

⊛

Example 2.0.2. Any uniform probability measure satisfies the axiom.

Proof. Let S be a finite non-empty sample space. We define the uniform probability measure as

$$\mathbb{P}(E) = \frac{|E|}{|S|}.$$

Fix any $E \subseteq S$, $0 \leq |E| \leq |S|$, so

$$0 \leq \frac{|E|}{|S|} \leq 1.$$

Also, $\mathbb{P}(S) = \frac{|S|}{|S|} = 1$. If $E_1, E_2, \ldots$ are mutually exclusive, then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) = \frac{\left|\bigcup_{i=1}^{\infty} E_i\right|}{|S|} = \frac{\sum_{i=1}^{\infty} |E_i|}{|S|} = \sum_{i=1}^{\infty} \frac{|E_i|}{|S|} = \sum_{i=1}^{\infty} \mathbb{P}(E_i).$$

Thus, any uniform probability measure satisfies the axiom. ⊛

2.1 Inclusion-Exclusion Principle

Proposition 2.1.1. For any two events $E, F \subseteq S$,

$$\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F).$$

Proof. Consider the mutually exclusive events $E \cap F, E \setminus F, F \setminus E$, we know

$$(E \cap F) \cup (E \setminus F) \cup (F \setminus E) = E \cup F.$$

Also,

$$(E \cap F) \cup (E \setminus F) = E \text{ and } (E \cap F) \cup (F \setminus E) = F.$$

Hence,

$$\begin{aligned}
\mathbb{P}(E \cup F) &= \mathbb{P}(E \cap F) + \mathbb{P}(E \setminus F) + \mathbb{P}(F \setminus E) \\
&= \mathbb{P}(E \cap F) + \mathbb{P}(E \setminus F) + \mathbb{P}(F \setminus E) + \mathbb{P}(E \cap F) - \mathbb{P}(E \cap F) \\
&= \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F).
\end{aligned}$$

■

Corollary 2.1.1. For any $E, F \subseteq S$, we know

$$\mathbb{P}(E \cup F) \leq \mathbb{P}(E) + \mathbb{P}(F).$$

Lecture 4

5 March.

Example 2.1.1. We will hike if it doesn't rain and there is no earthquake. Otherwise we will stay in the city and eat food and get fat. Now if

$$\mathbb{P}(\text{rain}) = 60\%, \quad \mathbb{P}(\text{earthquake}) = 5\%, \quad \mathbb{P}(\text{rainy earthquake}) = 17\%.$$

What is $\mathbb{P}(\text{fat})$?

Proof. If $E =$rain and $F =$earthquake, then

$$\mathbb{P}(E) = 60\%, \quad \mathbb{P}(F) = 5\%, \quad \mathbb{P}(E \cap F) = 1\%.$$

Thus,

$$\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F) = 64\%.$$

Remark 2.1.1. Even if we don't know $\mathbb{P}(E \cap F)$, we could have said

$$\mathbb{P}(E) = 60\% \leq \mathbb{P}(E \cup F) \leq 65\% = \mathbb{P}(E) + \mathbb{P}(F).$$

⊛

Proposition 2.1.2. Given any three events E, F, G in a probability space $(S, \mathbb{P})$,

$$\mathbb{P}(E \cup F \cup G) = \mathbb{P}(E) + \mathbb{P}(F) + \mathbb{P}(G) - \mathbb{P}(E \cap F) - \mathbb{P}(F \cap G) - \mathbb{P}(E \cap G) + \mathbb{P}(E \cap F \cap G).$$

Proof.

$$\begin{aligned}
\mathbb{P}(E \cup F \cup G) = \mathbb{P}((E \cup F) \cup G) &= \mathbb{P}(E \cup F) + \mathbb{P}(G) - \mathbb{P}((E \cup F) \cap G) \\
&= \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F) + \mathbb{P}(G) - \mathbb{P}((E \cap G) \cup (F \cap G)) \\
&= \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F) + \mathbb{P}(G) - (\mathbb{P}(E \cap G) + \mathbb{P}(F \cap G) - \mathbb{P}(E \cap F \cap G)) \\
&= \mathbb{P}(E) + \mathbb{P}(F) + \mathbb{P}(G) - \mathbb{P}(E \cap F) - \mathbb{P}(F \cap G) - \mathbb{P}(E \cap G) + \mathbb{P}(E \cap F \cap G).
\end{aligned}$$

■

Example 2.1.2. Rolling a 60-sided die, numbered 1 to 60, and is fair. We win if rolling a multiple of $2, 3, 5$, then $\mathbb{P}(\text{win}) = ?$

Proof. We know $S = [60]$ with uniform probability. If $E \subseteq S$, then $\mathbb{P}(E) = \frac{|E|}{60}$. Suppose

$$W = \{\text{win number}\} = \{x \in S : x \text{ is divisible by } 2, 3, 5\},$$

and given $k \in \mathbb{N}$, let

$$E_k = \{x \in S : x \text{ is divisible by } k\},$$

then $W = E_2 \cup E_3 \cup E_5$. Note that

$$\mathbb{P}(E_2) = \frac{30}{60} = \frac{1}{2}, \quad \mathbb{P}(E_3) = \frac{20}{60} = \frac{1}{3}, \quad \mathbb{P}(E_5) = \frac{12}{60} = \frac{1}{5}.$$

Thus, we can compute $\mathbb{P}(W)$ by

$$\mathbb{P}(W) = \mathbb{P}(E_2) + \mathbb{P}(E_3) + \mathbb{P}(E_5) - \mathbb{P}(E_2 \cap E_3) - \mathbb{P}(E_3 \cap E_5) - \mathbb{P}(E_2 \cap E_5) + \mathbb{P}(E_2 \cap E_3 \cap E_5),$$

and note that if k and l are relatively prime, then $E_k \cap E_l = E_{kl}$. Thus,

$$\mathbb{P}(W) = \frac{1}{2} + \frac{1}{3} + \frac{1}{5} - \frac{1}{6} - \frac{1}{10} - \frac{1}{15} + \frac{1}{30} = \frac{22}{30}.$$

⊛

Theorem 2.1.1. Let $(S, \mathbb{P})$ be a probability space, and let $E_1, E_2, \ldots, E_n$ be arbitrary events ($E_i \subseteq S$). Then,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{r=1}^{n} (-1)^{r+1} \sum_{1 \leq i_1 < i_2 < \cdots < i_r \leq n} \mathbb{P}(E_{i_1} \cap E_{i_2} \cap \cdots \cap E_{i_r}) = \sum_{\varnothing \neq I \subseteq [n]} (-1)^{|I|+1} \mathbb{P}\left(\bigcap_{i \in I} E_i\right).$$

Proof. Define

$$F_I = \left(\bigcap_{i \in I} E_i\right) \cap \left(\bigcap_{i \notin I} E_i^C\right) \text{ for all } I \subseteq [n].$$

Observe

$$E_1 \cup E_2 \cup \cdots \cup E_n = \bigcup_{\substack{I \subseteq [n] \\ I \neq \varnothing}} F_I.$$

Now since $F_I \cap F_J = \varnothing$ for $I, J \subseteq [n]$ where I, J are distinct. Thus,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) = \mathbb{P}\left(\bigcup_{\substack{I \subseteq [n] \\ I \neq \varnothing}} F_I\right) = \sum_{\substack{I \subseteq [n] \\ I \neq \varnothing}} \mathbb{P}(F_I).$$

Furthermore, for any $I \subseteq [n]$,

$$\bigcap_{i \in I} E_i = \bigcup_{\substack{J \subseteq [n] \\ I \subseteq J}} F_J.$$

Thus,

$$\mathbb{P}\left(\bigcap_{i \in I} E_j\right) = \sum_{I \subseteq J \subseteq [n]} \mathbb{P}(F_J).$$

This gives

$$\sum_{\varnothing\neq I\subseteq[n]}(-1)^{|I|+1}\mathbb{P}\left(\bigcap_{i\in I}E_i\right)=\sum_{\varnothing\neq I\subseteq[n]}(-1)^{|I|+1}\sum_{I\subseteq J\subseteq[n]}\mathbb{P}(F_J)=\sum_{\varnothing\neq J\subseteq[n]}\mathbb{P}(F_J)\left[\sum_{\varnothing\neq I\subseteq J}(-1)^{|I|+1}\right]$$
$$=\sum_{\varnothing\neq J\subseteq[n]}\mathbb{P}(F_J)=\mathbb{P}\left(\bigcup_{i=1}^{n}E_i\right).$$

Remark 2.1.2. For fixed $J\subseteq[n]$,

$$\sum_{\varnothing\neq I\subseteq J}(-1)^{|I|+1}=1$$

since

$$\sum_{I\subseteq J}(-1)^{|I|+1}=\sum_{r=0}^{|J|}\binom{|J|}{r}(-1)^{r+1}=(-1)\sum_{r=0}^{|J|}\binom{|J|}{r}(-1)^{r}1^{|J|-r}=0.$$

■

Example 2.1.3. If there is a party, and everyone brings a hat. If now everyone put their hats together and then randomly gets back one, and we want to compute the probability of somebody gets his hat.

Proof. Now if $S=S_n$ and $\mathbb{P}$ is uniform and E is the set of events that somebody gets their hat. Let

$$E_i=\{\text{person } i \text{ gets his hat}\}=\{\pi\in S_n:\pi(i)=i\}.$$

Then, $E=\bigcup_{i=1}^{n}E_i$. Thus, by inclusion-exclusion principle,

$$\mathbb{P}(E)=\mathbb{P}\left(\bigcup_{i=1}^{n}E_i\right)=\sum_{\varnothing\neq I\subseteq[n]}(-1)^{|I|+1}\mathbb{P}\left(\bigcap_{i\in I}E_i\right).$$

Note that $\bigcap_{i\in I}E_i$ is the set of events that all people in I gets their hats, so

$$\left|\bigcap_{i\in I}E_i\right|=(n-|I|)!$$

Hence, since $\mathbb{P}$ is uniform,

$$\mathbb{P}(E)=\sum_{\varnothing\neq I\subseteq[n]}(-1)^{|I|+1}\frac{(n-|I|)!}{n!}=\sum_{r=1}^{n}\binom{n}{r}(-1)^{r+1}\frac{(n-r)!}{n!}=\sum_{r=1}^{n}\frac{(-1)^{r+1}}{r!}.$$

Hence,

$$\mathbb{P}\left(\text{nobody gets their hat}\right)=1-\sum_{r=1}^{n}\frac{(-1)^{r+1}}{r!}=\sum_{r=0}^{n}\frac{(-1)^{r}}{r!}\to\frac{1}{e}\text{ as }n\to\infty.$$

⊛

Lecture 5

10 March.

Proposition 2.1.3 (Union bound). Let $E_1, E_2, \ldots, E_n$ be events in a probability space. Then,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) \leq \sum_{i=1}^{n} \mathbb{P}(E_i).$$

Proof. Let $F_i = E_i \setminus \bigcup_{j<i} E_j$, then $\bigcup_{j=1}^{i} F_j = \bigcup_{j=1}^{i} E_j$.(Since $F_j \subseteq E_j$ for all j, we have $\bigcup_{j=1}^{i} F_j \subseteq \bigcup_{j=1}^{i} E_j$. If $x \in \bigcup_{j=1}^{i} E_j$, let j_0 be the smallest index s.t. $x \in E_{j_0}$, then $x \in F_{j_0} = E_{j_0} \setminus \bigcup_{\ell<j_0} E_\ell$, so $x \in \bigcup_{j=1}^{i} F_j$.) Note that F_j's are pairwise disjoint. Thus,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{n} F_i\right) = \sum_{i=1}^{n} \mathbb{P}(F_i) \leq \sum_{i=1}^{n} \mathbb{P}(E_i)$$

by the axiom of probability. ■

Example 2.1.4. Let $\pi \in S_n$ be a uniformly random permutation. Let E_i be the event that $\{\pi(i) = i\}$, then

$$\mathbb{P}(\bigcup_{i=1}^{n} E_i) \leq \sum_{i=1}^{n} \mathbb{P}(E_i) = \sum_{i=1}^{n} \frac{1}{n} = 1.$$

Theorem 2.1.2 (Bonferroni). Let $E_1, E_2, \ldots, E_n$ be events in a probability space $(S, \mathbb{P})$. Let $1 \leq t \leq n$. Then

$$\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) \left\{ \begin{matrix} \leq \\ \geq \end{matrix} \right\} \sum_{r=1}^{t} (-1)^{r+1} \sum_{1 \leq i_1 < i_2 < \cdots < i_r \leq n} \mathbb{P}(E_{i_1} \cap E_{i_2} \cap \cdots \cap E_{i_r}) \text{ if } t \text{ is } \begin{cases} \text{odd} \\ \text{even} \end{cases}.$$

Proof. We induct on t.

- Base case: $t = 1$, which can be proved by union bound.
- Induction step: As before, we have $\bigcup_{i=1}^{n} E_i = \bigcup_{i=1}^{n} F_i$, where $F_i = E_i \setminus \bigcup_{j<i} E_j$. Then

$$\begin{aligned}
\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{n} F_i\right) &= \sum_{i=1}^{n} \mathbb{P}(F_i) = \sum_{i=1}^{n} \mathbb{P}\left(E_i \setminus \bigcup_{j<i} E_j\right) \\
&= \sum_{i=1}^{n} \left[\mathbb{P}(E_i) - \mathbb{P}\left(E_i \cap \left(\bigcup_{j<i} E_j\right)\right)\right] \\
&= \sum_{i=1}^{n} \mathbb{P}(E_i) - \sum_{i=1}^{n} \mathbb{P}\left(E_i \cap \left(\bigcup_{j<i} E_j\right)\right) \\
&= \sum_{i=1}^{n} \mathbb{P}(E_i) - \sum_{i=1}^{n} \mathbb{P}\left(\bigcup_{j<i} (E_i \cap E_j)\right).
\end{aligned}$$

We use the induction hypothesis for t to bound $\mathbb{P}\left(\bigcup_{j<i}(E_i \cap E_j)\right)$.

If t is odd, then

$$\mathbb{P}\left(\bigcup_{j<i}(E_i \cap E_j)\right) \leq \sum_{r=1}^{t}(-1)^{r+1} \sum_{1\leq j_1<j_2<\cdots<j_r<i} \mathbb{P}\left(\bigcap_{k=1}^{r}(E_i \cap E_{j_k})\right)$$
$$= \sum_{i=1}^{t}(-1)^{r+1} \sum_{1\leq j_1<j_2<\cdots<j_r<i} \mathbb{P}(E_{j_1} \cap E_{j_2} \cap \cdots \cap E_{j_r} \cap E_i).$$

Thus,

$$\sum_{i=1}^{n}\mathbb{P}\left(\bigcup_{j<i}(E_i \cap E_j)\right) \leq \sum_{i=1}^{n}\sum_{r=1}^{t}(-1)^{r+1} \sum_{1\leq j_1<j_2<\cdots<j_r<i} \mathbb{P}\left(E_{j_1} \cap E_{j_2} \cap \cdots \cap E_{j_r} \cap E_i\right)$$
$$= \sum_{r=2}^{t+1}(-1)^{r} \sum_{1\leq j_1<j_2<\cdots<j_r\leq n} \mathbb{P}(E_{j_1} \cap \cdots \cap E_{j_r}).$$

Thus,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n}\mathbb{P}(E_i) - \sum_{i=1}^{n}\mathbb{P}\left(\bigcup_{j<i}(E_j \cap E_i)\right)$$
$$\geq \sum_{i=1}^{n}\mathbb{P}(E_i) - \sum_{r=2}^{t+1}(-1)^{r} \sum_{1\leq j_1<j_2<\cdots<j_r\leq n} \mathbb{P}(E_{j_1} \cap \cdots \cap E_{j_r})$$
$$= \sum_{r=1}^{t+1}(-1)^{r+1} \sum_{1\leq i_1<\cdots<i_r\leq n} \mathbb{P}(E_{i_1} \cap E_{i_2} \cap \cdots \cap E_{i_r}).$$

If t is even, then we can prove similarly. ■

Example 2.1.5. Let $E_i = \{\pi \in S_n : \pi(i) = i\}$, and $S = S_n$, then

$$\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) \geq \sum_{i=1}^{n}\mathbb{P}(E_i) - \sum_{1\leq i<j\leq n} \mathbb{P}(E_1 \cap E_j) = \sum_{i=1}^{n}\frac{1}{n} - \sum_{1\leq i<j\leq n}\frac{1}{n(n-1)}$$
$$= 1 - \binom{n}{2}\frac{1}{n(n-1)} = 1 - \frac{1}{2} = \frac{1}{2}.$$

Chapter 3

Continuity of Probability

Definition 3.0.1. Given a probability space $(S, \mathbb{P})$, we say a sequence of events $E_1, E_2, \ldots$ is increasing if $E_1 \subseteq E_2 \subseteq \ldots$ and is decreasing if $E_1 \supseteq E_2 \supseteq \ldots$.

Definition 3.0.2. Let $E_1 \subseteq E_2 \subseteq \ldots$ be an increasing sequence of events in a probability space. Then we define the limit $\lim_{n\to\infty} E_n = \bigcup_{n=1}^{\infty} E_n$. (Note that $\bigcup_{i=1}^{n} E_i = E_n$.) If $E_1 \supseteq E_2 \supseteq \ldots$ is a decreasing sequence of events, then we define $\lim_{n\to\infty} E_n = \bigcap_{n=1}^{\infty} E_n$. (Note that $E_n = \bigcap_{i=1}^{n} E_i$.)

Remark 3.0.1. If $E_1, E_2, \ldots$

- is increasing, then
$$E_1 \subseteq E_2 \subseteq \cdots \subseteq E_n \subseteq \cdots \subseteq \lim_{n\to\infty} E_n.$$
- is decreasing, then
$$E_1 \supseteq E_2 \supseteq \cdots \supseteq E_n \supseteq \cdots \supseteq \lim_{n\to\infty} E_n.$$

Proposition 3.0.1. If $E_1, E_2, \ldots$ is an increasing or decreasing sequence of events in a probability space $(S, \mathbb{P})$, then
$$\mathbb{P}\left(\lim_{n\to\infty} E_n\right) = \lim_{n\to\infty} \mathbb{P}(E_n),$$
i.e. probability is continuous.

Proof. We first show the increasing case. For each n, let $F_n = E_n \setminus \bigcup_{i=1}^{n-1} E_i$. Then, for any n,
$$\bigcup_{i=1}^{n} F_i = \bigcup_{i=1}^{n} E_i = E_n.$$
Also,
$$\bigcup_{n=1}^{\infty} F_n = \bigcup_{n=1}^{\infty} E_n = \lim_{n\to\infty} E_n.$$
Thus, since F_i's are mutually disjoint,
$$\mathbb{P}(\lim_{n\to\infty} E_n) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} E_n\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} F_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(F_n) = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbb{P}(F_i) = \lim_{n\to\infty} \mathbb{P}\left(\bigcup_{i=1}^{n} F_i\right)$$
$$= \lim_{n\to\infty} \mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) = \lim_{n\to\infty} \mathbb{P}(E_n).$$

Now we show the decreasing case. If $E_1 \supseteq E_2 \supseteq \ldots$ is decreasing, then

$$E_1^C \subseteq E_2^C \subseteq \ldots \text{ is increasing.}$$

Thus,

$$\lim_{n\to\infty} \mathbb{P}\left(E_n^C\right) = \mathbb{P}\left(\lim_{n\to\infty} E_n^C\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} E_n^C\right) = \mathbb{P}\left(\left(\bigcap_{n=1}^{\infty} E_n\right)^C\right) = 1 - \mathbb{P}\left(\bigcap_{n=1}^{\infty} E_n\right).$$

Also,

$$\lim_{n\to\infty} \mathbb{P}\left(E_n^C\right) = \lim_{n\to\infty} 1 - \mathbb{P}(E_n),$$

so

$$\lim_{n\to\infty} 1 - \mathbb{P}(E_n) = 1 - \mathbb{P}\left(\bigcap_{n=1}^{\infty} E_n\right) \Rightarrow \lim_{n\to\infty} \mathbb{P}(E_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} E_n\right).$$

■

Definition 3.0.3. Given any infinite sequence of events $E_1, E_2, E_3, \ldots$ in a probability space $(S, \mathbb{P})$, we define:

$$\limsup_{n\to\infty} E_n = \lim_{n\to\infty} \bigcup_{i=n}^{\infty} E_i = \bigcap_{n=1}^{\infty} \left(\bigcup_{i=n}^{\infty} E_i\right).$$

Notice the sequence $\bigcup_{i=n}^{\infty} E_i$ for $n \in \mathbb{N}$ is decreasing in n. Also, we define

$$\liminf_{n\to\infty} E_n = \lim_{n\to\infty} \bigcap_{i=n}^{\infty} E_i = \bigcup_{n=1}^{\infty} \left(\bigcap_{i=n}^{\infty} E_i\right).$$

Notice that $\bigcap_{i=n}^{\infty} E_i$ is increasing.

Remark 3.0.2. If $x \in \bigcup_{i=n}^{\infty} E_i$, then it means $\exists i \geq n$ s.t. $x \in E_i$. Hence, if $x \in \bigcap_{n=1}^{\infty} \left(\bigcup_{i=n}^{\infty} E_i\right)$, then it means for all $n \geq 1$, $\exists i \geq n$ s.t. $x \in E_i$, i.e. x is in infinitely many of the events E_i. Thus, $\limsup_{n\to\infty} E_n$ means infinitely many of the events E_n occurs, while $\liminf_{n\to\infty} E_n$ means only finitely many of the events do not occur since $x \in \bigcup_{n=1}^{\infty} \left(\bigcap_{i=n}^{\infty} E_i\right)$ means $\exists n \geq 1$ s.t. $\forall i \geq n$, $x \in E_i$.

Theorem 3.0.1 (The first Borel-Cantelli Lemma)**.** Let $(S, \mathbb{P})$ be a probability space, and let $E_1, E_2, \ldots$ be a sequence of events. If $\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty$, then $\mathbb{P}\left(\limsup_{n\to\infty} E_n\right) = 0$.

Proof.

$$0 \leq \mathbb{P}\left(\limsup_{n\to\infty} E_n\right) = \mathbb{P}\left(\lim_{n\to\infty} \bigcup_{i=n}^{\infty} E_i\right) = \lim_{n\to\infty} \mathbb{P}\left(\bigcup_{i=n}^{\infty} E_i\right) \leq \lim_{n\to\infty} \sum_{i=n}^{\infty} \mathbb{P}(E_i) = 0$$

since $\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty$. ■

Lecture 6

Example 3.0.1. Now the restaurant runs a promotion. If the customer toss a lucky coin and gets 8 heads in a row, then he/she can eat for free, then 12 March.

$$\mathbb{P}\left(\text{customer } i \text{ eat for free}\right) = \frac{1}{2^8}.$$

Now the i-th customer gets free food if they toss i heads in a row, and suppose

$$E_i = \{i\text{-th customer gets free food}\},$$

then $\mathbb{P}(E_i) = \frac{1}{2^i}$. Thus,

$$\sum_{i=1}^{\infty} \mathbb{P}(E_i) = \sum_{i=1}^{\infty} \frac{1}{2^i} = 1 < \infty.$$

Thus, $\mathbb{P}(\limsup_{n\to\infty} E_n) = 0$. This means almost certainly, only finitely many customers get free food.

Example 3.0.2. There are four coins, which we toss n times. Let H_n be the number of heads. Let E_n be the events that

$$H_n > \left(\frac{1}{2} + \varepsilon\right) n, \text{ for some } \varepsilon > 0.$$

We know $S = \{H, T\}^n$ and

$$\begin{aligned} E_n \subseteq S &= \left\{w \in \{H,T\}^n : w \text{ has } \geq \left(\frac{1}{2} + \varepsilon\right) n \text{ heads}\right\} \\ &= \bigcup_{r=\left(\frac{1}{2}+\varepsilon\right)n} F_r, \text{ where } F_r = \{w : w \text{ has exactly } r \text{ heads}\}, \end{aligned}$$

so

$$\mathbb{P}(E_n) = \sum_{r=\left(\frac{1}{2}+\varepsilon\right)n} \mathbb{P}(F_r).$$

Note that

$$\mathbb{P}(F_r) = \frac{|F_r|}{|S_n|},$$

where $|S_n| = 2^n$ and $|F_r| = \binom{n}{r}$. Thus,

$$\mathbb{P}(F_r) = \frac{\binom{n}{r}}{2^n},$$

and thus

$$\mathbb{P}(E_n) = \sum_{r=\left(\frac{1}{2}+\varepsilon\right)n} \frac{\binom{n}{r}}{2^n} = 2^{-n} \sum_{r=\left(\frac{1}{2}+\varepsilon\right)n}^{n} \binom{n}{r}.$$

Definition 3.0.4 (The binary entropy function). $H(x) = -x\log_2 x - (1-x)\log_2(1-x)$ for $0 \leq x \leq 1$.

We use the fact that for $\varepsilon > 0$,

$$\sum_{r=\left(\frac{1}{2}+\varepsilon\right)n}^{n} \binom{n}{r} \approx 2^{H\left(\frac{1}{2}+\varepsilon\right)n} \text{ and } H(x) \leq 1 \text{ with equality only when } x = \frac{1}{2}.$$

Thus, we can let $H\left(\frac{1}{2}+\varepsilon\right) = 1 - \delta$ where $\delta > 0$, then

$$\mathbb{P}(E_n) \lesssim 2^{-n} 2^{(1-\delta)n} = 2^{-\delta n},$$

so

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) \lesssim \sum_{n=1}^{\infty} 2^{\delta n} < \infty \text{ since } 2^{-\delta} < 1.$$

Thus, by First Borrel-Cantelli lemma, $\mathbb{P}(\limsup_{n\to\infty} E_n) = 0$, i.e.

$$\mathbb{P}\left(\exists N \text{ s.t. } \forall n \geq N, \text{ number of heads} < \left(\frac{1}{2} + \varepsilon\right) n\right) = 1.$$

Let

$$h_n = \frac{\text{number of heads in } n \text{ toses}}{n},$$

we showed that for any $\varepsilon > 0$ with probability 1, there is some N s.t. if $n \geq N$, then $h_n \leq \frac{1}{2} + \varepsilon$, so

$$\limsup_{n\to\infty} h_n \leq \frac{1}{2}.$$

By symmetry, we can show that for

$$t_n = \frac{\text{number of tails in } n \text{ toses}}{n},$$

$\limsup_{n\to\infty} t_n \leq \frac{1}{2}$. Since $h_n + t_n = 1$, and $\limsup_{n\to\infty} t_n \leq \frac{1}{2}$ gives $\liminf_{n\to\infty} h_n \geq \frac{1}{2}$, so $\lim_{n\to\infty} h_n = \frac{1}{2}$.

Chapter 4

Conditional Probabilities and Independence

Definition 4.0.1. Let $(S, \mathbb{P})$ be a probability space, and let $F \subseteq S$ be an event with $\mathbb{P}(F) > 0$. The conditional probability of an event E given F is

$$\mathbb{P}(E \mid F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)}.$$

Proposition 4.0.1.

(1) $0 \leq \mathbb{P}(E \mid F) \leq 1$ for every $E \subseteq S$.

(2) $\mathbb{P}(F \mid F) = 1$.

(3) If E and F are disjoint, i.e. $E \cap F = \varnothing$, then

$$\mathbb{P}(E \mid F) = \frac{\mathbb{P}(\varnothing)}{\mathbb{P}(F)} = 0.$$

(4) $\mathbb{P}(E \cap F) = \mathbb{P}(E \mid F)\mathbb{P}(F)$.

(5) If $E_1, E_2, E_3, \ldots$ are mutually exclusive, then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i \mid F\right) = \sum_{i=1}^{\infty} \mathbb{P}(E_i \mid F).$$

Corollary 4.0.1. $(S, \mathbb{P}(\cdot \mid F))$ is also a probability space, where:

- $0 \leq \mathbb{P}(E \mid F) \leq 1$.
- $\mathbb{P}(S \mid F) = 1$.
- Countable additivity.

Example 4.0.1. n people at n party, each with their own hat. At the end, every one gets a random hat. What is the probability that exactly r people get their own hats?

Proof. Let $E_r = \{\text{exactly } r \text{ people get their own hats}\}$. Then, $E_r = \bigcup_{\substack{R \subseteq [n] \\ |R| = r}} E_R$, where

$$E_R = \{\text{people in } R \text{ get their own hats and people not in } R \text{ not}\}.$$

Thus,

$$\mathbb{P}(E_r) = \mathbb{P}\left(\bigcup_{\substack{R\subseteq[n]\\|R|=r}} E_R\right) = \sum_{\substack{R\subseteq[n]\\|R|=r}} \mathbb{P}(E_R).$$

By symmetry, E_R has the same probability for every R. Thus,

$$\mathbb{P}(E_r) = \binom{n}{r}\mathbb{P}\left(E_{\{n-r+1,n-r+2,\dots,n\}}\right).$$

Note that

$$E_{\{n-r+1,\dots,n\}} = \underbrace{\{\text{people in } n-r+1,\dots,n \text{ get their hats}\}}_{F} \cap \underbrace{\{\text{nobody in } 1,\dots,n-r \text{ get their hats}\}}_{E}.$$

Thus, $\mathbb{P}(E\cap F) = \mathbb{P}(E \mid F)\mathbb{P}(F)$, where

$$\mathbb{P}(F) = \frac{|F|}{|S_n|} = \frac{(n-r)!}{n!}$$

and

$$\mathbb{P}(E \mid F) = \mathbb{P}(\text{dearrangement of } n-r \text{ hats}) = \sum_{k=0}^{n-r}\frac{(-1)^k}{k!}$$

by our previous argument, so

$$\mathbb{P}(E\cap F) = \frac{(n-r)!}{n!}\sum_{k=0}^{n-r}\frac{(-1)^k}{k!}.$$

⊛

4.1 The Law of Total Probability

Definition 4.1.1. Given a set S, we say $E_1, E_2, \dots, E_n$ partition S if

$$E_1 \uplus E_2 \uplus \dots \uplus E_n = S.$$

Theorem 4.1.1 (Law of Total Probability). Let $(S, \mathbb{P})$ be a probability space, and let $S = F_1 \uplus F_2 \uplus \dots \uplus F_n$ be a partition, then for any event E,

$$\mathbb{P}(E) = \sum_{i=1}^{n}\mathbb{P}(E \mid F_i)\mathbb{P}(F_i).$$

Proof.

$$E = E\cap S = E\cap\left(\bigcup_{i=1}^{n}F_i\right) = \bigcup_{i=1}^{n}(E\cap F_i),$$

so

$$\mathbb{P}(E) = \sum_{i=1}^{n}\mathbb{P}(E\cap F_i) = \sum_{i=1}^{n}\mathbb{P}(E \mid F_i)\mathbb{P}(F_i).$$

■

Lecture 7

17 March.

Example 4.1.1. A professor is grading homework while watching Man City vs Real Madroid. If Real Madroid win, professor is happy, and students pass with probability 80%. If they don't, professor is sad, and students pass with probability 50%. Now we know Real Madroid win with probability 67%. What is the probability that a student will pass?

Proof. Let

$$S = \left\{ (x, y) : x = \begin{cases} 1, & \text{if Madroid win;} \\ 0, & \text{otherwise.} \end{cases}, y = \begin{cases} 1, & \text{if student passes;} \\ 0, & \text{otherwise.} \end{cases} \right\},$$

Let $F_1 = \{\text{Real Madroid win}\}, F_2 = \{\text{Real Madroid lose}\}$, then $S = F_1 \cup F_2$. Now let $E = \{\text{student passes}\}$, then since we know

$$\mathbb{P}(F_1) = 0.67, \mathbb{P}(F_2) = 0.33, \mathbb{P}(E \mid F_1) = 0.8, \mathbb{P}(E \mid F_2) = 0.5,$$

then

$$\mathbb{P}(E) = \mathbb{P}(E \mid F_1)\mathbb{P}(F_1) + \mathbb{P}(E \mid F_2)\mathbb{P}(F_2) = 0.70.$$

⊛

Theorem 4.1.2 (Bayes' Rule)**.** Let $(S, \mathbb{P})$ be a probability space, and suppose we have partition $S = F_1 \cup F_2 \cup \ldots \cup F_n$, and an event $E \subseteq S$, then for any $1 \leq i \leq n$,

$$\mathbb{P}(F_i \mid E) = \frac{\mathbb{P}(E \mid F_i)\mathbb{P}(F_i)}{\sum_{j=1}^{n} \mathbb{P}(E \mid F_j)\mathbb{P}(F_j)}.$$

Proof.

$$\mathbb{P}(F_i \mid E) = \frac{\mathbb{P}(F_i \cap E)}{\mathbb{P}(E)} = \frac{\mathbb{P}(E \cap F_i)}{\mathbb{P}(E)} = \frac{\mathbb{P}(E \mid F_i)\mathbb{P}(F_i)}{\mathbb{P}(E)} = \frac{\mathbb{P}(E \mid F_i)\mathbb{P}(F_i)}{\sum_{j=1}^{n} \mathbb{P}(E \mid F_j)\mathbb{P}(F_j)}.$$

■

Example 4.1.2. A student passes their homeworks. What's the probability that Real Madroid won?

Proof.

$$\mathbb{P}(F_1 \mid E) = \frac{\mathbb{P}(E \mid F_1)\mathbb{P}(F_1)}{\mathbb{P}(E \mid F_1)\mathbb{P}(F_1) + \mathbb{P}(E \mid F_2)\mathbb{P}(F_2)} \approx 0.76.$$

⊛

Example 4.1.3. 1% of the population has COVID. A rapid test is 95% accurate. A random person tests positive for COVID. What is the probability they actually have COVID?

Proof. Let $E = \{\text{have COVID}\}$ and $F = \{\text{test positive}\}$. We want to calculate $\mathbb{P}(E \mid F)$. We know $\mathbb{P}(F \mid E) = 0.95$ and $\mathbb{P}(E) = 0.01$, and since $S = E \cup E^C$, so

$$\mathbb{P}(E \mid F) = \frac{\mathbb{P}(F \mid E)\mathbb{P}(E)}{\mathbb{P}(F \mid E)\mathbb{P}(E) + \mathbb{P}(F \mid E^C)\mathbb{P}(E^C)}.$$

Note that $\mathbb{P}(F \mid E^C) = 0.05$. Thus,

$$\mathbb{P}(E \mid F) \approx 0.16.$$

⊛

Remark 4.1.1. The definition of accuracy of the rapid test is "Given a random person, the probability of the rapid test giving the correct result."

Example 4.1.4. City has 3000000 populations, and the DNA test is 99.99% accurate. If we take a random person and get a match, what is the probability it was their DNA?

Proof. Let $E = \{\text{have a right person}\}$, and $F = \{\text{test gives a match}\}$. We want to calculate $\mathbb{P}(E \mid F)$. Note that $\mathbb{P}(E) = \frac{1}{3000000}$ and $\mathbb{P}(F \mid E) = 0.9999$, and $\mathbb{P}(F \mid E^C) = 0.0001$. Hence,

$$\mathbb{P}(E \mid F) = \frac{\mathbb{P}(F \mid E)\mathbb{P}(E)}{\mathbb{P}(F \mid E)\mathbb{P}(E) + \mathbb{P}(F \mid E^C)\mathbb{P}(E^C)} = \frac{0.9999 \times \frac{1}{3000000}}{0.9999 \times \frac{1}{3000000} + 0.0001 \times \frac{2999999}{3000000}} \approx 0.$$

⊛

4.2 Independence

Definition 4.2.1. Given a probability space $(S, \mathbb{P})$, we say two events are independent if

$$\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F).$$

Example 4.2.1. Roll two fair die. Let $E = \{\text{first roll is odd}\}$ and $F = \{\text{second roll is even}\}$. In this case

$$S = \{(x, y) : x, y \in [6]\}$$

and $\mathbb{P}$ is uniform probability, then

$$E = \{(x, y) \mid x \in \{1, 3, 5\}, y \in [6]\}, F = \{(x, y) \mid x \in [6], y \in \{2, 4, 6\}\}.$$

Hence, $|S| = 36$, $|E| = 18$, and $|F| = 18$. This gives $\mathbb{P}(E) = \mathbb{P}(F) = \frac{1}{2}$. Note that

$$E \cap F = \{(x, y) \mid x \in \{1, 3, 5\}, y \in \{2, 4, 6\}\},$$

so

$$\mathbb{P}(E \cap F) = \frac{1}{4} = \mathbb{P}(E)\mathbb{P}(F),$$

which means E and F are independent.

Proposition 4.2.1.

(1) If E, F are disjoint, then they are independent. $\Leftrightarrow$ $\mathbb{P}(E)$ or $\mathbb{P}(F)$ is 0.

(2) If $E \subseteq F$, and they are independent, then $\mathbb{P}(E) = 0$ or $\mathbb{P}(F) = 1$.

Proof.

(1) $0 = \mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F)$ if and only if $\mathbb{P}(E)$ or $\mathbb{P}(F)$ is 0.

(2) $\mathbb{P}(E) = \mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F)$, so $\mathbb{P}(E) = 0$ or $\mathbb{P}(F) = 1$.

■

Proposition 4.2.2. If E and F are independent, then so are E and F^C.

Proof.

$$\mathbb{P}(E) = \mathbb{P}(E \cap S) = \mathbb{P}(E \cap (F \cup F^C)) = \mathbb{P}((E \cap F) \cup (E \cap F^C)) = \mathbb{P}(E \cap F) + \mathbb{P}(E \cap F^C).$$

Hence,

$$\mathbb{P}(E \cap F^C) = \mathbb{P}(E) - \mathbb{P}(E \cap F) = \mathbb{P}(E) - \mathbb{P}(E)\mathbb{P}(F) = \mathbb{P}(E)(1 - \mathbb{P}(F)) = \mathbb{P}(E)\mathbb{P}(F^C).$$

■

Corollary 4.2.1. If E, F are independent, then E, F^C are independent, so E^C, F^C are independent, so E^C, F are independent.

Example 4.2.2. Now we roll two fair die. We can define $E = \{\text{first roll is odd}\}$ and $\mathbb{P}(E) = \frac{1}{2}$, while $F = \{\text{second roll is even}\}$ and $\mathbb{P}(F) = \frac{1}{2}$. Also, define $G = \{\text{sum is even}\}$, and we have $\mathbb{P}(G) = \frac{1}{2}$ since $|E| = 6 \times 3$ where it means for the first die we can roll any number, and the second die has 3 options.

Claim 4.2.1. E, G are independent.

Proof.

$$E \cap G = \{(x, y) : x \in \{1, 3, 5\}, y \in \{1, 3, 5\}\},$$

so $|E \cap G| = 9$ and thus

$$\mathbb{P}(E \cap G) = \frac{9}{36} = \frac{1}{4} = \mathbb{P}(E)\mathbb{P}(G).$$

This shows E and G are independent. ⊛

Similarly, F, G are independent. However,

$$0 = \mathbb{P}(E \cap F \cap G) \neq \mathbb{P}(E)\mathbb{P}(F)\mathbb{P}(G).$$

Definition 4.2.2. Given a sequence of events $E_1, E_2, \ldots$ in a probability space $(S, \mathbb{P})$. We say they are (mutually) independent if for any finite set $I \subseteq \mathbb{N}$,

$$\mathbb{P}\left(\bigcap_{i \in I} E_i\right) = \prod_{i \in I} \mathbb{P}(E_i).$$

Example 4.2.3. $\{E_1, E_2, E_3\}$ are independent if

$$\begin{aligned}
\mathbb{P}(E_i) &= \mathbb{P}(E_i) \\
\mathbb{P}(E_i \cap E_j) &= \mathbb{P}(E_i)\mathbb{P}(E_j) \\
\mathbb{P}(E_1 \cap E_2 \cap E_3) &= \mathbb{P}(E_1)\mathbb{P}(E_2)\mathbb{P}(E_3).
\end{aligned}$$

Remark 4.2.1. Pairwise independence menas any pair E_i, E_j are independent, which is different from mutually independence.

Example 4.2.4. Let $E_1, E_2, \ldots, E_n$ be mutually independent, then

$$\begin{aligned}
\mathbb{P}(E_1 \cup \cdots \cup E_n) &= \sum_{\varnothing \neq I \subseteq [n]} (-1)^{|I|+1} \mathbb{P}\left(\bigcap_{i \in I} E_i\right) = \sum_{\varnothing \neq I \subseteq [n]} (-1)^{|I|+1} \prod_{i \in I} \mathbb{P}(E_i) \\
&= -\sum_{\varnothing \neq I \subseteq [n]} \prod_{i \in I} (-\mathbb{P}(E_i)) = 1 - \sum_{I \subseteq [n]} \prod_{i \in I} (-\mathbb{P}(E_i)) = 1 - \prod_{i=1}^{n} (1 - \mathbb{P}(E_i)).
\end{aligned}$$

An alternative proof is by de Morgan's Law:

$$\left(\bigcup_{i=1}^{n} E_i\right)^C = \bigcap_{i=1}^{n} E_i^C.$$

Hence,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} E_i\right) = 1 - \mathbb{P}\left(\bigcap_{i=1}^{n} E_i^C\right) = 1 - \prod_{i=1}^{n} \mathbb{P}(E_i^C) = 1 - \prod_{i=1}^{n} (1 - \mathbb{P}(E_i)).$$

Lecture 8

19 March.

Example 4.2.5. If $\mathbb{P}(\text{sunny}) = 0.8$, $\mathbb{P}(\text{swim} \mid \text{sunny}) = 0.7$, $\mathbb{P}(\text{swim} \mid \text{not sunny}) = 0.3$, and $\mathbb{P}(\text{shark bite} \mid \text{swim}) = 0.5$ and $\mathbb{P}(\text{shark bite} \mid \text{not swim}) = 0.01$, then $\mathbb{P}(\text{shark bite}) = ?$

Proof. By the law of total probability,

$$\begin{aligned}\mathbb{P}(\text{shark bite}) &= \mathbb{P}(\text{shark bite} \mid \text{swim})\mathbb{P}(\text{swim}) + \mathbb{P}(\text{shark bite} \mid \text{not swim})\mathbb{P}(\text{not swim}) \\ &= 0.5\mathbb{P}(\text{swim}) + 0.01(1 - \mathbb{P}(\text{swim})) = 0.49\mathbb{P}(\text{swim}) + 0.01.\end{aligned}$$

By the law of total probability again,

$$\begin{aligned}\mathbb{P}(\text{swim}) &= \mathbb{P}(\text{swim} \mid \text{sunny})\mathbb{P}(\text{sunny}) + \mathbb{P}(\text{swim} \mid \text{not sunny})\mathbb{P}(\text{not sunny}) \\ &= 0.7 \times 0.8 + 0.3 \times (1 - 0.8) = 0.62.\end{aligned}$$

Hence, $\mathbb{P}(\text{shark bite}) = 0.49 \times 0.62 + 0.01 = 0.3138$. ⊛

Question. Given that one get bitten by a shark, what's the probability it was sunny?

Answer.

$$\begin{aligned}\mathbb{P}(\text{bite} \mid \text{sunny}) &= \mathbb{P}(\text{bite} \mid \text{swim, sunny})\mathbb{P}(\text{swim} \mid \text{sunny}) + \mathbb{P}(\text{bite} \mid \text{not swim, sunny})\mathbb{P}(\text{not swim} \mid \text{sunny}) \\ &= \mathbb{P}(\text{bite} \mid \text{swim})\mathbb{P}(\text{swim} \mid \text{sunny}) + \mathbb{P}(\text{bite} \mid \text{not swim})\mathbb{P}(\text{not swim} \mid \text{sunny}) \\ &= 0.5 \times 0.7 + 0.01 \times (1 - 0.7) = 0.353.\end{aligned}$$

Hence,

$$\mathbb{P}(\text{sunny} \mid \text{bite}) = \frac{\mathbb{P}(\text{sunny} \cap \text{bite})}{\mathbb{P}(\text{bite})} = \frac{\mathbb{P}(\text{bite} \mid \text{sunny}) \cdot \mathbb{P}(\text{sunny})}{\mathbb{P}(\text{bite})} = \frac{0.353 \times 0.8}{0.3138} \approx 0.899.$$

⊛

Example 4.2.6 (Monty hall problem)**.** Three doors, two have goats, one have car, if the contestant choose a door, then the host, who knows where is the car, open a door to reaveal a goat, then should the contestant switch?

Proof. Let $F_1 = \{\text{initial choice is car}\}$ and $F_2 = \{\text{initial choice is goat}\}$. Then $\mathbb{P}(F_1) = \frac{1}{3}$ and $\mathbb{P}(F_2) = \frac{2}{3}$. Now let $E = \{\text{wins the car if he switches}\}$ and $E' = \{\text{wins the car if he does not switch}\}$, then by the law of total probability,

$$\begin{aligned}\mathbb{P}(E') &= \mathbb{P}(E' \mid F_1)\mathbb{P}(F_1) + \mathbb{P}(E' \mid F_2)\mathbb{P}(F_2) \\ &= 1 \cdot \frac{1}{3} + 0 \cdot \frac{2}{3} = \frac{1}{3}.\end{aligned}$$

Also,

$$\begin{aligned}\mathbb{P}(E) &= \mathbb{P}(E \mid F_1)\mathbb{P}(F_1) + \mathbb{P}(E \mid F_2)\mathbb{P}(F_2) \\ &= 0 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = \frac{2}{3}.\end{aligned}$$

Hence, switching is a better choice. ⊛

Example 4.2.7 (Gambler's Ruin). A gambler goes to a casino with \$67. They need \$100 to pay debts. They play a game where you win \$1 with probability $\frac{1}{2}$ and you lose \$1 with probability $\frac{1}{2}$. As soon as they have \$100, they quit and are happy. But if they reach \$0, they have to stop and will die. What is the probability of being happy?

Proof. Given $0 \leq n \leq 100$, let

$$p_n = \mathbb{P}(\text{happy} \mid \text{start with } \$n).$$

Hence, $p_{100} = 1$ and $p_0 = 0$. Now we want to calculate p_{67}. Consider the first move, and apply the law of total probability, we have

$$p_{67} = p_{68} \times \frac{1}{2} + p_{66} \times \frac{1}{2}.$$

In fact,

$$\begin{aligned} p_{67} &= \mathbb{P}(\text{happy} \mid \text{start with } \$67 \text{ and win first round})\mathbb{P}(\text{win first round} \mid \text{start with } \$67) \\ &\quad + \mathbb{P}(\text{happy} \mid \text{start with } \$67 \text{ and lose first round})\mathbb{P}(\text{lose first round} \mid \text{start with } \$67) \end{aligned}$$

and by the independence of the rounds, the probability of happiness only depends on how much money we have, not on how we get there. By the same reasoning, for any $1 \leq n \leq 99$,

$$p_n = \frac{1}{2}p_{n+1} + \frac{1}{2}p_{n-1}.$$

Note that

$$p_1 = \frac{1}{2}p_2 + \frac{1}{2}p_0 = \frac{1}{2}p_2,$$

and

$$p_2 = \frac{1}{2}p_3 + \frac{1}{2}p_1 = \frac{1}{2}p_3 + \frac{1}{4}p_2,$$

so $p_2 = \frac{2}{5}p_3$. We can do induction on n to show

$$p_n = \frac{n}{n+1}p_{n+1} \text{ for } 1 \leq n \leq 99.$$

Hence,

$$p_n = \frac{n}{n+1}p_{n+1} = \frac{n}{n+1}\frac{n+1}{n+2}p_{n+2} = \frac{n}{n+2}p_{n+2} = \cdots = \frac{n}{n'}p_{n'}.$$

Hence,

$$p_n = \frac{n}{100}p_{100} = \frac{n}{100}.$$

⊛

4.3 Second Borel-Cantelli Lemma

Recall that if $E_1, E_2, \ldots$ is a countably infinite sequence of events in $(S, \mathbb{P})$, then

$$\limsup_{n\to\infty} E_n = \bigcap_{i=1}^{\infty}\left(\bigcup_{i=n}^{\infty} E_i\right),$$

while the first Borel-Cantelli lemma states that

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty \Rightarrow \mathbb{P}(\limsup_{n\to\infty} E_n) = 0.$$

Theorem 4.3.1 (Second Borel-Cantelli Lemma). Let $(S, \mathbb{P})$ be a probability space, and let $E_1, E_2, \dots$ be a countably infinite sequence of **mutually independent** events. Then, if $\sum_{n=1}^{\infty} \mathbb{P}(E_n) = \infty$, then

$$\mathbb{P}(\limsup_{n\to\infty} E_n) = 1.$$

Proof.

$$\begin{aligned}
\mathbb{P}(\limsup_{n\to\infty} E_n) &= \mathbb{P}\left(\lim_{n\to\infty}\left(\bigcup_{i=n}^{\infty} E_i\right)\right) = \lim_{n\to\infty} \mathbb{P}\left(\bigcup_{i=n}^{\infty} E_i\right) \\
&= \lim_{n\to\infty} 1 - \mathbb{P}\left(\bigcup_{i=n}^{\infty} E_i\right)^C = 1 - \lim_{n\to\infty} \mathbb{P}\left(\bigcap_{i=n}^{\infty} E_i^C\right) \\
&= 1 - \lim_{n\to\infty} \prod_{i=n}^{\infty} \mathbb{P}(E_i^C) = 1 - \lim_{n\to\infty} \prod_{i=n}^{\infty} (1 - \mathbb{P}(E_i)) = 1.
\end{aligned}$$

Since the convergence test for infinite product states that

$$\sum_{i=n}^{\infty} \mathbb{P}(E_i) = \infty \Rightarrow \prod_{i=n}^{\infty} (1 - \mathbb{P}(E_i)) = 0.$$

■

Remark 4.3.1. The convergence test for infinite product states that for any sequence $(p_i)_{i=1}^{\infty}$ with $0 \le p_i \le 1$ for all $i \in \mathbb{N}$, for any $n \in \mathbb{N}$,

$$\sum_{i=n}^{\infty} p_i = \infty \Rightarrow \prod_{i=n}^{\infty} (1 - p_i) = 0.$$

We can prove this by an inequality:

$$\log(1-x) \le -x \text{ for all } 0 \le x < 1.$$

Thus,

$$\sum_{i=n}^{N} \log(1 - p_i) \le -\sum_{i=n}^{N} p_i \Rightarrow \log\left(\prod_{i=n}^{N} (1 - p_i)\right) \le -\sum_{i=n}^{N} p_i$$

for all $N \ge n$. Hence, if we have $\sum_{i=n}^{\infty} p_i = \infty$, then $-\sum_{i=n}^{\infty} p_i = -\infty$. This shows

$$\log\left(\prod_{i=n}^{\infty} (1 - p_i)\right) = -\infty,$$

so

$$\prod_{i=n}^{\infty} (1 - p_i) = 0.$$

Chapter 5

Discrete Random Variables

Definition 5.0.1. Given a probability space $(S, \mathbb{P})$, a random variable is a function $X : S \to \mathbb{R}$. We say it is discrete if it only takes countably many values. The random variable determines a (usually simpler) probability space on the set of values it takes, $X(S)$. For any $x \in \mathbb{R}$,

$$\{X = x\} = \{w \in S : X(w) = x\} \subseteq S.$$

If X is discrete, let $x_1, x_2, \dots$ be the values it takes. Then,

$$S = \bigcup_{i=1}^{\infty} \{X = x_i\},$$

and we can define a distribution

$$p(x_i) = \mathbb{P}(X = x_i) = \mathbb{P}(\{w \in S : X(w) = x_i\}).$$

This defines a new probability space $(\{x_1, x_2, \dots\}, p)$, where p is called the probability mass function of X.

Example 5.0.1. Multiple choices test:

- 5 questions, each with 4 options, one of which is correct.
- You choose a uniform, random answer for each question independently.
- X = number of correct answers.

What is $p(k)$ for $0 \le k \le 5$?

Proof. Since our choices are uniformly random,

$$\begin{aligned} p(k) = \mathbb{P}(\{k \text{ answers correct}\}) &= \frac{\#\{\text{choices with exactly } k \text{ correct}\}}{\#\{\text{choices}\}} \\ &= \frac{\#\{\text{choices with } k \text{ correct}\}}{4^5} = \frac{\binom{5}{k}3^{5-k}}{4^5} = \binom{5}{k}\left(\frac{1}{4}\right)^k\left(\frac{3}{4}\right)^{5-k}. \end{aligned}$$

Alternatively,

$$\begin{aligned} \mathbb{P}(\text{getting exact } k \text{ right}) &= \mathbb{P}\Big(\bigcup_{\substack{R\subseteq[5]\\|R|=k}} \{\text{all answers in } R \text{ correct and all answer not in } R \text{ incorrect}\}\Big) \\ &= \sum_{\substack{R\subseteq[5]\\|R|=k}} \mathbb{P}\left(\{\text{all answers in } R \text{ correct and all answer not in } R \text{ incorrect}\}\right) \end{aligned}$$

If we let $E_i = \{\text{answer } i \text{ correct}\}$, then $\mathbb{P}(E_i) = \frac{1}{4}$ and $\mathbb{P}(E_i^C) = \frac{3}{4}$. Thus, we want to compute

$$\sum_{\substack{R\subseteq[5]\\|R|=k}} \mathbb{P}\left(\left(\bigcap_{i\in R} E_i\right) \cap \left(\bigcap_{i\notin R} E_i^C\right)\right) = \sum_{\substack{R\subseteq[5]\\|R|=k}} \prod_{i\in R} \mathbb{P}(E_i) \times \prod_{i\notin R} \mathbb{P}(E_i^C) = \binom{5}{k}\left(\frac{1}{4}\right)^k \left(\frac{3}{4}\right)^{5-k}.$$

⊛

Lecture 9

24 March.

Example 5.0.2 (Coupon collector problem)**.** Every purchase we make, we get a coupon, which is uniformly randomly chosen from one of n types, independent of all previous coupons. The goal is to complete a set: have at least one of every type of coupon. We can define a random variable T, which is the number of purchases before we have a full set. Then, what is the distribution of T? i.e. for $k \in \mathbb{N}$, what is $p(k) = \mathbb{P}(T = k)$?

Proof. Suppose

$$S = \{(s_1, s_2, \dots) : s_i \in [n] \text{ is the type of the } i\text{-th coupon}\}.$$

Then $\mathbb{P}(s_j = j) = \frac{1}{n}$ since coupons are uniform. By independence,

$$\mathbb{P}(\{s_i = j_i : 1 \le i \le k\}) = \frac{1}{n^k} \text{ for any } j_i \in [n].$$

Also,

$$T((s_1, s_2, \dots)) = \begin{cases} \min\{k : (s_1, \dots, s_k) \text{ contain all type form}\}, & \text{if such } k \text{ exists;} \\ -1, & \text{if such } k \text{ does not exist.} \end{cases}$$

Thus, $\mathbb{P}(T = k)$ for $k < n$ is 0 since we need at least n purchases to get n coupons. Now since the first n coupons are uniformly distributed over all possible sequences. Thus,

$$\begin{aligned} \# \text{ of possible sequences } &= n^n \text{ (product rule)} \\ \# \text{ of good sequences } &= n! \text{ (permutation of all types of coupon).} \end{aligned}$$

Thus,

$$\mathbb{P}(T = n) = \frac{n!}{n^n} \approx \left(\frac{1}{e}\right)^n.$$

Now we want to compute $\mathbb{P}(T = k)$ for $k > n$. We can first consider $\mathbb{P}(T > k)$. Let

$$E_i^{(k)} = \{\text{coupon } i \text{ missing from the first } k \text{ coupons}\}.$$

Then,

$$\{T > k\} := \bigcup_{i=1}^{n} E_i^{(k)}.$$

Note that for all i

$$\mathbb{P}\left(E_i^{(k)}\right) = \left(\frac{n-1}{n}\right)^k.$$

> **Question.** Are $E_1^{(k)}, E_2^{(k)}, \dots, E_n^{(k)}$ independent?

Answer. Since $\mathbb{P}\left(E_i^{(k)}\right) = \left(\frac{n-1}{n}\right)^k$ for all i, so

$$\prod_{i=1}^{n} \mathbb{P}(E_i^{(k)}) = \left(\frac{n-1}{n}\right)^{kn},$$

but

$$\mathbb{P}\left(\bigcap_{i=1}^{n} \mathbb{P}(E_i^{(k)})\right) = 0,$$

so the answer is no. ⊛

By inclusion-exclusion principle,

$$\begin{aligned}\mathbb{P}\left(\bigcup_{i=1}^{n} E_i^{(k)}\right) &= \sum_{\varnothing \neq I \subseteq [n]} (-1)^{|I|+1} \mathbb{P}\left(\bigcap_{i\in I} E_i^{(k)}\right) = \sum_{\varnothing \neq I \subseteq [n]} (-1)^{|I|+1} \left(\frac{n-|I|}{n}\right)^k \\ &= \sum_{r=1}^{n-1} \binom{n}{r} (-1)^{r+1} \left(\frac{n-r}{n}\right)^k.\end{aligned}$$

Thus,

$$\begin{aligned}\mathbb{P}(T=k) &= \mathbb{P}(T > k-1) - \mathbb{P}(T > k) \\ &= \sum_{r=1}^{n-1} \binom{n}{r} (-1)^{r+1} \left(\frac{n-r}{n}\right)^{k-1} - \sum_{r=1}^{n-1} \binom{n}{r} (-1)^{r+1} \left(\frac{n-r}{n}\right)^{k} \\ &= \sum_{r=1}^{n-1} \binom{n}{r} (-1)^{r+1} \left(\frac{n-r}{n}\right)^{k-1} \frac{r}{n}.\end{aligned}$$

⊛

Definition 5.0.2 (Expected value). Let $(S, \mathbb{P})$ be a probability space and let $X : S \to \mathbb{R}$ be a discrete random variable. Then the expected value of X

$$\mathbb{E}[X] = \sum_{x_i} x_i p(x_i) = \sum_{x_i} x_i \mathbb{P}(X = x_i).$$

Remark 5.0.1. Expected value is what we expect to see on average if we repeat the random experiment many times.

Suppose we repeat N times, and let n_i to be the number of times we see $X = x_i$, then the average value of X is

$$\frac{\sum_i x_i n_i}{N} = \sum_i x_i \frac{n_i}{N} \approx \sum_i x_i \mathbb{P}(X = x_i) \text{ when } N \to \infty.$$

Example 5.0.3. Two multiple choice questions, each with four options. If we choose answers uniformly at random, how many do we expect to get correct?

Proof. Let $X = \#$ of correct answers. Then,

x_i	$\mathbb{P}(X = x_i)$
0	$\left(\frac{3}{4}\right)^2$
1	$2\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)$
2	$\left(\frac{1}{4}\right)^2$

Thus,

$$\mathbb{E}[X] = \sum_i x_i \mathbb{P}(X = x_i) = 0 \cdot \frac{9}{16} + 1 \cdot \frac{6}{16} + 2 \cdot \frac{1}{16} = \frac{1}{3}.$$

⊛

Claim 5.0.1.

$$\mathbb{E}[X] = \sum_{s \in S} X(s)\mathbb{P}(\{s\})$$

Proof.

$$\begin{aligned}\mathbb{E}[X] &= \sum_{x_i} x_i \mathbb{P}(X = x_i) = \sum_{x_i} x_i \mathbb{P}(\{s : X(s) = x_i\}) \\ &= \sum_{x_i} x_i \mathbb{P}\left(\bigcup_{s:X(s)=x_i} \{s\}\right) = \sum_{x_i} x_i \sum_{\substack{s \in S \\ X(s)=x_i}} \mathbb{P}(\{s\}) = \sum_{s \in S} X(s)\mathbb{P}\left(\{s\}\right).\end{aligned}$$

⊛

5.1 Expected value of functions of random variable

Let $(S, \mathbb{P})$ be a probability space. Let $X : S \to \mathbb{R}$ be a random variable. Let $g : \mathbb{R} \to \mathbb{R}$ be a function. Then, $g(X) = g \circ X : S \to \mathbb{R}$ is another random variable.

Claim 5.1.1.

$$\mathbb{E}[g(X)] = \sum_{x_i} g(x_i)\mathbb{P}(X = x_i).$$

Proof. By the earlier claim,

$$\begin{aligned}E[g(X)] &= \sum_{s \in S} g(X(s))\mathbb{P}\left(\{s\}\right) = \sum_{x_i} \sum_{\substack{s \in S \\ X(s)=x_i}} g(X(s))\mathbb{P}\left(\{s\}\right) \\ &= \sum_{x_i} g(x_i) \sum_{\substack{s \in S \\ X(s)=x_i}} \mathbb{P}\left(\{s\}\right) = \sum_{x_i} g(x_i)\mathbb{P}(X = x_i).\end{aligned}$$

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Proposition 5.1.1. Let $X_1, X_2, \ldots, X_n : S \to \mathbb{R}$ be random variables on a probability space and $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{R}$. Then if

$$X = \sum_{i=1}^{n} \alpha_i X_i : S \to \mathbb{R}, \text{ we have } \mathbb{E}[X] = \sum_{i=1}^{n} \alpha_i \mathbb{E}[X_i].$$

This is called linearity of expectation.

Proof. By earlier claim,

$$\begin{aligned}\mathbb{E}[X] &= \sum_{s\in S} X(s)\mathbb{P}(\{s\}) = \sum_{s\in S}\left(\sum_{i=1}^{n}\alpha_i X_i(s)\right)\mathbb{P}(\{s\}) \\ &= \sum_{s\in S}\sum_{i=1}^{n}\alpha_i X_i(s)\mathbb{P}(\{s\}) = \sum_{i=1}^{n}\alpha_i\sum_{s\in S}X_i(s)\mathbb{P}(\{s\}) \\ &= \sum_{i=1}^{n}\alpha_i\mathbb{E}[X_i].\end{aligned}$$

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Remark 5.1.1. We do not require any independence.

Definition 5.1.1. Random variables $X_1, X_2, \ldots$ are independent if for any $x_1, x_2, \ldots$, the events

$$E_i = \{X_i = x_i\} \text{ for all } i$$

are independent.

Example 5.1.1. Multiple choice: each question has k options, one of each which is correct. Suppose there are n questions in total. If we guess uniformly at random, independently, what is the expected score?

Answer. Let X be the number of correct numbers. Then X takes values in $[n] \cup \{0\}$. For each $0 \le r \le n$,

$$\mathbb{P}(X = r) = \binom{n}{r}\left(\frac{1}{k}\right)^r\left(\frac{k-1}{k}\right)^{n-r}.$$

Thus,

$$\mathbb{E}[X] = \sum_{x_i} x_i\mathbb{P}(X = x_i) = \sum_{r=0}^{n} r\mathbb{P}(X = r) = \sum_{r=1}^{n} r\binom{n}{r}\left(\frac{1}{k}\right)^r\left(\frac{k-1}{k}\right)^{n-r} = \frac{n}{k}.$$

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Answer. Let

$$X_i = \begin{cases} 1, & \text{if } i\text{-th question is correct;} \\ 0, & \text{if } i\text{-th question is not correct,}\end{cases}$$

which is the indicator random variable for the event of getting i-th question correct. Then,

$$\begin{aligned}\mathbb{E}[X_i] &= 0\cdot\mathbb{P}(i\text{-th question is not correct}) + 1\cdot\mathbb{P}(i\text{-th question is correct}) \\ &= 0\cdot\frac{k-1}{k} + 1\cdot\frac{1}{k} = \frac{1}{k}.\end{aligned}$$

Then, the number of correct answers is given by

$$X = \sum_{i=1}^{n} X_i.$$

By linearity of expectation,

$$\mathbb{E}[X] = \sum_{i=1}^{n}\mathbb{E}[X_i] = \frac{n}{k}.$$

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Question. Three investment opportunities:

1. Guaranteed to get \$1 after a year.
2. We get \$1000 with probability $\frac{1}{1000}$ and \$0 at probability 0.
3. Get $\$\frac{2000}{999}$ with probability $\frac{999}{1000}$ and $\$-1000$ at probability $\frac{1}{1000}$.

Which should we choose?

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Let X_i be the random variable representing the earning of option i after a year. Then 26 March.

$$\begin{aligned}
\mathbb{E}[X_1] &= \$1 \cdot 1 = \$1. \\
\mathbb{E}[X_2] &= \$1000 \cdot \frac{1}{1000} + \$0 \cdot \frac{999}{1000} = \$1. \\
\mathbb{E}[X_3] &= \$\frac{2000}{999} \cdot \frac{999}{1000} - \$1000 \cdot \frac{1}{1000} = \$1.
\end{aligned}$$

Thus, three options all have same expectation, which means expectation can not reflect all. Hence, we need variance, which can measure how far a random typically is from its expectation.

Definition 5.1.2 (Variance). Let X be a random variable on a probability space $(S, \mathbb{P})$, and let $\mu = \mathbb{E}[X]$. We define the variance to be

$$\mathrm{Var}(X) = \mathbb{E}\left[(X-\mu)^2\right].$$

Thus,

$$\begin{aligned}
\mathrm{Var}(X_1) &= \mathbb{E}[(X_1-1)^2] = (1-1)^2 \cdot 1 = 0 \\
\mathrm{Var}(X_2) &= \mathbb{E}[(X_2-1)^2] = (1000-1)^2 \cdot \frac{1}{1000} + (0-1)^2 \cdot \frac{999}{1000} \approx 999. \\
\mathrm{Var}(X_3) &= \mathbb{E}[(X_3-1)^2] = \left(\frac{2000}{999} - 1\right)^2 \cdot \frac{999}{1000} + (-1000-1)^2 \cdot \frac{1}{1000} \approx 1003.
\end{aligned}$$

Claim 5.1.2. For any random variable,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

Proof.

$$\begin{aligned}
\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\
&= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\
&= \mathbb{E}[X^2] - \mathbb{E}[X]^2
\end{aligned}$$

since $\mu = \mathbb{E}[X]$. ⊛

Corollary 5.1.1. $\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$.

Proof.

$$0 \leq \mathbb{E}[(X-\mu)^2] = \mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

■

Remark 5.1.2. Sometimes it is easier to compute

$$\mathbb{E}[X(X-1)] = \mathbb{E}[X^2] - \mathbb{E}[X],$$

then

$$\mathrm{Var}(X) = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - \mathbb{E}[X]^2.$$

Definition 5.1.3. The standard deviation of a random variable X, denoted $\sigma(X)$, is

$$\sigma(X) = \sqrt{\mathrm{Var}(X)}.$$

We sometimes denote to variance by $\sigma^2(X)$.

Proposition 5.1.2 (Markov's Inequality). Let $X \geq 0$ be a non-negative random variable on a probability space $(S, \mathbb{P})$, and let $a > 0$ be a positive real number. Then,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof.

$$\begin{aligned}
\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x) &= \underbrace{\sum_{x:x<a} x\mathbb{P}(X = x)}_{\geq 0} + \sum_{x:x\geq a} x\mathbb{P}(X = x) \\
&\geq \sum_{x:x\geq a} a\mathbb{P}(X = x) = a \sum_{x:x\geq a} \mathbb{P}(X = x) = a\mathbb{P}(X \geq a).
\end{aligned}$$

■

Remark 5.1.3. Markov's inequality says that small expectation implies random variable is small with high probability, but Markov's inequality does not say if expectation is large, then random variable large with high probability. In fact, this is not always true: Consider

$$X_n = \begin{cases} n^2, & \text{with probability } \frac{1}{n}; \\ 0, & \text{with probability } 1 - \frac{1}{n}. \end{cases}$$

Then,

$$\mathbb{E}[X_n] = n^2 \cdot \frac{1}{n} = n \to \infty \text{ as } n \to \infty.$$

However,

$$\mathbb{P}(X_n \neq 0) = \frac{1}{n} \to 0 \text{ as } n \to \infty.$$

Theorem 5.1.1 (Chebyshev's inequality). Let X be a random variable with $\mu = \mathbb{E}[X]$ on a probability space $(S, \mathbb{P})$, and let $a > 0$ be a real number. Then,

$$\mathbb{P}(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}.$$

Proof. We can apply Markov's inequality to $Y = (X - \mu)^2 \geq 0$. Thus,

$$\mathbb{P}(Y \geq a^2) \leq \frac{\mathbb{E}[Y]}{a^2} = \frac{\mathbb{E}[(X-\mu)^2]}{a^2} = \frac{\mathrm{Var}(X)}{a^2}.$$

■

Remark 5.1.4. Chebyshev's inequality implies small variance implies X is close to its expectation with high probability.

Corollary 5.1.2. The probability that X is more than k standard deviation from its expectation is $\leq \frac{1}{k^2}$.

Proof.

$$\mathbb{P}(|X-\mu| \geq k\sigma(X)) \leq \frac{\operatorname{Var}(X)}{(k\sigma(X))^2} = \frac{\operatorname{Var}(X)}{k^2\sigma^2(X)} = \frac{1}{k^2}.$$

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Observation. Let X be a random variable, $a, b \in \mathbb{R}$. Then,

$$\mathbb{E}[aX+b] = a\mathbb{E}[X] + b$$

by linearity, and

$$\begin{aligned}\operatorname{Var}(aX+b) &= \mathbb{E}\left[((aX+b) - \mathbb{E}[aX+b])^2\right] = \mathbb{E}\left[((aX+b) - (a\mathbb{E}[X]+b))^2\right] \\ &= \mathbb{E}\left[a^2\left(X - \mathbb{E}[X]\right)^2\right] = a^2\mathbb{E}\left[(X-\mathbb{E}[X])^2\right] = a^2 \operatorname{Var}(X).\end{aligned}$$

5.2 Discrete Distribution

5.2.1 Bernoulli distribution

Definition 5.2.1 (Bernoulli distribution)**.** If $X \sim \operatorname{Ber}(p)$, i.e. X has the $\operatorname{Ber}(p)$ distribution, then

$$X = \begin{cases} 1, & \text{with probability } p; \\ 0, & \text{with probability } 1-p. \end{cases}$$

Then,

$$\mathbb{E}[X] = 1 \cdot p + 0 \cdot (1-p) = p$$

and

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathbb{E}[X] - \mathbb{E}[X]^2 = p - p^2 = p(1-p).$$

5.2.2 Binomial distribution

Definition 5.2.2 (Binomial distribution)**.** If we run n independent trials of a random experiment of which succeeds with probability p. We count the number of successes, where the possible values are $0, 1, 2, \ldots, n-1, n$. Thus, we can define $\operatorname{Bin}(n,p), n \in \mathbb{N}, 0 \leq p \leq 1$ to be the number of successes in the random experiment.

Example 5.2.1.

- # of heads in n coin tosses.
- # of odd rolls in n rolls of a die.
- # of errors when replicating DNA.
- # of supporters of a particular party in a poll.

The probability mass function of $\operatorname{Bin}(n,p)$ is

$$\mathbb{P}(X=k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Hence,

$$\mathbb{E}[X] = \sum_{k=0}^{n} k \cdot \mathbb{P}(X = k) = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}.$$

Alternatively,

$$X = \sum_{i=1}^{n} X_i, \text{ where } X_i = \begin{cases} 1, & \text{if } i\text{-th trial is successful;} \\ 0, & \text{if not,} \end{cases}$$

and each $X_i \sim \mathrm{Ber}(p)$. Thus,

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} p = np.$$

Recall the Binomial Theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Thus,

$$n(x+y)^{n-1} = \frac{\mathrm{d}}{\mathrm{d}x}(x+y)^n = \frac{\mathrm{d}}{\mathrm{d}x}\left[\sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}\right] = \sum_{k=0}^{n} k \binom{n}{k} x^{k-1} y^{n-k}.$$

Multiplying x on both sides, then

$$nx(x+y)^{n-1} = \sum_{k=0}^{n} k \binom{n}{k} x^k y^{n-k}.$$

Now substituting $x = p$ and $y = 1-p$, then

$$np = np(p + (1-p))^{n-1} = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k} = \mathbb{E}[X].$$

Now we compute $\mathrm{Var}(X)$. Note that

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

and

$$\mathbb{E}[X^2] = \sum_{k=0}^{n} k^2 \binom{n}{k} p^k (1-p)^{n-k}.$$

Differentiable with respect to x, then we have

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[nx(x+y)^{n-1}\right] = \frac{\mathrm{d}}{\mathrm{d}x}\left[\sum_{k=0}^{n} k \binom{n}{k} x^k y^{n-k}\right].$$

Hence,

$$n(x+y)^{n-1} + n(n-1)x(x+y)^{n-2} = \sum_{k=0}^{n} k^2 \binom{n}{k} x^{k-1} y^{n-k}.$$

Multiplying by x, we have

$$nx(x+y)^{n-1} + n(n-1)x^2(x+y)^{n-2} = \sum_{k=0}^{n} k^2 \binom{n}{k} x^k y^{n-k},$$

and substituting $x = p, y = 1-p$, we have

$$np + n(n-1)p^2 = \sum_{k=0}^{n} k^2 \binom{n}{k} p^k (1-p)^{n-k} = \mathbb{E}[X^2].$$

Thus,

$$\begin{aligned} \mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 &= np + n(n-1)p^2 - (np)^2 \\ &= np - np^2 = np(1-p) = n \cdot \mathrm{Var}(\mathrm{Ber}(p)). \end{aligned}$$

5.2.3 Poisson distribution

If there are rare independent events, and we count how many occurences in a given time period, then we can give a distribution, by $\text{Poi}(\lambda)$, where $\lambda \geq 0$ is a real number.

Example 5.2.2.

- earthquakes.
- # of customers eating store.
- # of phone calls.

λ = expected # of occurences.

Thus, possible values of $X \sim \text{Poi}(\lambda)$ is $0, 1, 2, \ldots$.

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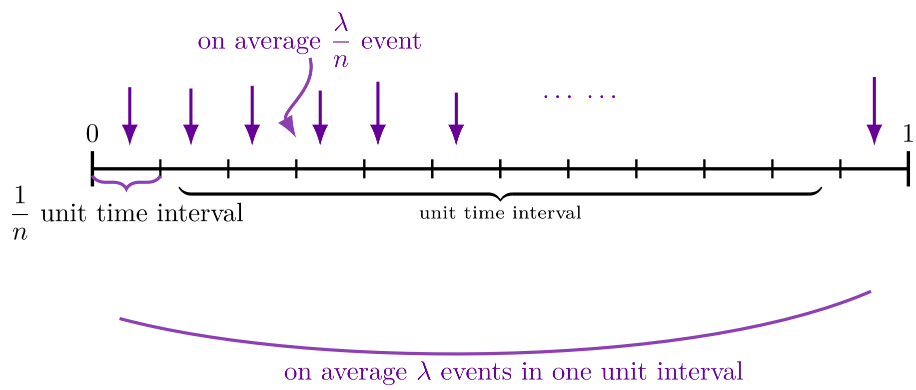
Definition 5.2.3 (Poisson distribution). Suppose $\lambda \in \mathbb{R}_{\geq 0}$, then we say $X \sim \text{Poi}(\lambda)$ if rare independent events that occurs on average λ times per unit interval of time, and we want to count how many times actually occur in a given interval. Hence, X takes value $0, 1, 2, 3, \ldots$. 31 March.

As we increase n (decrease the length of the subintervals), we expect each subinterval to have 0 or 1 events, with $\frac{\lambda}{n}$ on average. Note that whether or not a subinterval has an event are independent.



Hence,

$$X = \text{\# of events in unit interval} \gtrsim \text{\# of subintervals that have an event} \sim \text{Bin}(n, \frac{\lambda}{n}).$$

Hence,

$$\begin{aligned}
\mathbb{P}(X = k) &\simeq \lim_{n\to\infty} \mathbb{P}\left(\text{Bin}(n, \frac{\lambda}{n}) = k\right) = \lim_{n\to\infty} \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\
&= \lim_{n\to\infty} \frac{n(n-1)\ldots(n-k+1)}{k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} = \frac{\lambda^k e^{-\lambda}}{k!}.
\end{aligned}$$

Thus, the mass function:

$$\mathbb{P}(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots.$$

Claim 5.2.1. This is a valid distribution.

Proof. Since $\mathbb{P}(X = k) \geq 0$ for all k, and

$$\sum_{k=0}^{\infty} \mathbb{P}(X = k) = \sum_{k=0}^{\infty} \frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = e^{-\lambda}e^{\lambda} = 1,$$

and the countable additivity is trivial, so this is a valid distribution. ⊛

Now we compute the expectation value:

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k\mathbb{P}(X = k) = \sum_{k=1}^{\infty} k \cdot \frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^k}{(k-1)!} = \lambda e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!}$$
$$= \lambda e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = \lambda e^{-\lambda}e^{\lambda} = \lambda.$$

Now we compute the variance:

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathbb{E}[X^2 - X] + \mathbb{E}[X] - \mathbb{E}[X]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

Note that $\mathbb{E}[X^2 - X] = \mathbb{E}[X(X-1)]$, and can be computed similarly as $\mathbb{E}[X]$.

Remark 5.2.1. $\mathrm{Poi}(\lambda) \approx \mathrm{Bin}\left(n, \frac{\lambda}{n}\right)$ when n is large. We can notice the expectaion of $\mathrm{Bin}\left(n, \frac{\lambda}{n}\right)$ is $np = \lambda$ and the variance of $\mathrm{Bin}\left(n, \frac{\lambda}{n}\right)$ is $np(1-p) = \lambda\left(1 - \frac{\lambda}{n}\right) \to \lambda$ as $n \to \infty$.

Remark 5.2.2 (Poisson approximation)**.** If n is large, and $np \to \lambda \in \mathbb{R}$, then we can approximate $\mathrm{Bin}(n, p)$ by $\mathrm{Poi}(np)$.

Example 5.2.3 (The Poisson Paradigm)**.** Suppose we have a probability space $(S, \mathbb{P})$, and a sequence of events $E_1, E_2, \ldots, E_n$, with

$$\mathbb{P}(E_i) = p_i, \quad \text{for } 1 \leq i \leq n.$$

If the p_i's are all small, and the events E_i are independent or "weakly independent", i.e. $\mathbb{P}(E_i \mid E_j) \approx \mathbb{P}(E_i)$ for $i \neq j$, then setting

$$\lambda = p_1 + p_2 + \cdots + p_n,$$

if X is the number of events that occur, then X is approximately $\mathrm{Poi}(\lambda)$.

Example 5.2.4. n people leaving a party, each takes a random hat. What is the probability that exactly k people get their hats?

Proof. Define the events $E_i = \{i\text{-th person gets own hat}\}$, then $\mathbb{P}(E_i) = \frac{1}{n}$ for all i, which is small. Also,

$$\mathbb{P}(E_i \mid E_j) = \frac{\frac{1}{n(n-1)}}{\frac{1}{n}} = \frac{1}{n-1} \approx \frac{1}{n},$$

so by Poisson paradigm, $X = \#$ of people getting own hat is approximaely $\mathrm{Poi}(1)$, i.e.

$$\mathbb{P}(X = k) \approx \frac{e^{-1}1^k}{k!} = \frac{1}{k!e}.$$

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Example 5.2.5. Which of these is a random binary string?

(a) 00000000001111111111

(b) 01010101010101010101

(c) 00011110100001111001

(d) 00100110000110111110

Example 5.2.6. Toss a fair coin n times. Let L_n be the length of the longest sequence of consecutive heads. What will L_n typically be?

Proof. Since $S = \{H, T\}^n$, and $\mathbb{P}$ is uniform, so

$$\{L_n \geq k\} = \{\text{there is a sequence of } k \text{ consecutive heads}\} = E.$$

Hence, $E = \bigcup_{i=1}^{n-k+1} E_i$ where E_i is the set of there is a sequence of k heads in a row starting from i-th toss. Also, $\mathbb{P}(E_i) = 2^{-k}$, which is small if k is not too small. Note that

$$\mathbb{P}(E_{i+1} \mid E_i) = \frac{1}{2} \text{ (only need to see the last toss)},$$

which is not close to 2^{-k}. Hence, no weak independence here, and thus no Poisson paradigm.

Now we change a method to solve this problem.

$$E = \{L_n \geq k\} = \bigcup_{i=1}^{n-k+1} E_i'$$

where

$$E_i' = \begin{cases} \text{toss } i, i+1, \ldots, i+k-1 \text{ are heads and toss } i+k \text{ is tail}, & \text{if } 0 \leq i \leq n-k; \\ \text{toss } n-k+1, \ldots, n \text{ are heads}, & \text{if } i = n-k+1. \end{cases}$$

Hence,

$$\mathbb{P}(E_i') = \begin{cases} 2^{-(k+1)}, & \text{if } 1 \leq i \leq n-k; \\ 2^{-k}, & \text{if } i = n-k+1, \end{cases}$$

which are all small. Now

$$\mathbb{P}(E_i' \mid E_j') = \begin{cases} \mathbb{P}(E_i'), & \text{if } |i-j| > k+1; \\ 0, & \text{if } |i-j| \leq k. \end{cases}$$

Note that when k is big, $\mathbb{P}(E_i') \approx 0$, so under this setting we have (E_i') weakly independent. Now since $\mathbb{P}(E_i')$ is small for all i, Poisson paradigm applies. Now let

$$\lambda = \sum_{i=1}^{n-k+1} \mathbb{P}(E_i') = (n-k+2)2^{-(k+1)},$$

then $X_k :=$ number of i s.t. E_i' happens $\sim \text{Poi}((n-k+2)2^{-(k+1)})$. Then, $\{L_n \geq k\} = \{X_k \geq 1\}$ and thus $\{L_n < k\} = \{X_k = 0\}$. Hence,

$$\mathbb{P}(L_n < k) = \mathbb{P}(X = 0) \approx e^{-\frac{n-k+2}{2^{k+1}}}.$$

If $k \leq (1+\varepsilon)\log_2 n$ for some $\varepsilon > 0$, then

$$\frac{n-k+2}{2^{k+1}} \to \infty,$$

so $\mathbb{P}(L_n < k) \to 0$ as $n \to \infty$. If $k \geq (1+\varepsilon)\log_2 n$ for all $\varepsilon > 0$, then

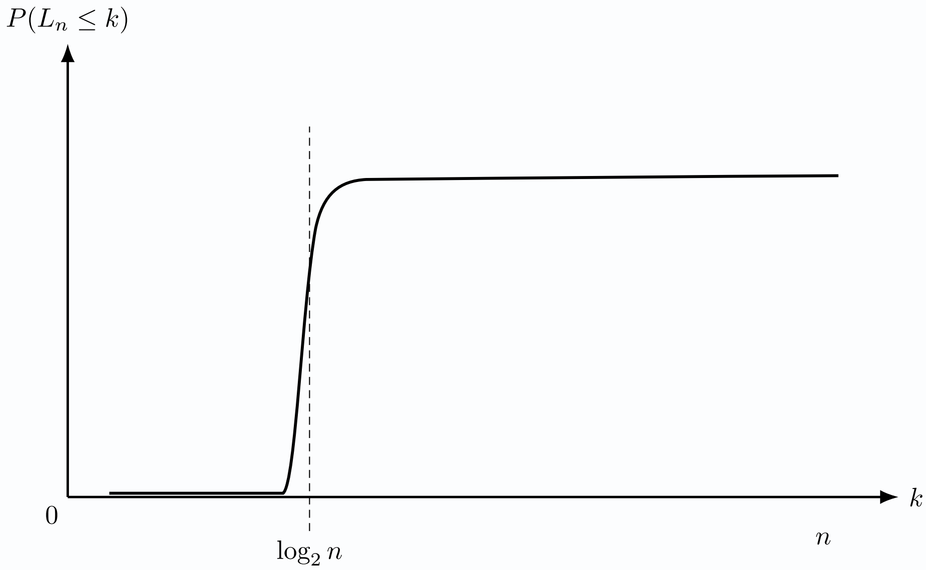
$$\frac{n-k+2}{2^{k+1}} \to 0,$$

then $\mathbb{P}(L_n < k) \to 1$. If $k = \log_2 n + c$ for some constant c, then

$$\frac{n-k+2}{2^{k+1}} = \frac{n - \log_2 n + 2 + c}{2^{c+1} n} \to 2^{-(c+1)}.$$

Thus,

$$\mathbb{P}(L_n < \log_2 n + c) \to e^{-2^{-(c+1)}}.$$



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5.2.4 Geometric distribution

Definition 5.2.4 (Geometric Distribution). $X \sim \text{Geom}(p)$ where $0 \leq p \leq 1$ when independent trials with success probability p and we want to count the number of trials until first success (inclusive). Hence, $\text{Geom}(p)$ takes values $1, 2, 3, 4, \ldots$.

Hence,

$$\mathbb{P}(X = k) = (1-p)^{k-1} p \geq 0.$$

We can check that

$$\sum_{k=1}^{\infty} \mathbb{P}(X = k) = \sum_{k=1}^{\infty} (1-p)^{k-1} p = p \sum_{k=1}^{\infty} (1-p)^{k-1} = p \cdot \frac{1}{1-(1-p)} = 1.$$

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Now we compute the expectation: 2 April.

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k\mathbb{P}(X = k) = \sum_{k=1}^{\infty} k(1-p)^{k-1} p = p \sum_{k=1}^{\infty} k(1-p)^{k-1}.$$

Note that

$$\sum_{k=1}^{\infty} x^k = \frac{x}{1-x},$$

so differentiating w.r.t. x, we have

$$\sum_{k=1}^{\infty} k x^{k-1} = \frac{(1-x) - x(-1)}{(1-x)^2} = \frac{1}{(1-x)^2}.$$

Hence, plugging in $x = 1 - p$,

$$\mathbb{E}[X] = p\sum_{k=1}^{\infty} k(1-p)^{k-1} = p \cdot \frac{1}{(1-(1-p))^2} = \frac{1}{p}.$$

Now we compute the variance:

$$\begin{aligned}\operatorname{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathbb{E}[X(X+1)] - \mathbb{E}[X] - \mathbb{E}[X]^2 \\ = \mathbb{E}[X(X+1)] - \frac{1}{p} - \frac{1}{p^2}.\end{aligned}$$

Note that

$$\mathbb{E}[X(X+1)] = \sum_{k=1}^{\infty} k(k+1)\mathbb{P}(X=k) = \sum_{k=1}^{\infty} k(k+1)(1-p)^{k-1}p = p\sum_{k=1}^{\infty} k(k+1)(1-p)^{k-1}.$$

Now we already have

$$\sum_{k=1}^{\infty} kx^{k-1} = \frac{1}{(1-x)^2}.$$

Differentiating on both side, we have

$$\sum_{k=1}^{\infty} k(k+1)x^{k-1} = \frac{2}{(1-x)^3}.$$

Thus,

$$\mathbb{E}[X(X+1)] = p \cdot \frac{2}{p^3} = \frac{2}{p^2}.$$

This gives

$$\operatorname{Var}(X) = \mathbb{E}[X(X+1)] - \frac{1}{p} - \frac{1}{p^2} = \frac{2}{p^2} - \frac{1}{p} - \frac{1}{p^2} = \frac{1}{p^2} - \frac{1}{p} = \frac{1-p}{p^2}.$$

Example 5.2.7 (Coupon collector problem)**.** Store has n types of coupons: Type 1, 2, ..., n. Every purchase gives uniformly random type of coupon.

- Q1. How many purchases needed to get a coupon of Type i, $1 \leq i \leq n$?
- Q2. How many purchases until we have all types?

Proof. Let $X_i = \#$ of purchases before first coupon of type i, $1 \leq i \leq n$. Then, $X_i \sim \operatorname{Geom}(\frac{1}{n})$. Hence, $\mathbb{P}(X_i = k) = \frac{1}{n}\left(1 - \frac{1}{n}\right)^{k-1}$, and $\mathbb{E}[X_i] = n$. Let $X = \#$ of purchases until we have all types of coupons. Then, $X = \max\{X_i : 1 \leq i \leq n\}$, but this is not linear, so it is not easy to compute.

For $i = 1, 2, \ldots, n$, let $y_i = \#$ of purchases needed to go from having $(i-1)$ different types of coupons to having i different types. Then, $X = \sum_{i=1}^{n} y_i$. Then, $y_i \sim \operatorname{Geom}\left(\frac{n-i+1}{n}\right)$ since y_i is the number of purchases we need to get one of the $n-i+1$ types of coupons that we haven't got. Thus,

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^{n} y_i\right] = \sum_{i=1}^{n}\mathbb{E}[y_i] = \sum_{i=1}^{n}\frac{n}{n-i+1} = n\sum_{i=1}^{n}\frac{1}{n-i+1} = n\sum_{i=1}^{n}\frac{1}{i} = nH_n.$$

⊛

Definition 5.2.5 (Negative Binomial Distribution)**.** We denote this distribution by $X \sim \operatorname{NB}(r, p)$, where $0 < p \leq 1$ and $r \in \mathbb{N}$. The setting is to repeat independent trial with success probability p, and count the number of trials until r-th success. Thus, X takes value $r, r+1, r+2, \ldots$.

Observation. $\operatorname{NB}(1, p) \sim \operatorname{Geom}(p)$.

Note that

$$\mathbb{P}(X = n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}$$

since we can choose $r-1$ trials to be success in the first $n-1$ trials, and the n-th trial must be success.

We can compute the expectation:

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{n=r}^{\infty} n\mathbb{P}(X = n) = \sum_{n=r}^{\infty} n\binom{n-1}{r-1} p^r (1-p)^{n-r} \\
&= \sum_{n=r}^{\infty} \frac{n(n-1)!}{(r-1)!(n-r)!} p^r (1-p)^{n-r} = \sum_{n=r}^{\infty} r\binom{n}{r} p^r (1-p)^{n-r} = r\sum_{n=r}^{\infty} \binom{n}{r} p^r (1-p)^{n-r} \\
&= \frac{r}{p} \sum_{n=r}^{\infty} \binom{(n+1)-1}{(r+1)-1} p^{r+1} (1-p)^{(n+1)-(r+1)} = \frac{r}{p} \sum_{n=r}^{\infty} \mathbb{P}(y = n) \quad (\text{if } y \sim \mathrm{NB}(r+1, p)) \\
&= \frac{r}{p}.
\end{aligned}$$

An alternative way to compute $\mathbb{E}[X]$: If $X \sim \mathrm{NB}(r, p)$, then we can write $X = X_1 + X_2 + \cdots + X_r$, where $X_i = \#$ of trials to go from $(i-1)$-th success to i-th success, where each $X_i \sim \mathrm{Geom}(p)$ is independent.

Thus,

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^{r} X_i\right] = \sum_{i=1}^{r} \mathbb{E}[X_i] = \sum_{i=1}^{r} \frac{1}{p} = \frac{r}{p}.$$

As for the variance,

$$\mathrm{Var}(X) = \frac{r(1-p)}{p^2} \quad [\mathrm{HW}].$$

Definition 5.2.6 (Hyper Geometric Distribution)**.** We denote hyper geometric distribution by $X \sim \mathrm{Hypergeometric}(N, m, n)$, where N, m, n are integers and $0 \leq m \leq N$, $1 \leq n \leq N$. The setting is: Population of size N, of which m are good, and we select a random subset of n members of the populations without replacement uniformly, and we count the number of good members in sample, so X takes value $0, 1, 2, \ldots, \min\{m, n\}$.

Observation. If we sample n people with replacement, then we could have a binomial distribution $X' \sim \mathrm{Bin}(n, \frac{m}{N})$, and thus

$$\mathbb{E}[X'] = \frac{nm}{N}, \quad \mathrm{Var}(X') = \frac{nm(N-m)}{N^2}.$$

Note that

$$\mathbb{P}(X = k) = \frac{\text{number of ways of picking } k \text{ good members and } n-k \text{ bad members}}{\text{number of ways of picking } n \text{ members}} = \frac{\binom{m}{k}\binom{N-m}{n-k}}{\binom{N}{n}}.$$

To compute the expectation, we can define

$$X_i = \begin{cases} 1, & \text{if } i\text{-th member in the sample is good;} \\ 0, & \text{if } i\text{-th member in the sample is bad.} \end{cases}$$

Then,

$$X = \sum_{i=1}^{n} X_i \Rightarrow \mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i].$$

Note that

$$\mathbb{E}[X_i] = \mathbb{P}(i\text{-th member is good}) = \frac{m}{N}$$

because every one is equally likely to be the i-th chosen person, and thus the probability of good member to be picked in i-th choice is $\frac{m}{N}$. Hence,

$$\mathbb{E}[X] = n \cdot \frac{m}{N}.$$

Now we compute the variance $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$, where

$$\begin{aligned}\mathbb{E}[X^2] &= \mathbb{E}\left[\left(\sum_{i=1}^{n} X_i\right)^2\right] = \mathbb{E}\left[\sum_{i,j} X_i X_j\right] = \sum_{i,j} \mathbb{E}[X_i X_j] \\ &= \sum_{i=1}^{n} \mathbb{E}[X_i^2] + 2 \sum_{1 \le i < j \le n} \mathbb{E}[X_i X_j] = \sum_{i=1}^{n} \mathbb{E}[X_i] + 2 \sum_{1 \le i < j \le n} \mathbb{E}[X_i X_j] \quad \text{since } X_i \text{ takes value } 0, 1.\end{aligned}$$

Note that

$$X_i X_j = \begin{cases} 1, & \text{if } i\text{-th and } j\text{-th people are good;} \\ 0, & \text{otherwise.} \end{cases}$$

Thus,

$$\mathbb{E}[X_i X_j] = \mathbb{P}\left(i\text{-th and } j\text{-th people are good}\right) = \frac{\#\text{ of pairs of good members}}{\#\text{ of pairs}} = \frac{\binom{m}{2}}{\binom{N}{2}}.$$

Thus,

$$\mathbb{E}[X^2] = n \cdot \frac{m}{N} + 2\binom{n}{2}\frac{\binom{m}{2}}{\binom{N}{2}} = \frac{nm}{N} + \frac{n(n-1)m(m-1)}{N(N-1)},$$

which shows

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{nm}{N} + \frac{n(n-1)m(m-1)}{N(N-1)} - \frac{n^2m^2}{N^2} = \frac{nm(N-m)(N-n)}{N^2(N-1)}.$$

Chapter 6

Continuous Random Variables

Lecture 13

6.1 Sigma Algebras

7 April.

Example 6.1.1 (Sigma algebras). If we have infinite sequence of tosses of a fair coin. Let $S = \{H, T\}^{\mathbb{N}}$, then what is $\mathbb{P}(w)$ for $w \in S$?

Proof. Since the coin is fair, so the probability should be uniform, and thus no one sequence should be more likely than another. Hence, for w, since each toss has $\frac{1}{2}$ probability, and due to the independence, we can multiply the probabilities: $\mathbb{P}(w)$ should be 0. However, $S = \bigcup_{w\in S} \{w\}$, so

$$1 = \mathbb{P}(S) = \mathbb{P}\left(\bigcup_{w\in S} \{w\}\right) = \sum_{w\in S} \mathbb{P}(\{w\}) = 0?$$

In fact, there is an error:

$$\mathbb{P}\left(\bigcup_{w\in S} \{w\}\right) \neq \sum_{w\in S} \mathbb{P}(\{w\}),$$

this is because probability only has "countable" additivity but S is uncountable. ⊛

Thus, we can define a uniform probability measure, where we wants:

(1) $\mathbb{P}(S) = 1$.

(2) $\mathbb{P}(\{w\}) = 0$ for every $w \in S$.

(3) If $E_1, E_2, \cdots \subseteq S$ are mutually exclusive, then $\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(E_i)$.

(4) For any $i \in \mathbb{N}$, let $T_i : S \to S$ be the map that flips the outcome of the i-th coin toss, and for $I \subseteq \mathbb{N}$, let $T_I : S \to S$ be the map that flips the outcomes of all toses in I. For every $E \subseteq S$, and any $I \subseteq \mathbb{N}$, then $\mathbb{P}(T_I(E)) = \mathbb{P}(E)$.

Proposition 6.1.1. No such $\mathbb{P}$ exists.

Proof. Define an equivalence relation on S. Say $w_1, w_2 \in S$ are equivalent if they only have finitely many differences, i.e.e $\exists n \in \mathbb{N}$ s.t. $(w_1)_i = (w_2)_i$ for all $i \geq n$. Then, we can partition S into equivalence classes $[w] = \{w' \in S : w' \sim w\}$. Let $E \subseteq S$ be an event that contains one member from every equivalence class. (by Axiom of choice)

Claim 6.1.1. $S = \bigcupdot_{\substack{I\subseteq\mathbb{N}\\|I|<\infty}} T_I(E)$.

Proof. For every $w \in S$, we have that $[w] \cap E = \{w'\}$ for some $w' \sim w$. Hence, $\exists$ finite st $I \subseteq \mathbb{N}$ of differences. Thus, $T_I(\{w'\}) = \{w\}$ and $T_I(\{w'\}) \subseteq T_I(E)$. Hence, $w \in \bigcup_{\substack{I \subseteq \mathbb{N} \\ |I| < \infty}} T_I(E)$.

Now we show the union is disjoint. If $T_I(E) \cap T_J(E) \neq \varnothing$ for some $I \neq J$, then $\exists w, w' \in E$ s.t.

$$T_I(\{w\}) = T_J(\{w'\}) = \{w''\}\,.$$

Hence, $w \sim w'' \sim w'$, so $w = w'$, so $I = J$ since this means flips toss in I and toss in J gets same outcome. ⊛

Note that $S = \bigcup_{\substack{I \subseteq \mathbb{N} \\ |I| < \infty}} T_I(E)$ and R.H.S. is a countable union. Thus,

$$1 = \mathbb{P}(S) = \mathbb{P}\left(\dot{\bigcup_{\substack{I \subseteq \mathbb{N} \\ |I| < \infty}}} T_I(E) \right) = \sum_{\substack{I \subseteq \mathbb{N} \\ |I| < \infty}} \mathbb{P}(T_I(E)) = \sum_{\substack{I \subseteq \mathbb{N} \\ |I| < \infty}} \mathbb{P}(E)$$

by property (4) above. However, if $\mathbb{P}(E) = 0$, then $\sum_{\substack{I \subseteq \mathbb{N} \\ |I| < \infty}} \mathbb{P}(E) = 0$, but if $\mathbb{P}(E) > 0$, then $\sum_{\substack{I \subseteq \mathbb{N} \\ |I| < \infty}} \mathbb{P}(E) = \infty$, so such $\mathbb{P}$ does not exist. ■

Definition 6.1.1. A sigma-algebra (σ-algebra) is a collection of subsets $\Sigma \subseteq 2^S$ satisfying

(1) $S \in \Sigma$.

(2) If $E \in \Sigma$, then $E^c \in \Sigma$.

(3) If $E_1, E_2, \cdots \in \Sigma$, then $\bigcup_{n=1}^{\infty} E_n \in \Sigma$.

Discussion and Motivation. The preceding example illustrates a fundamental obstruction in defining a probability measure on the entire power set 2^S when $S = \{H, T\}^{\mathbb{N}}$.

At first glance, it is natural to require that the probability measure $\mathbb{P}$ be *uniform*, in the sense that every infinite sequence $w \in S$ has the same probability. Since each coordinate is an independent fair coin toss, this suggests that $\mathbb{P}(\{w\}) = 0$ for all $w \in S$, while $\mathbb{P}(S) = 1$. This is not contradictory, because countable additivity does not extend to uncountable unions.

However, the proposition above shows that even if we only assume the following seemingly reasonable properties:

- normalization ($\mathbb{P}(S) = 1$),
- countable additivity,
- and invariance under finite coordinate flips,

we arrive at a contradiction. The key step is the construction (via the axiom of choice) of a set $E \subseteq S$ that selects exactly one representative from each equivalence class of sequences that differ in only finitely many coordinates. This set is highly non-constructive and exhibits pathological behavior.

The contradiction arises because S can be decomposed into a countable disjoint union of translates $T_I(E)$, all of which must have the same probability as E by invariance. This forces the total measure to be either 0 or ∞, contradicting $\mathbb{P}(S) = 1$.

The conclusion is that it is impossible to define a probability measure on all subsets of S that satisfies these natural properties. In particular, there exist subsets of S (such as the set E above) to which no consistent probability can be assigned.

This motivates the introduction of *σ-algebras*. Instead of assigning probabilities to all subsets of S, we restrict attention to a smaller collection $\Sigma \subseteq 2^S$ of *measurable sets*, which is closed under complements and countable unions. A probability measure is then defined only on Σ, thereby avoiding pathological sets and ensuring consistency of the theory.

"Fix" probability Now we can define a probability space to be a triple $(S, \Sigma, \mathbb{P})$, where S is the sample space, $\Sigma \subseteq 2^S$ is a sigma-algebra, and $\mathbb{P} : \Sigma \to \mathbb{R}_{\geq 0}$ assigns a probability to all events (members of Σ).

Definition 6.1.2. The Σ generated by intervals $[a, b]$ are called Borel σ-algebra.

Remark 6.1.1. If S is finite, we can take $\Sigma = 2^S$.

6.2 Introduction

Definition 6.2.1 (Cumulative distribution function)**.** Given a probability space and a random variable $X : S \to \mathbb{R}$, the cumulative distribution function, or cdf, $F_X : \mathbb{R} \to [0, 1]$ is given by

$$F_X(b) = \mathbb{P}(X \leq b).$$

Proposition 6.2.1.

(1) $\lim_{x\to\infty} F_X(x) = 1$, and $\lim_{x\to-\infty} F_X(x) = 0$.

(2) $F_X(x)$ is increasing in x.

(3) F_X is right-continuous: Let $\{x_n\}_{n\geq 1}$ be a sequence such that

$$x_n \searrow x,$$

which means:

1. $x_1 \geq x_2 \geq \cdots \geq x_n \geq \cdots$ (monotone decreasing),
2. $\lim\limits_{n\to\infty} x_n = x$.

Then the cumulative distribution function F_X is *right-continuous* at x, i.e.,

$$\lim_{n\to\infty} F_X(x_n) = F_X(x).$$

proof of (1).

$$\lim_{x\to\infty} F_X(x) = \mathbb{P}\left(\lim_{x\to\infty} \{X \leq x\}\right) = \mathbb{P}\left(\bigcup_{x\to\infty} \{X \leq x\}\right) = \mathbb{P}(X \in \mathbb{R}) = 1.$$

■

proof of (2). If $a < b$, then

$$F_X(b) - F_X(a) = \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a) = \mathbb{P}(a < X \leq b) \geq 0.$$

■

proof of (3). The sets $\{X \leq x_n\}$ form a decreasing sequence of events:

$$\{X \leq x_1\} \supseteq \{X \leq x_2\} \supseteq \cdots$$

and

$$\bigcap_{n=1}^{\infty} \{X \leq x_n\} = \{X \leq x\}.$$

By the continuity of probability from above, we have

$$\lim_{n\to\infty} \mathbb{P}(X \leq x_n) = \mathbb{P}\left(\lim_{n\to\infty} \{X \leq x_n\}\right) = \mathbb{P}\Big(\bigcap_{n=1}^{\infty} \{X \leq x_n\}\Big) = \mathbb{P}(X \leq x) = F_X(x).$$

Intuition: The symbol $x_n \searrow x$ indicates a sequence approaching x *from the right*, e.g., $x_n = x + \frac{1}{n}$. ■

Note 6.2.1. F_X does not have to be left-continuous: If $b_n \nearrow b$, then $\{X \leq b_n\} \nearrow \{X < b\}$. Thus,

$$\lim_{n\to\infty} F_X(b_n) = \mathbb{P}(X < b) = F_X(b) - \mathbb{P}(X = b).$$

Definition 6.2.2 (Continuoius random variable). A random variable X is said to be (absolutely) continuous if there is some function $f_X : \mathbb{R} \to \mathbb{R}_{\geq 0}$ s.t. the cumulative distribution function

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t.$$

Then, f_X is the probability density function of X, or pdf.

Remark 6.2.1. In general,

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$

If X is absolutely continuous, then

$$\mathbb{P}(a < X \leq b) = \int_{-\infty}^{b} f_X(t)\,\mathrm{d}t - \int_{-\infty}^{a} f_X(t)\,\mathrm{d}t = \int_{a}^{b} f_X(t)\,\mathrm{d}t.$$

Thus, $f_X(x)$ is a measure of how likely the random variable is to be around x. Hence,

$$\mathbb{P}(x - \varepsilon < X \leq x + \varepsilon) = \int_{x-\varepsilon}^{x+\varepsilon} f_X(t)\,\mathrm{d}t \approx 2\varepsilon f_X(x)$$

as $\varepsilon \searrow 0$ if f_X is "nice". Hence, by taking $\varepsilon \searrow 0$, we know $\mathbb{P}(X = x) = 0$ for nice $f_X(x)$. Actually, continuity if $f_X(x)$ can guarantee this thing.

Remark 6.2.2. By FTC, $f_X(x) = F'_X(x)$.

Example 6.2.1. Now if

$$f(x) = \begin{cases} \dfrac{C}{x^3}, & \text{if } x \geq 1; \\ 0, & \text{if } x < 1. \end{cases}$$

(a) What does C have to be for f to be the pdf of a random variable of X?

(b) What is $\mathbb{P}(a \leq X \leq b)$ for $1 \leq a \leq b$?

Proof.

(a) Since

$$1 = \mathbb{P}(-\infty < X < \infty) = \int_{-\infty}^{\infty} f(t)\,\mathrm{d}t = \int_{1}^{\infty} \frac{C}{t^3}\,\mathrm{d}t = \left[\frac{-C}{2t^2}\right]_1^{\infty} = \frac{C}{2},$$

so $C = 2$.

(b) For $1 \leq a \leq b$,

$$\mathbb{P}(a \leq X \leq b) = \int_{a}^{b} f(t)\,\mathrm{d}t = \frac{1}{a^2} - \frac{1}{b^2}.$$

⊛

Definition 6.2.3 (Expectation and Variance of continuous random variable)**.** In the discrete setting, we define $\mathbb{E}[X] = \sum_{x_i} x_i \mathbb{P}(X = x_i)$, while in the continuous setting, we define $\mathbb{E}[X] = \int_{-\infty}^{\infty} t f_X(t)\,\mathrm{d}t$, and we define the variance $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$.

Lecture 14

9 April.

Example 6.2.2. If X is a continuous random variable with

$$f_X(t) = \begin{cases} \dfrac{C}{t^3}, & \text{if } t \geq 1; \\ 0, & \text{if } t < 1. \end{cases}$$

(1) What is C?

(2) What are $\mathbb{E}[X]$ and $\mathrm{Var}(X)$?

Proof.

(1) We have $\int_{-\infty}^{\infty} f_X(t)\,\mathrm{d}t = 1$, so

$$1 = \int_{-\infty}^{1} f_X(t)\,\mathrm{d}t + \int_{1}^{\infty} f_X(t)\,\mathrm{d}t = \int_{1}^{\infty} f_X(t)\,\mathrm{d}t = \frac{C}{2},$$

so $C = 2$.

(2)

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} t f_X(t)\,\mathrm{d}t = \int_{1}^{\infty} t f_X(t)\,\mathrm{d}t = 2.$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \int_{-\infty}^{\infty} t^2 f_X(t)\,\mathrm{d}t - 4 = \int_{1}^{\infty} t^2 \cdot \frac{2}{t^3}\,\mathrm{d}t - 4 = \infty,$$

so $\mathrm{Var}(X)$ does not exist.

⊛

Remark 6.2.3. The divergence of the variance or even the expectation can also occur for discrete random variables when the probability of large values doesn't decay very quickly, e.g. $\mathbb{P}(X = k) = \frac{c}{k^2}$ for $k \in \mathbb{N}$.

6.3 Continuous Probability Distribution

6.3.1 Uniform Distribution

Definition 6.3.1 (Uniform Distribution)**.** Suppose $\alpha < \beta \in \mathbb{R}$, then we say $X \sim \mathrm{Unif}(\alpha, \beta)$ if we want to choose a uniformly random point in $[\alpha, \beta]$, and X is the outcome. Note that the probability of a subinterval of $[\alpha, \beta]$ only depends on its width.

PDF

$$f_X(t) = \begin{cases} C, & \text{if } t \in [\alpha, \beta]; \\ 0, & \text{if } t \notin [\alpha, \beta] \end{cases}$$

since the probability of every point in $[\alpha, \beta]$ to be chosen is same. Now we compute C. Note that

$$1 = \int_{-\infty}^{\infty} f_X(t)\,\mathrm{d}t = \int_{\alpha}^{\beta} C\,\mathrm{d}t = C(\beta - \alpha),$$

so $C = \frac{1}{\beta - \alpha}$.

CDF

$$F_X(x) = \mathbb{P}(X \le x) = \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t = \begin{cases} 0, & \text{if } x \le \alpha; \\ \displaystyle\int_{\alpha}^{x} \frac{1}{\beta - \alpha}\,\mathrm{d}t = \frac{x - \alpha}{\beta - \alpha}, & \text{if } \alpha \le x \le \beta; \\ 1, & \text{if } x \ge \beta. \end{cases}$$

Expectation

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} t f_X(t)\,\mathrm{d}t = \int_{\alpha}^{\beta} \frac{t}{\beta - \alpha}\,\mathrm{d}t = \frac{\alpha + \beta}{2}.$$

Variance We know $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$, where

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} t^2 f_X(t)\,\mathrm{d}t = \int_{\alpha}^{\beta} \frac{t^2}{\beta - \alpha}\,\mathrm{d}t = \frac{\beta^2 + \alpha\beta + \alpha^2}{3}.$$

$$\mathbb{E}[X]^2 = \left(\frac{\alpha + \beta}{2}\right)^2 = \frac{\beta^2 + 2\alpha\beta + \alpha^2}{4}.$$

Hence,

$$\mathrm{Var}(X) = \frac{\beta^2 + \alpha\beta + \alpha^2}{3} - \frac{\beta^2 + 2\alpha\beta + \alpha^2}{4} = \frac{(\beta - \alpha)^2}{12}.$$

Theorem 6.3.1 (Simulating continuous random variables)**.** Let X be a continuous random variable with cdf $F_X : \mathbb{R} \to [0, 1]$. Let $U \sim \mathrm{Unif}(0, 1)$. Let $Y = F_X^{-1}(U) \in \mathbb{R}$. Then Y has the same distribution as X.

Proof. We want to show $F_Y = F_X$. Then,

$$F_Y(x) = \mathbb{P}(Y \le x) = \mathbb{P}(F_X^{-1}(U) \le x) = \mathbb{P}(U \le F_X(x)) = F_U(F_X(x)) = \frac{F_X(x) - 0}{1 - 0} = F_X(x)$$

because F_X is increasing and $U \sim \mathrm{Unif}(0, 1)$. ■

6.3.2 Exponential Distribution

Setting: A bus arrives at a bus stop on average every 18 minutes. We arrive at the bus stop. How long do we have to wait for the next bus?

If X is our waiting time, we want X to satisfy memoryless property:

$$\mathbb{P}(X \ge x + y \mid X \ge x) = \mathbb{P}(X \ge y).$$

Theorem 6.3.2. The only continuous memoryless distribution is the exponential distribution.

Proof. Let X be a memoryless continuous distribution. Define $g(x) = \mathbb{P}(X \ge x)$. Then,

$$g(x + y) = \mathbb{P}(X \ge x + y) = \mathbb{P}(X \ge x + y \mid X \ge x)\mathbb{P}(X \ge x) = \mathbb{P}(X \ge y)\mathbb{P}(X \ge x) = g(y)g(x).$$

Thus, $g(x + y) = g(x)g(y)$ for all $x, y \ge 0$. Now define $\lambda = -\ln(g(1))$, i.e. $g(1) = e^{-\lambda}$. Since $g(1) = \mathbb{P}(X \ge 1) \in [0, 1]$, so $\lambda \in [0, \infty)$. Let $n \in \mathbb{N}$, then

$$g(n) = g(1 + 1 + \cdots + 1) = g(1)^n = e^{-n\lambda}.$$

Let $\frac{r}{s} \in \mathbb{Q}_{\geq 0}$, then

$$g(r) = g\left(\frac{r}{s} + \frac{r}{s} + \cdots + \frac{r}{s}\right) = g\left(\frac{r}{s}\right)^s,$$

so $g\left(\frac{r}{s}\right) = g(r)^{\frac{1}{s}} = (e^{-r\lambda})^{\frac{1}{s}} = e^{-\lambda\frac{r}{s}}$. Now let $x \in \mathbb{R}$, let q_n be a sequence of rational numbers with $q_n \searrow x$. Then,

$$\begin{aligned} g(x) &= \mathbb{P}(X \geq x) = 1 - \mathbb{P}(X \leq x) = 1 - \lim_{n\to\infty} \mathbb{P}(X \leq q_n) = \lim_{n\to\infty} \mathbb{P}(X > q_n) = \lim_{n\to\infty} g(q_n) \\ &= \lim_{n\to\infty} e^{-\lambda q_n} = e^{-\lambda x}. \end{aligned}$$

Remark 6.3.1. This proof only holds for positive real numbers, otherwise we cannot use technique like $g(\underbrace{1 + 1 + \cdots + 1}_{r}) = g(1)^r$.

■

Notation. We say $X \sim \text{Exp}(\lambda)$ for $\lambda \in \mathbb{R}_{>0}$ if X is of exponential distribution.

CDF

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} 1 - e^{-\lambda x}, & \text{if } x \geq 0; \\ 0, & \text{if } x < 0. \end{cases}$$

PDF

$$f_X(t) = F_X'(t) = \begin{cases} \lambda e^{-\lambda t}, & \text{if } t \geq 0; \\ 0, & \text{if } t < 0. \end{cases}$$

Expectation

$$\begin{aligned} \mathbb{E}[X] &= \int_{-\infty}^{\infty} t f_X(t) \,\mathrm{d}t = \int_0^{\infty} t\lambda e^{-\lambda t} \,\mathrm{d}t \\ &= [t(-e^{-\lambda t})]_0^{\infty} - \int_0^{\infty} (-e^{-\lambda t}) \,\mathrm{d}t = \int_0^{\infty} e^{-\lambda t} \,\mathrm{d}t = \left[\frac{-e^{-\lambda t}}{\lambda}\right]_0^{\infty} = \frac{1}{\lambda}. \end{aligned}$$

Hence, $\mathbb{E}[X] = \frac{1}{\lambda}$.

Variation Since $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$, where

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} t^2 f_X(t) \,\mathrm{d}t = \int_0^{\infty} t^2 \lambda e^{-\lambda t} \,\mathrm{d}t = \frac{2}{\lambda^2}.$$

Thus,

$$\mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

Example 6.3.1. A bus arrives on average every 15 minutes. What is the probability that we have to wait at least 5 minutes for the bus?

Proof. $X \sim \text{Exp}(\lambda)$, and $\mathbb{E}[X] = \frac{1}{\lambda} = 15$, so $\lambda = \frac{1}{15}$. Since

$$\mathbb{P}(X \geq x) = e^{-\lambda x},$$

so $\mathbb{P}(X \geq 5) = e^{-5\lambda} = e^{-\frac{1}{3}}$. ⊛

Example 6.3.2. A passenger can take one of two buses. The first arrives on average every 15 minutes, the second on average every 18 minutes. If they take the first bus arrives, and the buses run independently, how long would they have to wait on average?

Proof. Let X_1 be the waiting time for the first bus, X_2 be the waiting time for the second, then $X_1 \sim \text{Exp}(\frac{1}{15})$ and $X_2 \sim \text{Exp}(\frac{1}{18})$, then if W is the waiting time for the passenger, then $W = \min(X_1, X_2)$. Hence,

$$\begin{aligned} F_W(x) &= \mathbb{P}(W \leq x) = \mathbb{P}(\min(X_1, X_2) \leq x) = 1 - \mathbb{P}(\min(X_1, X_2) > x) \\ &= 1 - \mathbb{P}(X_1 > x \text{ and } X_2 > x) = 1 - \mathbb{P}(X_1 > x)\mathbb{P}(X_2 > x) \\ &= 1 - (e^{-\lambda_1 x})(e^{-\lambda_2 x}) = 1 - e^{-(\lambda_1+\lambda_2)x}, \end{aligned}$$

where $\lambda_1 = \frac{1}{15}$ and $\lambda_2 = \frac{1}{18}$. Thus,

$$W \sim \text{Exp}(\lambda_1 + \lambda_2) = \text{Exp}\left(\frac{1}{15} + \frac{1}{18}\right) = \text{Exp}\left(\frac{11}{90}\right).$$

Hence, $\mathbb{E}[W] = \frac{90}{11}$.

Remark 6.3.2. More generally, if $X_i \sim \text{Exp}(\lambda_i)$ for $1 \leq i \leq n$, and these are independent, then

$$X = \min(X_1, X_2, \ldots, X_n) \sim \text{Exp}\left(\sum_{i=1}^{n} \lambda_i\right).$$

⊛

Example 6.3.3 (Waiting Time Paradox). The *waiting time paradox* says that if buses do *not* arrive at perfectly regular intervals, then a passenger who shows up at a random time tends to land in a *longer-than-average* gap between buses. Because of that sampling bias, the time the passenger waits can be larger than "half of the average gap," and the bus interval that the passenger finds themself inside is also typically larger than the population-average interval.

What is going on in your example?

Let

$$X = \text{the time from now until the next bus arrives.}$$

If we model the bus arrivals as a Poisson process with rate $\lambda = 1/10$ per minute, then

$$X \sim \text{Exp}(1/10), \qquad \mathbb{E}[X] = 10 \text{ minutes.}$$

Now suppose that, at the moment you observe the system,

- the last bus left 8 minutes ago, and
- the next bus will come in 2 minutes.

So the two buses around you are exactly 10 minutes apart.
The apparent paradox is this:

(1) By the memoryless property, the future waiting time from *now* has mean 10 minutes.

(2) By time-reversal symmetry of the Poisson process, the elapsed time since the previous bus also has mean 10 minutes.

(3) So it looks like

$$\mathbb{E}[\text{backward gap}] + \mathbb{E}[\text{forward gap}] = 10 + 10 = 20,$$

but the bus interval in the picture is only 10 minutes.

Where the paradox lies

The mistake is that these expectations are *not conditioned on the event that the surrounding gap equals* 10 *minutes.* They are unconditional averages over *all* possible observation times.

There are really two different random quantities:

- B = time since the previous bus (backward recurrence time),
- F = time until the next bus (forward recurrence time).

For a Poisson process,

$$B \sim \text{Exp}(1/10), \qquad F \sim \text{Exp}(1/10), \qquad \mathbb{E}[B] = \mathbb{E}[F] = 10.$$

But once you are told that in this particular realization $B = 8$ and $F = 2$, there is nothing random left in that local picture: the surrounding interval is simply

$$B + F = 8 + 2 = 10.$$

So the incorrect step is mixing

$$\mathbb{E}[B] + \mathbb{E}[F]$$

with

$$B + F \text{ for a particular observed sample.}$$

Those are different objects.

Why $\mathbb{E}[B + F] = 20$ is actually correct

For a Poisson bus process, the bus interval containing a *random observation time* is not distributed like an ordinary interarrival time. It is *length-biased*: long gaps are more likely to contain your arrival time.

If the ordinary interarrival time is

$$T \sim \text{Exp}(1/10), \qquad \mathbb{E}[T] = 10,$$

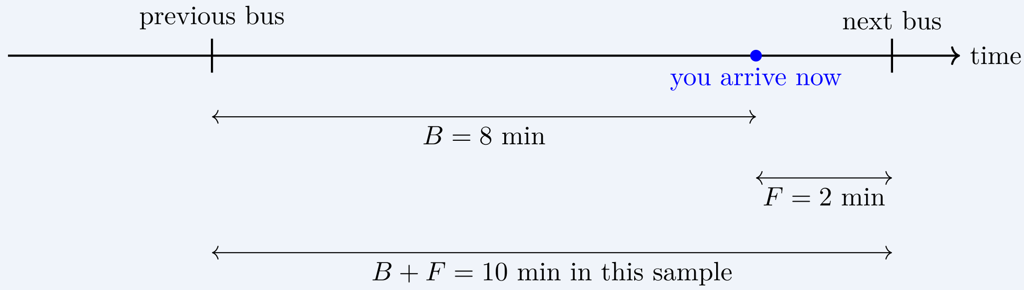
then the interval containing a random observation time has mean

$$\mathbb{E}[B + F] = 20.$$

That does *not* contradict the fact that an ordinary gap between consecutive buses has average length 10. It only says that when you sample by picking a random time, you are more likely to land inside a long gap than a short one.

For the exponential case, one can verify directly that

$$\mathbb{E}[B + F] = \mathbb{E}[B] + \mathbb{E}[F] = 10 + 10 = 20.$$



In this drawing, the local sample gives $8 + 2 = 10$. But if you repeat the experiment many times and choose your observation time uniformly at random on a long timeline, then the average containing interval is 20 minutes, not 10 minutes, because long intervals are overrepresented.

Short resolution

The paradox disappears once we separate:

- **ordinary bus gap:** mean 10 minutes,
- **gap seen by a random observer:** mean 20 minutes,
- **your particular observed sample:** here it is exactly $8 + 2 = 10$ minutes.

So the statement "$10 + 10 = 10$" is not a real contradiction. The left side is an *unconditional expectation*; the right side is the length of *one realized interval* in your example.

Lecture 15

21 April.

Example 6.3.4. The *friendship paradox* is the surprising fact that, in many social networks, **your friends tend to have more friends than you do, on average**. This does not mean every single friend is more popular than you. Instead, it means that when we sample people by following friendship links, highly connected people are more likely to be seen, because they appear in many people's friend lists.

A small graph intuition

Consider the graph in Figure 6.1. Each node is a person, and an edge means a friendship. Node C is a hub with degree 4, while the other four nodes have degree 1.

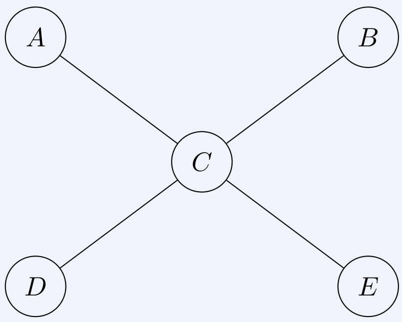


Figure 6.1: A simple network where one person is much more connected than the others.

The average degree of all people in this network is

$$\frac{1+1+4+1+1}{5} = \frac{8}{5} = 1.6.$$

Now ask: if we pick a person and look at their friends, how many friends do those friends typically have?

For A, the only friend is C, who has degree 4. The same is true for B, D, and E. Only C sees low-degree neighbors. Because C is listed as a friend by four different people, C gets counted over and over when we average over "friends of people." This pushes the average upward.

In fact, the average degree of a randomly chosen *friend* in this graph is

$$\frac{1^2+1^2+4^2+1^2+1^2}{1+1+4+1+1} = \frac{20}{8} = 2.5,$$

which is larger than 1.6.

Why the paradox happens

The key idea is **sampling bias**. People with many friends are more likely to show up when we traverse edges, because each friendship gives them another chance to be observed. In network terms, nodes with larger degree receive more weight.

If the degrees are $k_1, k_2, \ldots, k_n$, then:

$$\text{average degree of a person} = \frac{1}{n}\sum_{i=1}^{n} k_i,$$

while the average degree of a person reached by following a random friendship edge is

$$\frac{\sum_{i=1}^{n} k_i^2}{\sum_{i=1}^{n} k_i}.$$

This second quantity is usually larger, because squaring gives extra weight to large degrees. That is the mathematical reason the friendship paradox appears.

Takeaway

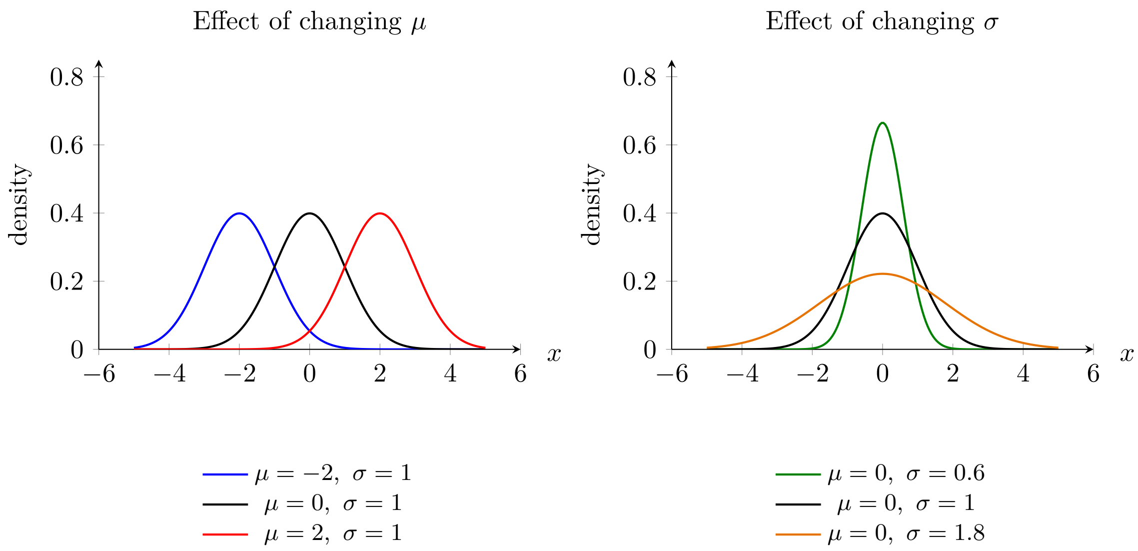
The friendship paradox is not really about friendship in particular; it is a general fact about networks. Whenever highly connected nodes are overrepresented in the way we sample, our local view of the network makes others look more connected than the population average.

6.3.3 Normal Distribution

We denote $X \sim N(\mu, \sigma^2)$ for some $\mu \in \mathbb{R}$ (mean) and $\sigma \in \mathbb{R}_{>0}$ (standard deviation) for the normal distribution.

What happens when the values change?

- **Larger μ:** the whole curve moves to the **right**.
- **Smaller μ:** the whole curve moves to the **left**.
- **Larger σ:** the curve becomes **wider and flatter**.
- **Smaller σ:** the curve becomes **narrower and taller**.



pdf

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}},$$

which has $f_X(x) > 0$ for all x, so for any $a < b$, $\mathbb{P}(a \leq x \leq b) > 0$.

Claim 6.3.1.

$$\int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1.$$

Proof. Let $I = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\,\mathrm{d}x$, and let $z = \frac{x-\mu}{\sigma}$, so we have

$$I = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2}} \sigma\,\mathrm{d}z = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}}\,\mathrm{d}z.$$

Hence,

$$I^2 = \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}}\,\mathrm{d}z\right)\left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{y^2}{2}}\,\mathrm{d}y\right) = \frac{1}{2\pi} \iint_{\mathbb{R}^2} e^{-\frac{z^2+y^2}{2}}\,\mathrm{d}z\,\mathrm{d}y.$$

Now we can substitute $z = r\cos\theta$ and $y = r\sin\theta$, and

$$\left\|\begin{pmatrix} \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix}\right\| = \left\|\begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\right\| = r\cos^2\theta + r\sin^2\theta = r.$$

Thus, $\mathrm{d}z\,\mathrm{d}y = r\mathrm{d}r\,\mathrm{d}\theta$. Hence,

$$I^2 = \frac{1}{2\pi} \int_0^{2\pi} \int_0^{\infty} e^{-\frac{(r\cos\theta)^2 + (r\sin\theta)^2}{2}} r\,\mathrm{d}r\,\mathrm{d}\theta = 1.$$

This gives $I = 1$. ⊛

Definition 6.3.2. We say $N(0,1)$ is the *standard normal distribution.*

Note that if $Z \sim N(0,1)$, then

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}.$$

Claim 6.3.2. If $X \sim N(\mu, \sigma^2)$, then if $Z = \frac{X-\mu}{\sigma}$, we have $Z \sim N(0,1)$.

Proof.

$$\mathbb{P}(Z \leq z) = \mathbb{P}\left(\frac{X-\mu}{\sigma} \leq z\right) = \mathbb{P}(X \leq \mu + z\sigma) = \int_{-\infty}^{\mu+z\sigma} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\,\mathrm{d}x.$$

Now we change the variable, let $t = \frac{x-\mu}{\sigma}$, i.e. $x = \mu + t\sigma$, then we have

$$\mathbb{P}(Z \leq z) = \int_{-\infty}^{\mu+z\sigma} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\,\mathrm{d}x = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}\,\mathrm{d}t = \mathbb{P}(N(0,1) \leq z).$$

⊛

Expectation Note that if $Z \sim N(0,1)$, then

$$\mathbb{E}[Z] = \int_{-\infty}^{\infty} t f_Z(t)\,\mathrm{d}t = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t e^{-\frac{t^2}{2}}\,\mathrm{d}t = \frac{1}{\sqrt{2\pi}}\left[-e^{-\frac{t^2}{2}}\right]_{-\infty}^{\infty} = 0.$$

Now if $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma}$, i.e. $X = \mu + \sigma Z$, so

$$\mathbb{E}[X] = \mu + \sigma\mathbb{E}[Z] = \mu.$$

Variance If $Z \sim N(0,1)$, then

$$\mathbb{E}[Z^2] = \int_{-\infty}^{\infty} t^2 f_Z(t)\,\mathrm{d}t = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t \cdot \left[t e^{-\frac{t^2}{2}}\right]\,\mathrm{d}t,$$

and we can integrate by parts:

$$\begin{aligned}\mathbb{E}[Z^2] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t\left(t e^{-\frac{t^2}{2}}\right)\,\mathrm{d}t = \frac{1}{\sqrt{2\pi}}\left[\left[t(-e^{-\frac{t^2}{2}})\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}}\,\mathrm{d}t\right] = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}\,\mathrm{d}t \\ &= \int_{-\infty}^{\infty} f_Z(t)\,\mathrm{d}t = 1.\end{aligned}$$

Hence,

$$\mathrm{Var}(Z) = \mathbb{E}[Z^2] - \mathbb{E}[Z]^2 = 1 - 0 = 1.$$

Thus, for $X \sim N(\mu, \sigma^2)$, we have $Z = \frac{X-\mu}{\sigma}$, i.e. $Z\sigma + \mu = X$. Hence,

$$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[(Z\sigma)^2] = \sigma^2\mathbb{E}[Z^2] = \sigma^2.$$

Example 6.3.5. Suppose we have two envelops, one has $\$x$ and the other has $\$2x$, and we choose one, and look inside, and we can choose whether or not to switch the envelop we choose. What can we do to maximize our chances of getting more money?

Proof. Suppose the envelop we get has $\$y$, then there is 50% of chance that $y = x$ and the other has $\$2y$, while there is 50% of chance that $y = 2x$ and the other has $\frac{1}{2}y$. Thus,

$$\mathbb{E}[\text{payout}] = \frac{1}{2}(\$2y) + \frac{1}{2}(\$\frac{1}{2}y) = \$\frac{5}{4}y.$$

We can beat 50% chance of getting the bigger envelop. The method is to let $Z \sim N(0,1)$ and sample z from Z. If $y < z$, then switch the envelop, and if $y \geq z$, then keep the envelop. There are three cases:

- Case 1: $z \leq x$, always keep envelop, we have 50% of winning.
- Case 2: $x < z \leq 2x$, always get right right envelop, so we have 100% of winning.
- Case 3: $2x < z$, always switch, so 50% of winning.

By the law of probability,

$$\begin{aligned}
\mathbb{P}(\text{win}) &= \mathbb{P}(\text{win} \mid z < x)\mathbb{P}(z < x) + \mathbb{P}(\text{win} \mid x < z < 2x)\mathbb{P}(x < z < 2x) + \mathbb{P}(\text{win} \mid 2x < z)\mathbb{P}(2x < z) \\
&= 50\% \cdot \mathbb{P}(z \leq x) + 100\% \cdot \mathbb{P}(x < z \leq 2x) + 50\% \cdot \mathbb{P}(2x < z) \\
&= 50\%(1 - \mathbb{P}(x < z \leq 2x)) + 100\% \cdot \mathbb{P}(x < z \leq 2x) \\
&= 50\% + 50\% \cdot \mathbb{P}(x < z \leq 2x) > 50\%.
\end{aligned}$$

⊛

Recall that if $X \sim N(\mu, \sigma^2)$, $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, then if we let $Z = \frac{X-\mu}{\sigma}$, then $Z \sim N(0,1)$, and we call Z to be the z-score.

CDF

$$F_Z(z) = \mathbb{P}(Z \leq z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} \, \mathrm{d}t := \Phi(z),$$

which cannot be expressed by elementary functions.

Remark 6.3.3.

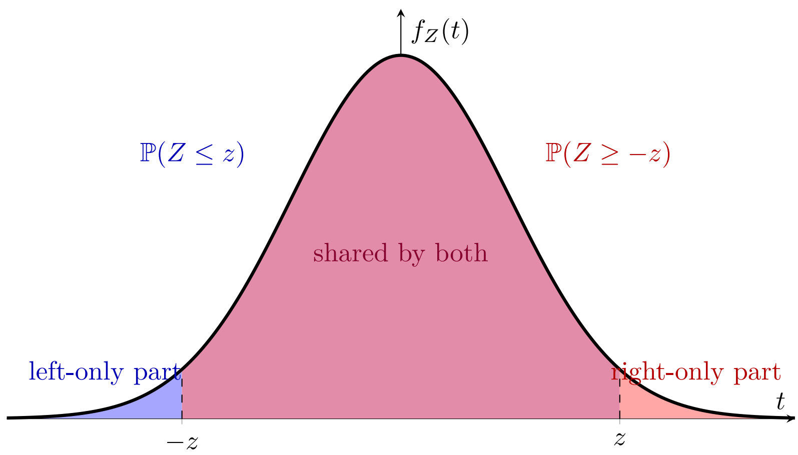
(i) $\mathbb{P}(a < Z \leq b) = \mathbb{P}(Z \leq b) - \mathbb{P}(Z \leq a) = \Phi(b) - \Phi(a)$.

(ii) $\mathbb{P}(Z > a) = 1 - \mathbb{P}(Z \leq a) = 1 - \Phi(a)$.

(iii) By symmetry of this distribution,

$$\Phi(z) = \mathbb{P}(Z \leq z) = \mathbb{P}(Z \geq -z) = 1 - \mathbb{P}(Z \leq -z) = 1 - \Phi(-z).$$

Thus, it suffices to know $\Phi(z)$ for $z \geq 0$.



(iv) If $X \sim N(\mu, \sigma^2)$,

$$\mathbb{P}(X \leq x) = \mathbb{P}\left(Z \leq \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right).$$

Example 6.3.6. Transmitting binary data over a noise channel. If we send 1, then the receiver gets $1 + X$; if we send 0, then the receiver gets $0 + X = X$, where $X \sim N(0, 0.3^2)$. Our decoding algorithm is, when the receiver receives a number ≥ 0.5, then they will see it as 1, otherwise they see it as 0. Then, what is the probability of mistaking a bit?

Proof. Suppose the signal sent is $S \in \{0,1\}$, then

$$\begin{aligned}\mathbb{P}(\text{error}) &= \mathbb{P}(\text{error} \mid S=0)\mathbb{P}(S=0) + \mathbb{P}(\text{error} \mid S=1)\mathbb{P}(S=1)\\ &= \mathbb{P}(X \geq 0.5)\mathbb{P}(S=0) + \mathbb{P}(X < -0.5)\mathbb{P}(S=1)\\ &= \mathbb{P}(X < -0.5)\mathbb{P}(S=0) + \mathbb{P}(X < -0.5)\mathbb{P}(S=1) \quad \text{(by symmetry)}\\ &= \mathbb{P}(X < -0.5) = \mathbb{P}\left(\frac{X-0}{0.3} < \frac{-0.5}{0.3}\right) = \Phi\left(-1.67\right).\end{aligned}$$

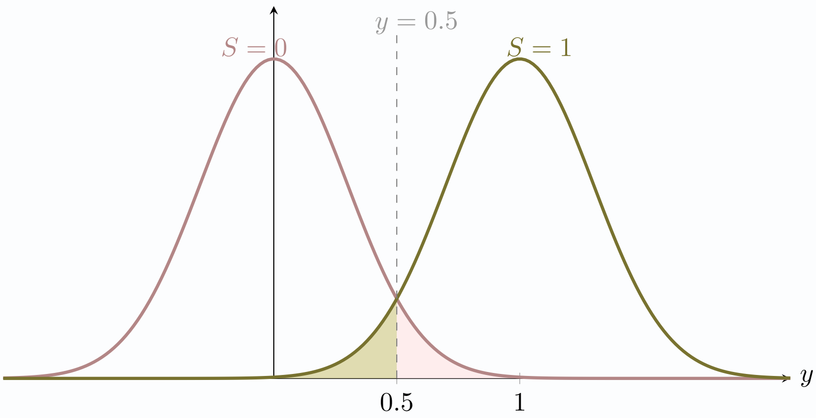


Figure 6.2: Colored area represents misreading signal

⊛

Lecture 16

Approximating Binomials If $np(1-p) \to \infty$, then $\text{Bin}(n,p)$ looks normal. 23 April.

Theorem 6.3.3 (de Moivre-Laplace). Let $S_n \sim \text{Bin}(n,p)$, and let $p \in (0,1)$ be fixed, then for any fixed $a < b$,

$$\mathbb{P}\left(a \leq \frac{S_n - np}{\sqrt{np(1-p)}} \leq b\right) \to \Phi(b) - \Phi(a) \text{ as } n \to \infty,$$

i.e. $\frac{S_n - np}{\sqrt{np(1-p)}}$ behaves like $N(0,1)$, which shows S_n behaves like $N(np, np(1-p))$.

Remark 6.3.4. This approximation already works well if $np(1-p) \geq 10$.

Continuity correction If $S_n \sim \text{Bin}(n,p)$, and we estimate it with a normal distribution, then we should correct for the continuity:

$$\{S_n = k\} \leftrightarrow \left\{k - \frac{1}{2} < S_n \leq k + \frac{1}{2}\right\}.$$

Example 6.3.7. Tossing a fair coin 40 times. What is the probability of seeing 20 heads?

Answer. Number of heads $S_n \sim \text{Bin}\left(40, \frac{1}{2}\right)$, then

$$\mathbb{P}(S_n = 20) = \binom{40}{20}\left(\frac{1}{2}\right)^{20}\left(\frac{1}{2}\right)^{20} = 0.1254.$$

For the normal approximation:

$$\begin{aligned}\mathbb{P}(S_n = 20) = \mathbb{P}(19.5 \le S_n \le 20.5) &= \mathbb{P}\left(\frac{-0.5}{\sqrt{10}} \le \frac{S_n - 20}{\sqrt{10}} \le \frac{0.5}{\sqrt{10}}\right) \\ &= \Phi\left(\frac{0.5}{\sqrt{10}}\right) - \Phi\left(\frac{-0.5}{\sqrt{10}}\right) = \Phi\left(\frac{0.5}{\sqrt{10}}\right) - \left(1 - \Phi\left(\frac{0.5}{\sqrt{10}}\right)\right) = 2\Phi\left(\frac{0.5}{\sqrt{10}}\right) - 1 \\ &\approx 0.1272.\end{aligned}$$

⊛

Example 6.3.8. There is a country with population of N. The government wants to estimate the level of support for a new initiative. Suppose $p\%$ of the population are in support, and the government polls n people (with replacement) uniformly at random. Then, number of positive responses $S_n \sim \text{Bin}(n, p)$, then what should n be to estimate p to within $\pm 1\%$ error with probability $\ge 95\%$?

Answer.

$$\begin{aligned}\{\text{Estimating } p \text{ to within } 1\%\} = \left\{\left|\frac{S_n}{n} - p\right| \le 0.01\right\} &= \{np - 0.01n \le S_n \le np + 0.01n\} \\ &= \{-0.01n \le S_n - np \le 0.01n\} \\ &= \left\{\frac{-0.01n}{\sqrt{np(1-p)}} \le \frac{S_n - np}{\sqrt{np(1-p)}} \le \frac{0.01n}{\sqrt{np(1-p)}}\right\}.\end{aligned}$$

Note that

$$\frac{0.01n}{\sqrt{np(1-p)}} = \frac{0.01\sqrt{n}}{\sqrt{p(1-p)}} \ge 0.02\sqrt{n},$$

so

$$\begin{aligned}\mathbb{P}(\text{accurate}) &= \mathbb{P}\left(\frac{-0.01n}{\sqrt{np(1-p)}} \le \frac{S_n - np}{\sqrt{np(1-p)}} \le \frac{0.01n}{\sqrt{np(1-p)}}\right) \\ &\ge \mathbb{P}\left(-0.02\sqrt{n} \le \frac{S_n - np}{\sqrt{np(1-p)}} \le 0.02\sqrt{n}\right) \\ &\approx \Phi(0.02\sqrt{n}) - \Phi(-0.02\sqrt{n}) = 2\Phi(0.02\sqrt{n}) - 1 \ge 0.95,\end{aligned}$$

Hence, we want $n \ge 9604$. ⊛

6.3.4 Gamma Distribution

Setting: Consider the Poisson process. That is, we have a sequence of occurrences, and:

1. The waiting time between successive occurrences is exponentially distributed with rate λ.
2. The occurrences are independent of each others. Note that the waiting time for the first occurrence follows a $\text{Exp}(\lambda)$ and the number of occurrences in a unit of time follows a $\text{Poi}(\lambda)$. Now, we care about the waiting time of the n-th occurrence.

Now let X_n = timestamp of the n-th event. Then, $X_1 \sim \text{Exp}(\lambda)$, and we know

$$\begin{aligned}F_{X_n}(t) = \mathbb{P}(X_n \le t) &= \mathbb{P}(\text{at least } n \text{ events occur in } [0, t]) = \mathbb{P}(\text{Poi}(t\lambda) \ge n) \\ &= \sum_{j=n}^{\infty} \mathbb{P}(\text{Poi}(t\lambda) = j) = \sum_{j=n}^{\infty} e^{-t\lambda}\frac{(t\lambda)^j}{j!}.\end{aligned}$$

Thus,

$$f_{X_n}(t) = \frac{\mathrm{d}}{\mathrm{d}t} F_{X_n}(t) = \sum_{j=n}^{\infty} \frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{e^{-\lambda t}(t\lambda)^j}{j!}\right] = \sum_{j=n}^{\infty} \frac{1}{j!}\left[-\lambda e^{-t\lambda}(t\lambda)^j + e^{-t\lambda}\lambda j (t\lambda)^{j-1}\right]$$
$$= \left(-\lambda \sum_{j=n}^{\infty} e^{-t\lambda}\frac{(t\lambda)^j}{j!}\right) + \left(\lambda \sum_{j=n}^{\infty} e^{-t\lambda}\frac{(t\lambda)^{j-1}}{(j-1)!}\right)$$
$$= \left(-\lambda \sum_{j=n}^{\infty} e^{-t\lambda}\frac{(t\lambda)^j}{j!}\right) + \left(\lambda \sum_{j=n-1}^{\infty} e^{-t\lambda}\frac{(t\lambda)^{j}}{j!}\right) = \frac{\lambda e^{-t\lambda}(t\lambda)^{n-1}}{(n-1)!},$$

so we have

$$f_{X_n}(t) = \frac{\lambda e^{-t\lambda}(t\lambda)^{n-1}}{(n-1)!} \quad (t \geq 0).$$

We can check that

$$f_{X_1}(t) = \lambda e^{-t\lambda},$$

which is an exponential pdf, so it makes sense.

Definition 6.3.3 (Gamma Distribution)**.** If we have $\lambda \in \mathbb{R}_{>0}$ and $\alpha \in \mathbb{R}_{>0}$, then $X \sim \Gamma(\alpha, \lambda)$ iff

$$f_X(x) = \begin{cases} \dfrac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)}, & \text{if } x \geq 0; \\ 0, & \text{if } x < 0. \end{cases}$$

where

$$\Gamma(\alpha) := \int_0^{\infty} e^{-y} y^{\alpha-1}\,\mathrm{d}y.$$

Note that $\Gamma(1) = 1$. In general, for $\alpha > 1$,

$$\Gamma(\alpha) = \int_0^{\infty} e^{-y}y^{\alpha-1}\,\mathrm{d}y = \left[-e^{-y}y^{\alpha-1}\right]_0^{\infty} + \int_0^{\infty} (\alpha-1)e^{-y}y^{\alpha-2}\,\mathrm{d}y = (\alpha-1)\Gamma(\alpha-1).$$

Hence,

$$\Gamma(1) = 1, \quad \Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1),$$

and by induction we can show that

$$\Gamma(n) = (n-1)! \quad \forall n \in \mathbb{N}.$$

Hence,

$$\Gamma(n, \lambda) = \text{timestamps for the } n\text{-th event in the Poisson process with rate } \lambda.$$

Expectation

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,\mathrm{d}x = \int_0^{\infty} x\frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)}\,\mathrm{d}x = \frac{1}{\Gamma(\alpha)}\int_0^{\infty} e^{-\lambda x}(\lambda x)^{\alpha}\,\mathrm{d}x$$
$$= \frac{1}{\Gamma(\alpha)}\int_0^{\infty} e^{-y}y^{\alpha}\cdot\frac{1}{\lambda}\,\mathrm{d}y = \frac{1}{\lambda\Gamma(\alpha)}\int_0^{\infty} e^{-y}y^{(\alpha+1)-1}\,\mathrm{d}y = \frac{\Gamma(\alpha+1)}{\lambda\Gamma(\alpha)} = \frac{\alpha}{\lambda}.$$

Special case: $\alpha = n \in \mathbb{N}$

$$\Gamma(n, \lambda) = \text{timestamps for the } n\text{-th event in the Poisson process with } \mathrm{Exp}(\lambda) \text{ waiting time.}$$

Suppose $X \sim \Gamma(n, \lambda)$, then $X = W_1 + W_2 + \cdots + W_n$, where W_i is the waiting time between $(i-1)$-th and i-th event. Hence, $W_i \sim \mathrm{Exp}(\lambda)$.

Hence,

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^{n} W_i\right] = \sum_{i=1}^{n}\mathbb{E}[W_i] = \sum_{i=1}^{n}\frac{1}{\lambda} = \frac{n}{\lambda}.$$

Variance If $S \sim \Gamma(\alpha, \lambda)$, then
$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2,$$
and
$$\begin{aligned}\mathbb{E}[X^2] &= \int_0^\infty x^2 \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)}\,\mathrm{d}x = \frac{1}{\lambda\Gamma(\alpha)}\int_0^\infty e^{\lambda x}(\lambda x)^{\alpha+1}\,\mathrm{d}x \\ &= \frac{1}{\lambda^2\Gamma(\alpha)}\int_0^\infty e^{-y}y^{\alpha+1}\,\mathrm{d}y \\ &= \frac{\Gamma(\alpha+2)}{\lambda^2\Gamma(\alpha)} = \frac{(\alpha+1)\alpha\Gamma(\alpha)}{\lambda^2\Gamma(\alpha)} = \frac{(\alpha+1)\alpha}{\lambda^2}.\end{aligned}$$
Thus,
$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{(\alpha+1)\alpha}{\lambda^2} - \frac{\alpha^2}{\lambda^2} = \frac{\alpha}{\lambda^2}.$$

Lecture 17

6.4 Functions of continuous random variables

28 April.

Let X be a continuous random variable with pdf f_X and $X : S \to \mathbb{R}$. Let $Y = g(X)$ for some $g : \mathbb{R} \to \mathbb{R}$. Then Y is also a random variable $Y = g \circ X : S \to \mathbb{R}$.

Question. What can we say about its distribution?

To solve this problem, we can

1. Find the cdf $F_Y(y) := \mathbb{P}(Y \le y)$.
2. Differentiate to find $f_Y(y) = F_Y'(y)$.

Example 6.4.1. Let $Y = X^2$, then
$$F_Y(y) = \mathbb{P}(Y \le y) = \mathbb{P}(X^2 \le y) = \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} f_X(x)\,\mathrm{d}x$$

Theorem 6.4.1. Let $Y = g(X)$, where g is differentiable and strictly increasing. Then, if X has pdf f_X, the pdf of Y is
$$f_Y(y) = \begin{cases} 0, & \text{if } y \notin g(\mathbb{R}); \\ f_X\left(g^{-1}(y)\right)\dfrac{\mathrm{d}}{\mathrm{d}y}\left(g^{-1}(y)\right), & \text{if } y \in g(\mathbb{R}). \end{cases}$$

Remark 6.4.1.
$$\frac{\mathrm{d}}{\mathrm{d}y}g^{-1}(y) = \frac{1}{g'\left(g^{-1}(y)\right)}.$$

Proof. We first find cdf of Y:
$$\mathbb{P}(Y \le y) = \mathbb{P}(g(X) \le y) = \begin{cases} \mathbb{P}(X \le g^{-1}(y)) = F_X\left(g^{-1}(y)\right), & \text{if } y \in g(\mathbb{R}); \\ 0, & \text{if } y \le \inf g(\mathbb{R}); \\ 1, & \text{if } y \ge \sup g(\mathbb{R}). \end{cases}$$
where $F_X(x) = \mathbb{P}(X \le x)$ is the cdf of X. To find the density function of Y, we can differentiate:
$$f_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y}\mathbb{P}(Y \le y) = \begin{cases} 0, & \text{if } y \notin g(\mathbb{R}); \\ f_X(g^{-1}(y)) \cdot \dfrac{\mathrm{d}}{\mathrm{d}y}\left(g^{-1}(y)\right), & \text{if } y \in g(\mathbb{R}). \end{cases}$$

by chain rule. ■

Chapter 7

Jointly Distributed Random Variables

Example 7.0.1. Suppose $X \sim \text{Bin}(n, p)$, $Y \sim \text{Geom}(p)$, where $0 \le p \le 1$ and $n \in \mathbb{N}$. What is $\mathbb{P}(X = k, Y = n)$ and $\mathbb{E}[X + Y]$?

Answer. By linearity of expectation: $\mathbb{E}[X+Y] = \mathbb{E}[X]+\mathbb{E}[Y] = np+\frac{1}{p}$. However, $\mathbb{P}(X = k, Y = n)$ may not equal to $\mathbb{P}(X = k)\mathbb{P}(Y = n)$.

Note that X and Y needs not to be independent. To compute $\mathbb{P}(X = k, Y = n)$, suppose we have a p-biased coin, and X is the number of heads in n tosses, while Y is the number of tosses until the first head. If we toss n times, count heads, then start over and count tosses to first head, then they are independent, and we have

$$\mathbb{P}(X = k, Y = n) = \mathbb{P}(X = k)\mathbb{P}(Y = n).$$

But if we use the same tosses to measure both X and Y, i.e. tosses until first heads / n tosses, whichever comes last, then variables depend on each other, so

$$\mathbb{P}(X = k, Y = n) = \mathbb{P}(X = k)\mathbb{P}(Y = n)$$

need not be true. ⊛

Definition 7.0.1 (Joint cdf)**.** Given random variables X, Y in a probability space $(S, \mathbb{P})$, we define the joint cummulative distribution function

$$F_{X,Y}(a, b) := \mathbb{P}\left(\{X \le a\} \cap \{Y \le b\}\right) = \mathbb{P}(X \le a, Y \le b).$$

Definition 7.0.2 (Marginal Distribution)**.** Given variables X, Y on a probability space $(S, \mathbb{P})$ with joint cdf $F_{X,Y}$, the marginal distributions are

$$F_X(x) = \mathbb{P}(X \le x) = \mathbb{P}(X \le x, Y \le \infty) = F_{X,Y}(x, \infty) = \lim_{y\to\infty} F_{X,Y}(x, y).$$

Similarly,

$$F_Y(y) = F_{X,Y}(\infty, y) = \lim_{x\to\infty} F_{X,Y}(x, y).$$

Definition 7.0.3. Given discrete random variables X, Y on a probability space $(S, \mathbb{P})$, we define the joint probability mass function

$$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y).$$

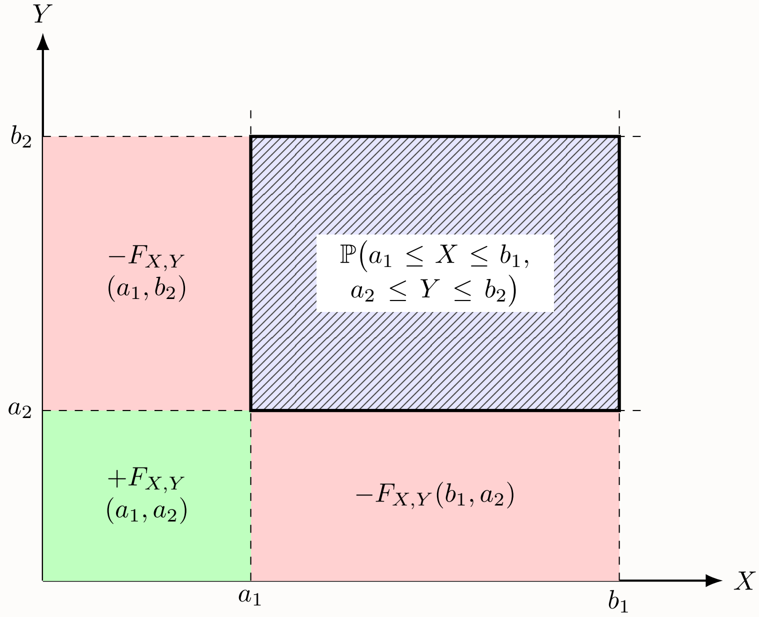
Note that we have the marginals

$$p_X(x) = \sum_y p_{x,y}(x,y), \quad p_Y(y) = \sum_x p_{X,Y}(x,y).$$

Lemma 7.0.1. For any random variables X, Y,

$$\mathbb{P}(\{a_1 \le X \le b_1\} \cap \{a_2 \le Y \le b_2\}) = F_{X,Y}(a_1,a_2) + F_{X,Y}(b_1,b_2) - F_{X,Y}(a_1,b_2) - F_{X,Y}(a_2,b_1).$$

Proof. We can prove it by the following graph.



■

Example 7.0.2. In a country,

15 % of families have 0 children
30 % of families have 1 children
35 % of families have 2 children
20 % of families have 3 children

Each child is equally likely to be a boy or a girl, independently. Let $X = \#$ of boys in a uniformly random family, and $Y = \#$ of girls in a uniformly random family. What is the distribution of X, Y? What is $\mathbb{E}[X]$?

Proof.

$$p_{X,Y}(i,j) = \mathbb{P}(X = i, Y = j) = \mathbb{P}(X = i, Y = j, X + Y = i + j),$$

since

$$\mathbb{P}(X = i, Y = j, X+Y = i+j) = \mathbb{P}(X+Y = i+j \mid X = i, Y = j)\mathbb{P}(X = i, Y = j) = \mathbb{P}(X = i, Y = j).$$

Also,

$$\begin{aligned}\mathbb{P}(X = i, Y = j, X + Y = i + j) &= \mathbb{P}(X = i, Y = j \mid X + Y = i + j)\mathbb{P}(X + Y = i + j) \\ &= \mathbb{P}(\text{Bin}\left(i + j, \frac{1}{2}\right) = i)\mathbb{P}(X + Y = i + j)\end{aligned}$$

$X\backslash Y$	0	1	2	3	$p_X(x)$
0	0.15	$\binom{1}{0}\left(\frac{1}{2}\right)^1(0.30) =$ 0.15	$\binom{2}{0}\left(\frac{1}{2}\right)^2(0.35) =$ 0.0875	$\binom{3}{0}\left(\frac{1}{2}\right)^3(0.20) =$ 0.025	0.4125
1	$\binom{1}{1}\left(\frac{1}{2}\right)^1(0.30) =$ 0.15	$\binom{2}{1}\left(\frac{1}{2}\right)^2(0.35) =$ 0.175	$\binom{3}{1}\left(\frac{1}{2}\right)^3(0.20) =$ 0.075	0	0.4
2	$\binom{2}{2}\left(\frac{1}{2}\right)^2(0.35) =$ 0.0875	$\binom{3}{2}\left(\frac{1}{2}\right)^3(0.20) =$ 0.075	0	0	0.1625
3	$\binom{3}{3}\left(\frac{1}{2}\right)^3(0.20) =$ 0.025	0	0	0	0.025

Hence,

$$\mathbb{E}[X] = \sum_{x=0}^{3} x\mathbb{P}(X = x) = 0.4 + 2 \times 0.1625 + 3 \times 0.025 = 0.8.$$

⊛

Definition 7.0.4. Given continuous random variables X, Y on a probability space $(S, \mathbb{P})$, we say they have a joint probability density function $f_{X,Y}(x, y)$ if for any (measurable) set $C \subseteq \mathbb{R}^2$,

$$\mathbb{P}((x, y) \in C) = \iint_C f_{X,Y}(x, y)\, \mathrm{d}x\, \mathrm{d}y.$$

We can observe that

$$F_{X,Y}(x, y) \coloneqq \mathbb{P}(X \le x, Y \le y) = \mathbb{P}\left((x, y) \in (-\infty, x] \times (-\infty, y]\right) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u, v)\, \mathrm{d}v\, \mathrm{d}u.$$

Thus,

$$f_{X,Y}(x, y) = \frac{\partial^2}{\partial u \partial v} F_{X,Y}(u, v)\Big|_{\substack{u=x\\v=y}}.$$

Example 7.0.3. Let X, Y have the joint density

$$f_{X,Y}(x, y) = \begin{cases} e^{-(x+y)}, & \text{if } x, y \ge 0; \\ 0, & \text{if } x < 0 \text{ or } y < 0. \end{cases}$$

What is $\mathbb{P}(X > Y)$? What is $\mathbb{P}(X \le t)$ for $t \in \mathbb{R}$? What is the distribution of $\frac{X}{Y}$?

Proof.

(a) What is $\mathbb{P}(X > Y)$? Let $C = \{(x, y) : X > Y\}$, then

$$\begin{aligned}
\mathbb{P}(X > Y) = \mathbb{P}((x, y) \in C) &= \iint_C f_{X,Y}(u, v)\, \mathrm{d}u\, \mathrm{d}v = \int_{-\infty}^{\infty} \int_{-\infty}^{u} f_{X,Y}(u, v)\, \mathrm{d}v\, \mathrm{d}u \\
&= \int_0^{\infty} \int_0^{u} e^{-(u+v)}\, \mathrm{d}v\, \mathrm{d}u = \int_0^{\infty} e^{-u} \int_0^{u} e^{-v}\, \mathrm{d}v\, \mathrm{d}u \\
&= \int_0^{\infty} e^{-u} \left(\left[-e^{-v}\right]_0^u\right) \mathrm{d}u = \int_0^{\infty} e^{-u}\left(1 - e^{-u}\right) \mathrm{d}u \\
&= \int_0^{\infty} e^{-u} - e^{-2u}\, \mathrm{d}u = \frac{1}{2}.
\end{aligned}$$

(b) What is $\mathbb{P}(X \le t)$ for $t \in \mathbb{R}$?

$$\begin{aligned}\mathbb{P}(X \le t) &= \iint_{\{u \le t\}} f_{X,Y}(u,v)\,\mathrm{d}u\,\mathrm{d}v = \int_{-\infty}^{\infty}\int_{-\infty}^{t} f_{X,Y}(u,v)\,\mathrm{d}u\,\mathrm{d}v \\ &= \int_0^{\infty}\int_0^{t} e^{-(u+v)}\,\mathrm{d}u\,\mathrm{d}v = 1 - e^{-t}.\end{aligned}$$

(c) What is $F_{\frac{X}{Y}}(t)$?

$$\begin{aligned}F_{\frac{X}{Y}}(t) &= \mathbb{P}\left(\frac{X}{Y} \le t\right) = \mathbb{P}(X \le tY) = \iint_{\{u \le tv\}} f_{X,Y}(u,v)\,\mathrm{d}u\,\mathrm{d}v \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{tv} f_{X,Y}(u,v)\,\mathrm{d}u\,\mathrm{d}v = \int_0^{\infty}\int_0^{tv} e^{-(u+v)}\,\mathrm{d}u\,\mathrm{d}v = 1 - \frac{1}{t+1}.\end{aligned}$$

⊛

Lecture 18

April 30.

Example 7.0.4. A dart lands at a uniformly random point (X, Y) on a dartboard of radius R.

(a) Find the joint pdf $f_{X,Y}(x,y)$.

(b) Find the marginal pdf of X.

(c) What is the expectation of D, the distance to the center of the dashboard?

Proof.

(a) Since (X, Y) is uniform on the board,

$$f_{X,Y}(x,y) = \begin{cases} C, & \text{if } x^2 + y^2 \le R^2; \\ 0, & \text{if } x^2 + y^2 > R^2. \end{cases}$$

Thus,

$$1 = \iint_{\mathbb{R}^2} f_{X,Y}(u,v)\,\mathrm{d}u\,\mathrm{d}v = \iint_{\{u^2+v^2 \le R^2\}} C\,\mathrm{d}u\,\mathrm{d}v = C\pi R^2.$$

Hence, $C = \frac{1}{\pi R^2}$.

(b)

$$\begin{aligned}F_X(t) = \mathbb{P}(X \le t) &= \iint_{\{u \le t\}} f_{X,Y}(u,v)\,\mathrm{d}u\,\mathrm{d}v \\ &= \begin{cases} 0, & \text{if } t < -R; \\ \displaystyle\iint_{\{u \le t\}} f_{X,Y}(u,v)\,\mathrm{d}u\,\mathrm{d}v, & \text{if } -R \le t \le R; \\ 1, & \text{if } t > R. \end{cases}\end{aligned}$$

If $-R \le t \le R$,

$$\iint_{\{u \le t\}} f_{X,Y}(u,v)\,\mathrm{d}u\,\mathrm{d}v = \int_{-R}^{t}\int_{-\sqrt{R^2-u^2}}^{\sqrt{R^2-u^2}} \frac{1}{\pi R^2}\,\mathrm{d}v\,\mathrm{d}u = \int_{-R}^{t} \frac{2}{\pi R^2}\sqrt{R^2-u^2}\,\mathrm{d}u.$$

Hence, X has the pdf

$$f_X(t) = \begin{cases} \dfrac{2}{\pi R^2}\sqrt{R^2 - t^2}, & \text{if } t \in [-R, R]; \\ 0, & \text{if } t \notin [-R, R]. \end{cases}$$

(c) Since for $0 \le t \le R$ we know

$$F_D(t) = \mathbb{P}(D \le t) = \mathbb{P}(D^2 \le t^2) = \mathbb{P}(X^2 + Y^2 \le t^2) = \iint_{\{u^2+v^2 \le t^2\}} f_{X,Y}(u, v)\, \mathrm{d}u\, \mathrm{d}v$$
$$= \iint_{\{u^2+v^2 \le t^2\}} \frac{1}{\pi R^2}\, \mathrm{d}u\, \mathrm{d}v = \frac{t^2}{R^2}.$$

Hence, D has the pdf

$$f_D(t) = F_D'(t) = \begin{cases} \dfrac{2t}{R^2}, & \text{if } 0 \le t \le R; \\ 0, & \text{otherwise.} \end{cases}$$

Hence,

$$\mathbb{E}[D] = \int_{\mathbb{R}} t f_D(t)\, \mathrm{d}t = \int_0^R \frac{2t^2}{R^2}\, \mathrm{d}t = \frac{2}{3}R.$$

⊛

7.1 Independent Random Variables

Definition 7.1.1. We say that two random variables X, Y on a probability space X, Y are independent if for all (measurable) $A, B \subseteq \mathbb{R}$, we have

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B).$$

Proposition 7.1.1. Random variables X and Y are independent if and only if

$$F_{X,Y}(x, y) = F_X(x)F_Y(y)$$

for all $x, y \in \mathbb{R}$.

Proof.

($\Rightarrow$) Suppose that X and Y are independent. Then

$$F_{X,Y}(x, y) = \mathbb{P}(X \le x,\ Y \le y) = \mathbb{P}(X \in (-\infty, x],\ Y \in (-\infty, y])$$
$$= \mathbb{P}(X \in (-\infty, x])\, \mathbb{P}(Y \in (-\infty, y]) = F_X(x)F_Y(y).$$

($\Leftarrow$) Assume that

$$F_{X,Y}(x, y) = F_X(x)F_Y(y)$$

for all $x, y \in \mathbb{R}$.

We first prove that

$$\mathbb{P}(X \in A,\ Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$$

whenever

$$A = (a_1, a_2], \qquad B = (b_1, b_2].$$

Indeed,

$$\begin{aligned}
\mathbb{P}(X \in A,\ Y \in B) &= \mathbb{P}(a_1 < X \leq a_2,\ b_1 < Y \leq b_2) \\
&= F_{X,Y}(a_2, b_2) - F_{X,Y}(a_1, b_2) - F_{X,Y}(a_2, b_1) + F_{X,Y}(a_1, b_1) \\
&= F_X(a_2)F_Y(b_2) - F_X(a_1)F_Y(b_2) - F_X(a_2)F_Y(b_1) + F_X(a_1)F_Y(b_1) \\
&= \big(F_X(a_2) - F_X(a_1)\big)\big(F_Y(b_2) - F_Y(b_1)\big) \\
&= \mathbb{P}(a_1 < X \leq a_2)\,\mathbb{P}(b_1 < Y \leq b_2) \\
&= \mathbb{P}(X \in A)\mathbb{P}(Y \in B).
\end{aligned}$$

Since the collection of half-open intervals $(a, b]$ generates the Borel σ-algebra on $\mathbb{R}$, it follows from the monotone class theorem (or the π-λ theorem) that

$$\mathbb{P}(X \in E,\ Y \in F) = \mathbb{P}(X \in E)\mathbb{P}(Y \in F)$$

for all Borel sets $E, F \subseteq \mathbb{R}$.

Therefore X and Y are independent.

■

Corollary 7.1.1. Suppose that the random variables X and Y admit a joint density function $f_{X,Y}$ and marginal density functions f_X and f_Y. Then X and Y are independent if and only if

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

for almost every $(x, y) \in \mathbb{R}^2$.

Remark 7.1.1. For discrete random variables or X, Y continuous and have joint pdf,

$$F_{X,Y}(x, y) = F_X(x)F_Y(y) \Leftrightarrow p_{X,Y}(x, y) = p_X(x)p_Y(y).$$

Example 7.1.1. Atletico Madroid v.s. Arsenal. Suppose the number of penalties $\sim \mathrm{Poi}(\lambda)$. When a penalty is awarded, it is for Atletico with probability p, and Arsenal with probability $1 - p$, independently of all other penalties. Let X be the number of penalties for Atl Madroid, Y be the number of penalties for Ars.

$$\begin{aligned}
p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y) &= \mathbb{P}(\{X = x, Y = y\} \cap \{X + Y = x + y\}) \\
&= \mathbb{P}(X = x, Y = y \mid X + Y = x + y)\mathbb{P}(X + Y = x + y).
\end{aligned}$$

Since $X + Y \sim \mathrm{Poi}(\lambda)$,

$$\mathbb{P}(X + Y = x + y) = e^{-\lambda}\frac{\lambda^{x+y}}{(x + y)!}.$$

Since each penalty goes to Atl Madroid with probability p, independently, if we condition on there being $x + y$ penalties, in total, $X \sim \mathrm{Bin}(x + y, p)$. Hence,

$$\begin{aligned}
\mathbb{P}(X = x, Y = y \mid X + Y = x + y) = \mathbb{P}(X = x \mid X + Y = x + y) &= \mathbb{P}(\mathrm{Bin}(x + y, p) = x) \\
&= \binom{x + y}{x} p^x (1 - p)^{(x+y)-x}.
\end{aligned}$$

Hence,

$$\begin{aligned}
p_{X,Y}(x,y) = \mathbb{P}(X=x, Y=y) &= \binom{x+y}{x} p^x (1-p)^y \cdot e^{-\lambda} \frac{\lambda^{x+y}}{(x+y)!} \\
&= \frac{e^{-\lambda} p^x (1-p)^y \lambda^{x+y}}{x! y!} = e^{-\lambda} \left(\frac{p^x \lambda^x}{x!} \right) \left(\frac{(1-p)^y \lambda^y}{y!} \right) \\
&= e^{-\lambda(p+(1-p))} \left(\frac{p^x \lambda^x}{x!} \right) \left(\frac{(1-p)^y \lambda^y}{y!} \right) \\
&= \left(e^{-\lambda p} \frac{p^x \lambda^x}{x!} \right) \left(e^{-\lambda(1-p)} \frac{(1-p)^y \lambda^y}{y!} \right) = p_X(x) p_Y(y).
\end{aligned}$$

This gives $X \sim \text{Poi}(\lambda p)$ and $Y \sim \text{Poi}(\lambda(1-p))$ and X, Y are independent.

Remark 7.1.2. Last equality holds since we can use

$$p_X(x) = \sum_{y=0}^{\infty} p_{X,Y}(x,y), \quad p_Y(y) = \sum_{x=0}^{\infty} p_{X,Y}(x,y).$$

Proposition 7.1.2. If X, Y are continuous random variables on a probability space $(S, \mathbb{P})$, and their joint pdf function $f_{X,Y}(x,y) = g(x)h(y)$ for some function g, h of x, y, respectively, then X and Y are independent.

Proof. Let $c_1 = \int_{\mathbb{R}} g(x)\,\mathrm{d}x$ and $c_2 = \int_{\mathbb{R}} h(y)\,\mathrm{d}y$. Then,

$$\begin{aligned}
1 &= \iint_{\mathbb{R}^2} f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y = \iint_{\mathbb{R}^2} g(x)h(y)\,\mathrm{d}x\,\mathrm{d}y \\
&= \left(\int_{\mathbb{R}} g(x)\,\mathrm{d}x \right) \left(\int_{\mathbb{R}} h(y)\,\mathrm{d}y \right) = c_1 c_2.
\end{aligned}$$

Also,

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,\mathrm{d}y = \int_{\mathbb{R}} g(x)h(y)\,\mathrm{d}y = c_2 g(x).$$

Similarly, $f_Y(y) = c_1 h(y)$. Hence,

$$f_{X,Y}(x,y) = g(x)h(y) = 1 \cdot g(x)h(y) = c_1 c_2 g(x) h(y) = f_X(x) f_Y(y).$$

■

Example 7.1.2. Suppose

- X, Y are continuous random variables with differentiable pdf f_X, f_Y.
- X, Y are independent.
- joint density function only depends on $X^2 + Y^2$, i.e. $f_{X,Y}(x,y) = g(x^2+y^2)$ for some function g.

Then, X and Y are normally distributed.

Proof.

$$f_X(x) f_Y(y) = f_{X,Y}(x,y) = g(x^2+y^2)$$

by independence. If we differentiate w.r.t. x, we have

$$f_X'(x) f_Y(y) = 2x g'(x^2+y^2).$$

Hence,

$$\frac{f_X'(x)}{f_X(x)} = \frac{2xg'(x^2+y^2)}{g(x^2+y^2)} \Rightarrow \frac{f_X'(x)}{xf_X(x)} = \frac{2g'(x^2+y^2)}{g(x^2+y^2)}.$$

Now we give a claim:

Claim 7.1.1.

$$\frac{f_X'(x_1)}{x_1 f_X(x_1)} = \frac{f_X'(x_2)}{x_2 f_X(x_2)} \quad \forall x_1, x_2 \in \mathbb{R}.$$

Proof. Find y_1, y_2 s.t. $x_1^2 + y_1^2 = x_2^2 + y_2^2$, i.e. $y_1 = x_2$ and $y_2 = x_1$, then

$$\frac{f_X'(x_1)}{x_1 f_X(x_1)} = \frac{2g'(x_1^2+y_1^2)}{g(x_1^2+y_1^2)} = \frac{2g'(x_2^2+y_2^2)}{g(x_2^2+y_2^2)} = \frac{f_X'(x_2)}{x_2 f_X(x_2)}.$$

Hence,

$$\frac{f_X'(x)}{xf_X(x)} \equiv c \text{ for some } c \in \mathbb{R}.$$

⊛

Now we know

$$\frac{f_X'(x)}{f_X(x)} = Cx,$$

then integrate it, we have

$$\ln f_X(x) = C'x^2 + C''.$$

Hence,

$$f_X(x) = Ae^{C'x^2}$$

for som constants A, C'. Now since

$$\int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1,$$

we have $C' < 0$, so $\exists \sigma \in \mathbb{R}_{>0}$ s.t. $C' = \frac{-1}{2\sigma^2}$. This gives

$$f_X(x) = Ae^{-\frac{x^2}{2\sigma^2}},$$

and by $\int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1$ and f_X is proposional to f_Z where $Z \sim N(0, \sigma^2)$, we have $A = \frac{1}{\sigma\sqrt{2\pi}}$. Hence, $X \sim N(0, \sigma^2)$. By symmetry, we know $Y \sim N(0, \overline{\sigma}^2)$. Hence,

$$f_{X,Y}(x,y) = f_X(x)f_Y(y) = \frac{1}{\sigma\overline{\sigma}2\pi}e^{-\frac{x^2}{2\sigma^2} - \frac{y^2}{2\overline{\sigma}^2}},$$

and since this only depends on $x^2 + y^2$, we must have $\sigma = \overline{\sigma}$.

⊛

Lecture 19

May 5.

Example 7.1.3. Suppose (X, Y) is uniform on the unit square $S = [0,1]^2$. Are X and Y independent?

Proof. Note that

$$f_{X,Y}(x,y) = \begin{cases} C, & \text{if } (x,y) \in S; \\ 0, & \text{if } (x,y) \neq S. \end{cases}$$

Now by $\iint f_{X,Y}\,\mathrm{d}x\,\mathrm{d}y = 1$, we have $C = 1$. Hence,

$$f_{X,Y}(x,y) = 1_{(x,y)\in S}(x,y) = 1_{0\leq x\leq 1}(x) \cdot 1_{0\leq y\leq 1}(y) = f_X(x) \cdot f_Y(y).$$

⊛

Example 7.1.4. Let (X, Y) be uniformly distributed over the triangle with vertices $(0,0),(1,0),(1,1)$. Are X, Y independent?

Proof. No. Since $\mathbb{P}(0 < X < \frac{1}{2}, \frac{1}{2} < Y < 1) = 0$ but

$$\mathbb{P}(0 < X < \frac{1}{2}) \cdot \mathbb{P}(\frac{1}{2} < Y < 1) = \frac{1}{4} \cdot \frac{1}{4} \neq 0.$$

⊛

7.2 Sums of Independent Random Variables

Question. Suppose X, Y are independent continuous random variables. What is the distribution of $X + Y$?

Answer. Calculate the cdf of $X + Y$:

$$F_{X+Y}(z) = \mathbb{P}(X + Y \leq z) = \mathbb{P}(\{(x,y) : x + y \leq z\}) = \iint_{\{x+y\leq z\}} f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y$$
$$= \iint_{\{x+y\leq z\}} f_X(x) f_Y(y)\,\mathrm{d}x\,\mathrm{d}y = \int_{-\infty}^{\infty}\int_{-\infty}^{z-x} f_X(x) f_Y(y)\,\mathrm{d}y\,\mathrm{d}x$$

Now let $v = y + x$, then

$$\int_{-\infty}^{\infty}\int_{-\infty}^{z-x} f_X(x) f_Y(y)\,\mathrm{d}y\,\mathrm{d}x = \int_{-\infty}^{\infty}\int_{-\infty}^{z} f_X(x) f_Y(v-x)\,\mathrm{d}v\,\mathrm{d}x.$$

Thus,

$$F_{X+Y}(z) = \int_{-\infty}^{z}\int_{-\infty}^{\infty} f_X(x) f_Y(v-x)\,\mathrm{d}x\,\mathrm{d}v.$$

Differentiating w.r.t. z,

$$f_{X+Y}(z) = F_X'(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\,\mathrm{d}x.$$

⊛

Definition 7.2.1. Given functions f, g, their convolution is

$$h(z) = \int_{-\infty}^{\infty} f(x) g(z-x)\,\mathrm{d}x.$$

Proposition 7.2.1. Let X and Y be independent continuous random variables with density functions f_X and f_Y. Then the density of the sum $X + Y$ is the convolution of f_X and f_Y.

Corollary 7.2.1. If X, Y are independent continuous random variables, then

$$f_{X+Y}(x) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\,\mathrm{d}x = \int_{-\infty}^{\infty} f_X(z-y) f_Y(y)\,\mathrm{d}y.$$

Example 7.2.1. Let $X, Y \sim \mathrm{Unif}([0,1])$ and independent. What is the distribution of $X + Y$?

Proof. By proposition,

$$\begin{aligned} f_{X+Y}(z) &= \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\,\mathrm{d}x = \int_{-\infty}^{\infty} 1_{[0,1]}(x)\cdot 1_{[0,1]}(z-x)\,\mathrm{d}x \\ &= \int_{\max(0,z-1)}^{\min(z,1)} 1\,\mathrm{d}x = \min(z,1) - \max(0,z-1) \\ &= \begin{cases} 0, & \text{if } z<0; \\ z, & \text{if } 0\le z\le 1; \\ 2-z, & \text{if } 1\le z\le 2; \\ 0, & \text{if } z\ge 2. \end{cases} \end{aligned}$$

⊛

Question. Let $X_1, X_2, \ldots, X_n \sim \mathrm{Unif}([0,1])$ independent, what is the distribution of $S_n = \sum_{i=1}^n X_i$ over the region $[0,1]$?

Answer. Do induction on n.

- $n=1$: $f_{S_1}(z) = 1$ for $0\le z\le 1$.
- $n=2$: $f_{S_2}(z) = z$ for $0\le z\le 1$ (last example).
- $n=3$: $S_3 = X_1+X_2+X_3 = S_2+X_3$, then for $0\le z\le 1$,

$$f_{S_3}(z) = \int_0^z f_{S_2}(z)\overbrace{f_{X_3}(z-x)}^{1}\,\mathrm{d}x = \int_0^z x\,\mathrm{d}x = \frac{z^2}{2}.$$

Claim 7.2.1. $f_{S_n}(z) = \frac{z^{n-1}}{(n-1)!}$ for $0\le z\le 1$.

Proof. We have shown for $n=1,2,3$ it is true (Base case). Now since $S_{n+1} = S_n + X_{n+1}$,

$$f_{S_{n+1}}(z) = \int_0^z f_{S_n}(x)\overbrace{f_{X_{n+1}}(z-x)}^{1}\,\mathrm{d}x = \int_0^z f_{S_n}(x)\,\mathrm{d}x = \int_0^z \frac{x^{n-1}}{(n-1)!}\,\mathrm{d}x = \frac{z^n}{n!}.$$

⊛

⊛

Example 7.2.2. We have 2 cakes in one fridge. Every day we eat a uniformly random amount, between 0 cake and 1 cake, independently of all previous days. How long does the first cake last?

Proof. Let X_i be the portion of cake eaten on day i. Thus, $X_i \sim \mathrm{Unif}([0,1])$ independently. Let S_n be the total amount eaten after n days. Thus, $S_n = X_1+X_2+\cdots+X_n$. Let D be the number of days it takes to finish the first cake. Then

$$\{D=n\} = \{S_n\ge 1, S_{n-1}<1\}.$$

Also, $\{D>n\} = \{S_n<1\}$. Thus,

$$\mathbb{P}(D>n) = \mathbb{P}(S_n<1) = \int_0^1 f_{S_n}(x)\,\mathrm{d}x = \int_0^1 \frac{x^{n-1}}{(x-1)!}\,\mathrm{d}x = \left[\frac{x^n}{n!}\right]_0^1 = \frac{1}{n!}.$$

Thus,

$$\mathbb{P}(D=n)=\mathbb{P}(D>n-1)-\mathbb{P}(D>n)=\frac{1}{(n-1)!}-\frac{1}{n!}=\frac{n-1}{n!}.$$

Hence,

$$\mathbb{E}[D]=\sum_{n=1}^{\infty}n\mathbb{P}(D=n)=\sum_{n=1}^{\infty}n\cdot\frac{n-1}{n!}=\sum_{n=2}^{\infty}\frac{1}{(n-2)!}=\sum_{k=0}^{\infty}\frac{1}{k!}=e.$$

⊛

Lecture 20

Recall that $X\sim\Gamma(\alpha,\lambda)$ if May 7.

$$f_X(x)=\begin{cases}\dfrac{\lambda^\alpha x^{\alpha-1}e^{-\lambda x}}{\Gamma(\alpha)}, & \text{if } x\geq 0;\\ 0, & \text{if } x<0.\end{cases}$$

Example 7.2.3. Let $X\sim\Gamma(\alpha,\lambda)$, $Y\sim\Gamma(\beta,\lambda)$, where $\alpha,\beta,\lambda>0$, and X,Y are independent. What is the distribution of $X+Y$?

Proof.

$$\begin{aligned}f_{X+Y}(z)&=\int_{-\infty}^{\infty}f_X(x)f_Y(z-z)\,\mathrm{d}x=\int_0^z\frac{\lambda^\alpha x^{\alpha-1}e^{-\lambda x}}{\Gamma(\alpha)}\cdot\frac{\lambda^\beta(z-x)^{\beta-1}e^{-\lambda(z-x)}}{\Gamma(\beta)}\,\mathrm{d}x\\&=\frac{\lambda^{\alpha+\beta}e^{-\lambda z}}{\Gamma(\alpha)\Gamma(\beta)}\int_0^z x^{\alpha-1}(z-x)^{\beta-1}\,\mathrm{d}x=\frac{\lambda^{\alpha+\beta}e^{-\lambda z}}{\Gamma(\alpha)\Gamma(\beta)}\int_0^1(zu)^{\alpha-1}(z-zu)^{\beta-1}z\,\mathrm{d}u\\&=\frac{\lambda^{\alpha+\beta}z^{\alpha+\beta-1}e^{-\lambda z}}{\Gamma(\alpha)\Gamma(\beta)}\int_0^1 u^{\alpha-1}(1-u)^{\beta-1}\,\mathrm{d}u=\frac{C_{\alpha,\beta}}{\Gamma(\alpha)\Gamma(\beta)}\lambda^{\alpha+\beta}z^{\alpha+\beta-1}e^{-\lambda z}\\&=C\cdot\frac{\lambda^{\alpha+\beta}z^{\alpha+\beta-1}e^{-\lambda z}}{\Gamma(\alpha+\beta)},\end{aligned}$$

which is propotional to $f_Z(z)$ where $Z\sim\Gamma(\alpha+\beta,\lambda)$. Since f_{X+Y} and f_Z are both pdf,

$$\int_{-\infty}^{\infty}f_{X+Y}(t)\,\mathrm{d}t=\int_{-\infty}^{\infty}f_Z(t)\,\mathrm{d}t=1,$$

so we have $C=1$, and thus $X+Y\sim\Gamma(\alpha+\beta,\lambda)$.

Summary: If $X\sim\Gamma(\alpha,\lambda)$, $Y\sim\Gamma(\beta,\lambda)$ are independent, $X+Y\sim\Gamma(\alpha+\beta,\lambda)$. ⊛

Corollary 7.2.2. If $X_i\sim\mathrm{Exp}(\lambda)$ for $1\leq i\leq n$, are independent, then $\sum_{i=1}^n X_i\sim\Gamma(n,\lambda)$.

Proof. Since $\mathrm{Exp}(\lambda)=\Gamma(1,\lambda)$, so this is true. ■

Example 7.2.4. Let $Z_i\sim N(0,1)$, $1\leq i\leq n$, be independent. What is the distribution of $\sum_{i=1}^n Z_i^2$?

Proof. We can first solve a subproblem.

Subproblem: What is the distribution of Z_i^2?

$$F_{Z_i^2}(a) = \mathbb{P}(Z_i^2 \le a) = \mathbb{P}(-\sqrt{a} \le Z_i \le \sqrt{a}) = \int_{-\sqrt{a}}^{\sqrt{a}} f_{Z_i}(z)\,\mathrm{d}z = \int_{-\sqrt{a}}^{\sqrt{a}} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\,\mathrm{d}z$$
$$= 2\int_0^{\sqrt{a}} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\,\mathrm{d}z.$$

Hence,

$$f_{Z_i^2}(a) = \frac{\mathrm{d}}{\mathrm{d}a} F_{Z_i^2}(a) = 2\left[\frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\right]_{z=\sqrt{a}} \cdot \frac{\mathrm{d}}{\mathrm{d}a}(\sqrt{a}) = 2\frac{1}{\sqrt{2\pi}} e^{-\frac{a}{2}} \cdot \frac{1}{2} a^{-\frac{1}{2}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{a}{2}} a^{\frac{1}{2}-1}.$$

This means $Z_i^2 \sim \Gamma\left(\frac{1}{2}, \frac{1}{2}\right)$ (compare pdf of Z_i^2 and $\Gamma\left(\frac{1}{2}, \frac{1}{2}\right)$), so

$$\sum_{i=1}^n Z_i^2 \sim \Gamma\left(\frac{n}{2}, \frac{1}{2}\right).$$

Remark 7.2.1. This is called the chi-squared distribution with n degrees of freedom.

⊛

Question. Let $X_i \sim N(\mu_i, \sigma_i^2)$ be independent for $1 \le i \le n$. What is the distribution of $\sum_{i=1}^n X_i$?

Proposition 7.2.2.

$$\sum_{i=1}^N X_i \sim N\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right).$$

Proof. We can first show the $n = 2$ case implies general case, then show that when $n = 2$, the case $\mu_1 = \mu_2 = 0$ and $\sigma_1 = 1, \sigma_2 = \sigma$ implies the general case of $n = 2$, and finally prove the base case: $n = 2, \mu_1 = \mu_2 = 0$ and $\sigma_1 = 1, \sigma_2 = \sigma$.

1. Prove by induction on n: We know $X_1 + \cdots + X_n + X_{n+1} = (X_1 + \cdots + X_n) + X_{n+1}$ where

$$X_1 + \cdots + X_n \sim N\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right),$$

and $X_{n+1} \sim N(\mu_{n+1}, \sigma_{n+1}^2)$. Thus, if we can prove the $n = 2$ case, we have

$$X_1 + \cdots + X_n + X_{n+1} \sim N\left(\sum_{i=1}^n \mu_i + \mu_{n+1}, \sum_{i=1}^n \sigma_i^2 + \sigma_{n+1}^2\right) = N\left(\sum_{i=1}^{n+1} \mu_i, \sum_{i=1}^{n+1} \sigma_i^2\right).$$

2. Prove the specific case $n = 2, \mu_1 = \mu_2 = 0, \sigma_1 = 1, \sigma_2 = \sigma$ implies general cases for $n = 2$: If $X_1 \sim N(\mu_1, \sigma_1^2), X_2 \sim N(\mu_2, \sigma_2^2)$, then

$$X_1 + X_2 = \sigma_1\left(\frac{X_1 + X_2}{\sigma_1}\right) = \sigma_1\left(\frac{X_1 - \mu_1}{\sigma_1} + \frac{X_2 - \mu_2}{\sigma_1} + \frac{\mu_1 + \mu_2}{\sigma_1}\right)$$
$$= \sigma_1\left(\frac{X_1 - \mu_1}{\sigma_1} + \frac{\sigma_2}{\sigma_1} \cdot \frac{X_2 - \mu_2}{\sigma_2}\right) + (\mu_1 + \mu_2),$$

where

$$\frac{X_1 - \mu_1}{\sigma_1} \sim N(0, 1), \quad \frac{\sigma_2}{\sigma_1}\frac{X - \mu_2}{\sigma_2} \sim N\left(0, \frac{\sigma_2^2}{\sigma_1^2}\right).$$

If the $\mu_1 = \mu_2 = 0$ and $\sigma_1 = 1, \sigma_2 = \sigma$ case is true, then

$$\frac{X_1-\mu_1}{\sigma_1} + \frac{\sigma_2}{\sigma_1} \cdot \frac{X_2-\mu_2}{\sigma_2} \sim N\left(0, 1+\frac{\sigma_2^2}{\sigma_1^2}\right),$$

so

$$\sigma_1 \left(\frac{X_1-\mu_1}{\sigma_1} + \frac{\sigma_2}{\sigma_1} \cdot \frac{X_2-\mu_2}{\sigma_2}\right) \sim N(0, \sigma_1^2+\sigma_2^2),$$

which shows

$$X_1 + X_2 = \sigma_1 \left(\frac{X_1-\mu_1}{\sigma_1} + \frac{\sigma_2}{\sigma_1} \cdot \frac{X_2-\mu_2}{\sigma_2}\right) + (\mu_1+\mu_2) \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2).$$

3. Prove the $n = 2, X_1 \sim N(0,1), X_2 \sim N(0,\sigma^2)$ case. Suppose $X_1 \sim N(0,1), X_2 \sim N(0,\sigma^2)$ are independent, then since we have

$$f_{X_1}(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad f_{X_2}(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}},$$

the density of $X_1 + X_2$ is the convolution of these densities:

$$\begin{aligned} f_{X_1+X_2}(z) &= \int_{-\infty}^{\infty} f_{X_1}(x) f_{X_2}(z-x)\,\mathrm{d}x = \int_{-\infty}^{\infty} f_{X_1}(z-y) f_{X_2}(y)\,\mathrm{d}y \\ &= \frac{1}{2\pi\sigma} \int_{-\infty}^{\infty} e^{-\frac{(z-y)^2}{2}} \cdot e^{-\frac{y^2}{2\sigma^2}}\,\mathrm{d}y. \end{aligned}$$

Note that

$$\begin{aligned} -\frac{(z-y)^2}{2} - \frac{y^2}{2\sigma^2} &= -\frac{z^2}{2} - \frac{\sigma^2+1}{2\sigma^2}\left(y^2 - \frac{2\sigma^2 zy}{\sigma^2+1}\right) \\ &= -\frac{z^2}{2} - \frac{\sigma^2+1}{2\sigma^2}\left(y - \frac{\sigma^2}{\sigma^2+1}z\right)^2 + \frac{\sigma^2}{2(\sigma^2+1)}z^2, \end{aligned}$$

so

$$\begin{aligned} f_{X_1+X_2}(z) &= \frac{1}{2\pi\sigma} e^{-\frac{z^2}{2} + \frac{\sigma^2}{2(\sigma^2+1)}z^2} \int_{-\infty}^{\infty} e^{-\frac{\sigma^2+1}{2\sigma^2}\left(y - \frac{\sigma^2}{\sigma^2+1}z\right)^2}\,\mathrm{d}y \\ &= \frac{1}{2\pi\sigma} e^{-\frac{z^2}{2} + \frac{\sigma^2}{2(\sigma^2+1)}z^2} \underbrace{\int_{-\infty}^{\infty} e^{-\frac{\sigma^2+1}{2\sigma^2}u^2}\,\mathrm{d}u}_{\text{some constant}} \\ &= C' e^{-\frac{z^2}{2} + \frac{\sigma^2}{2(\sigma^2+1)}z^2} \\ &= C' e^{-\frac{(\sigma^2+1)z^2 - \sigma^2 z^2}{2(\sigma^2+1)}} = C' e^{-\frac{z^2}{2(\sigma^2+1)}} \end{aligned}$$

This is proportional to the pdf of $N(0, 1+\sigma^2)$, so $X_1 + X_2 \sim N(0, 1+\sigma^2)$. ■

Example 7.2.5. We say a random variable X is log-normally distributed if $\log X$ is normally distributed. There is a stock whose price after n weeks is S_n. The ratios $\frac{S_n}{S_{n-1}}$ are independent and log-normally with $\mu = 0.0165$ and $\sigma = 0.0730$, then

(1) What is the probability that $S_n > S_{n-1}$ for the 52 weeks of a year?

(2) What is the probability that $S_n > S_{n-1}$ for at least 30 weeks in the year?

(3) What is the probability that the stock will have gained value over the course of the year? (過一年會漲)

Proof. Note that

$$\{S_n > S_{n-1}\} \equiv \left\{\frac{S_n}{S_{n-1}} > 1\right\} \equiv \left\{\log\frac{S_n}{S_{n-1}} > 0\right\},$$

where $\log \frac{S_n}{S_{n-1}} \sim N(0.0165, 0.0730^2)$. Let $X \sim N(0.0165, 0.0730^2)$, then

$$\mathbb{P}(X > 0) = \mathbb{P}\left(\frac{X - 0.0165}{0.0730} > \frac{-0.0165}{0.0730}\right) = \mathbb{P}(Z < \frac{0.0165}{0.0730}) = \Phi\left(\frac{0.0165}{0.0730}\right) = 0.5894,$$

where $Z \sim N(0,1)$ and by symmetry.

(1)

$$\begin{aligned}\mathbb{P}\left(\{S_1 > S_0\} \cap \{S_2 > S_1\} \cap \cdots \cap \{S_{52} > S_{51}\}\right) &= \mathbb{P}\left(\left\{\frac{S_1}{S_0} > 1\right\} \cap \left\{\frac{S_2}{S_1} > 1\right\} \cap \cdots \cap \left\{\frac{S_{52}}{S_{51}} > 1\right\}\right) \\ &= \mathbb{P}\left(\frac{S_1}{S_0} > 1\right)\mathbb{P}\left(\frac{S_2}{S_1} > 1\right)\ldots\mathbb{P}\left(\frac{S_{52}}{S_{51}} > 1\right) \\ &= \Phi\left(\frac{0.0165}{0.0730}\right)^{52} \\ &\approx 1.15 \times 10^{-12}\end{aligned}$$

by independence.

(2) Let $Y = \#$ of weeks in the year with $S_n > S_{n-1}$. Then, $Y \sim \mathrm{Bin}\left(52, 0.5814\right)$ since each ratio $\frac{S_n}{S_{n-1}}$ are independent. Hence,

$$\mathbb{P}(Y \geq 30) = \sum_{k=0}^{52}\binom{52}{k}0.5894^k \cdot 0.4106^{52-k}.$$

Using normal approximation:

$$\mathrm{Bin}(52, 0.5894) \sim N(52 \times 0.5894, 52 \times 0.5894 \times 0.4106).$$

Thus,

$$\mathbb{P}(Y \geq 30) = \mathbb{P}(Y \geq 29.5) = \mathbb{P}\left(\frac{Y - 52 \times 0.5894}{52 \times 0.5894 \times 0.4106} \geq \frac{29.5 - 52 \times 0.5894}{52 \times 0.5894 \times 0.4106}\right) \approx 0.627.$$

(3)

$$\mathbb{P}(S_{52} > S_0) = \mathbb{P}\left(\log\left(\frac{S_{52}}{S_0}\right) > 0\right) = \mathbb{P}\left(\sum_{n=1}^{52}\log\left(\frac{S_n}{S_{n-1}}\right) > 0\right) \approx 0.9484.$$

This is because

$$\sum_{n=1}^{52}\log\left(\frac{S_n}{S_{n-1}}\right) \sim N(52 \times 0.0165, 52 \times 0.0730^2)$$

by the log-normally distributed condition and the previous example.

■

Lecture 21

Sum of independent discrete random variables

May 12.

Proposition 7.2.3. Let X, Y be independent discrete random variables with probability mass functions p_x, p_y. Then their sum has pmf

$$p_{x+y}(z) = \sum_x p_X(x)p_Y(z-x) = \sum_y p_X(z-y)p_Y(y).$$

Proof.

$$p_{X+Y}(z) = \mathbb{P}(X+Y=z),$$

and let $E = \{X+Y=z\}$, then the sample space S can be partitioned based on the value of X. Let $F_i = \{X = x_i\}$, where $\{x_1, x_2, \dots\}$ are the values of X takes. By the law of total probability,

$$\begin{aligned}\mathbb{P}(X+Y=z) &= \sum_i \mathbb{P}(X+Y=z \mid X=x_i)\mathbb{P}(X=x_i) = \sum_i \mathbb{P}(Y=z-x_i \mid X=x_i)\mathbb{P}(X=x_i) \\ &= \sum_i \mathbb{P}(Y=z-x_i)\mathbb{P}(X=x_i) = \sum_i p_X(x_i)p_Y(z-x_i) = \sum_x p_X(x)p_Y(z-x).\end{aligned}$$

■

Example 7.2.6 (Sums of Poisson random variables)**.** Suppose $X \sim \mathrm{Poi}(\lambda_1), Y \sim \mathrm{Poi}(\lambda_2)$ are independent, then what is the distribution of $X+Y$?

Proof.

$$\begin{aligned}p_{X+Y}(z) = \sum_{x=0}^{z} p_X(x)p_Y(z-x) &= \sum_{x=0}^{z} \frac{\lambda_1^x e^{-\lambda_1}}{x!}\frac{\lambda_2^{z-x}e^{-\lambda_2}}{(z-x)!} \\ &= e^{-(\lambda_1+\lambda_2)}\sum_{x=0}^{z}\frac{\lambda_1^x\lambda_2^{z-x}}{x!(z-x)!} = \frac{e^{-(\lambda_1+\lambda_2)}}{z!}\sum_{x=0}^{z}\frac{z!}{x!(z-x)!}\lambda_1^x\lambda_2^{z-x} \\ &= \frac{e^{-(\lambda_1+\lambda_2)}}{z!}(\lambda_1+\lambda_2)^z,\end{aligned}$$

so $X+Y \sim \mathrm{Poi}(\lambda_1+\lambda_2)$. ⊛

Chapter 8

Conditional Random Variables

Setting: Two random variables in a probability space, not necessarily independent. If we know the value of one, what does that tell us about the other?

8.1 Discrete Case

Definition 8.1.1. Let X, Y be discrete random variables. The conditional probability mass function of X given Y is given by

$$p_{X|Y}(x \mid y) = \mathbb{P}(X = x \mid Y = y).$$

By definition of conditional probability:

$$p_{X|Y}(x \mid y) = \mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} = \frac{p_{X,Y}(x,y)}{p_Y(y)}.$$

If X, Y are independent, $p_{X,Y}(x,y) = p_X(x)p_Y(y)$, so

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x,y)}{p_Y(y)} = \frac{p_X(x)p_Y(y)}{p_Y(y)} = p_X(x).$$

Example 8.1.1. Suppose $X \sim \text{Poi}(\lambda_1), Y \sim \text{Poi}(\lambda_2)$ are independent. What is the conditional distribution of X given $X + Y = n$?

Proof. Since $X + Y \sim \text{Poi}(\lambda_1 + \lambda_2)$ by the last example in last section, we have

$$\begin{aligned} p_{X|X+Y}(x \mid n) &= \frac{\mathbb{P}(X = x, X + Y = n)}{\mathbb{P}(X + Y = n)} = \frac{\mathbb{P}(X = x, Y = n - x)}{\mathbb{P}(X + Y = n)} = \frac{\mathbb{P}(X = x)\mathbb{P}(Y = n - x)}{\mathbb{P}(X + Y = n)} \\ &= \frac{\left(\frac{\lambda_1^x e^{-\lambda_1}}{x!}\right)\left(\frac{\lambda_2^{n-x} e^{-\lambda_2}}{(n-x)!}\right)}{\frac{(\lambda_1+\lambda_2)^n e^{-(\lambda_1+\lambda_2)}}{n!}} = \binom{n}{x}\left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^x \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-x}. \end{aligned}$$

This is the pmf of a binomial random variable: X conditioning on $X + Y = n \sim \text{Bin}\left(n, \frac{\lambda_1}{\lambda_1+\lambda_2}\right)$. ⊛

Example 8.1.2. We repeat a random experiment n times, independently, with probability of occurs p. Now let $X = \#$ of successes ($X \sim \text{Bin}(n,p)$), and σ = sequence of outcomes (success/failure) $\in \{S, F\}^n$. For example,if $n = 5$, then

$$\mathbb{P}(\sigma = SSFSF) = p \cdot p \cdot (1 - p) \cdot p \cdot (1 - p).$$

Then, what is the distribution of σ given $X = k$?

Proof. If s contains k S's, then

$$\mathbb{P}_{\sigma|X}(s \mid k) = \frac{p_{\sigma,X}(s,k)}{p_X(k)} = \frac{\mathbb{P}(\sigma = s, X = k)}{\mathbb{P}(X = k)} = \frac{\mathbb{P}(\sigma = s)}{\mathbb{P}(X = k)} = \frac{p^k(1-p)^{n-k}}{\binom{n}{k}p^k(1-p)^k} = \frac{1}{\binom{n}{k}}.$$

Thus, $\sigma \mid X = k$ is uniform on the set of sequences with k successes. ⊛

8.2 Continuous Random Variables Case

Suppose X, Y are continuous random variables in a probability space. If we have $Y = y$, what can we say about X?

It doesn't make sense to talk about $\mathbb{P}(X = x \mid Y = y)$ directly. Instead, we should look at intervals.

$$\begin{aligned}
\mathbb{P}\left(x \le X \le x + \delta_x \mid y \le Y \le y + \delta_y\right) &= \frac{\mathbb{P}(x \le X \le x + \delta_x, y \le Y \le Y + \delta_y)}{\mathbb{P}(y \le Y \le y + \delta_y)} \\
&= \frac{\int_x^{x+\delta_x} \int_y^{y+\delta_y} f_{X,Y}(u,v)\,\mathrm{d}v\,\mathrm{d}u}{\int_y^{y+\delta_y} f_Y(v)\,\mathrm{d}v} \\
&\approx \frac{f_{X,Y}(x,y)\delta_x\delta_y}{f_Y(y)\delta_y} \qquad \text{as } \delta_x, \delta_y \to 0 \\
&\to \frac{f_{X,Y}(x,y)}{f_Y(y)}\,\mathrm{d}x
\end{aligned}$$

Definition 8.2.1. The conditional probability density function of X given $Y = y$ is

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

and

$$\mathbb{P}(X \in A \mid Y = y) = \int_A f_{X|Y}(x \mid y)\,\mathrm{d}x.$$

Remark 8.2.1. If X, Y are independent:

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{f_X(x)f_Y(y)}{f_Y(y)} = f_X(x).$$

Example 8.2.1.

$$f_{X,Y}(x,y) = \begin{cases} \dfrac{e^{-\frac{x}{y}}e^{-y}}{y}, & \text{if } x, y > 0; \\ 0, & \text{otherwise.} \end{cases}$$

What is $\mathbb{P}(X > 1 \mid Y = y)$?

Proof.

$$f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,\mathrm{d}x = \begin{cases} 0, & \text{if } y \le 0; \\ \displaystyle\int_0^\infty \frac{e^{-\frac{x}{y}}e^{-y}}{y}\,\mathrm{d}x = e^{-y}, & \text{if } y > 0. \end{cases}$$

Thus, for $x, y > 0$,

$$f_{X|Y}(x \mid y) = \frac{\left(\frac{e^{-\frac{x}{y}}e^{-y}}{y}\right)}{e^{-y}} = \frac{e^{-\frac{x}{y}}}{y}.$$

Hence,

$$\mathbb{P}(X > 1 \mid Y = y) = \int_1^\infty f_{X\mid Y}(x \mid y)\,\mathrm{d}x = \int_1^\infty \frac{e^{-\frac{x}{y}}}{y}\,\mathrm{d}x = e^{-\frac{1}{y}}.$$

⊛

8.3 Mixed case

Suppose X is a continuous random variable, and N is a discrete random variable in the same probability space. What is the conditional distribution of X given $N = n$?

$$\begin{aligned}\mathbb{P}(x \le X \le x + \delta_x \mid N = n) &= \frac{\mathbb{P}(x \le X \le x + \delta_x, N = n)}{\mathbb{P}(N = n)} = \frac{\mathbb{P}(N = n \mid x \le X \le x + \delta_x)\mathbb{P}(x \le X \le x + \delta_x)}{\mathbb{P}(N = n)}\\ &\to \frac{\mathbb{P}(N = n \mid X = x) f_X(x)}{\mathbb{P}(N = n)} \quad \text{as } \delta_x \to 0.\end{aligned}$$

Example 8.3.1. Now we take a coin whose probability of heads is unknown, i.e. $\mathbb{P}(\text{head}) = P \sim \mathrm{Unif}([0,1])$. We toss the coin n times and observe k heads. Given this information, what can we say about the distribution of P?

Proof.

$$f_{P\mid N}(p \mid k) = \frac{\mathbb{P}(N = k \mid P = p) f_P(p)}{\mathbb{P}(N = k)} = \frac{\binom{n}{k} p^k (1-p)^{n-k} \cdot 1}{\mathbb{P}(N = k)} = cp^k(1-p)^{n-k}$$

for some c independent of p since

$$\int_0^1 f_{P\mid N}(p \mid k)\,\mathrm{d}p = 1,$$

and thus we can calculate c. This distribution is called the beta distribution. ⊛

Lecture 22

Let X_1, X_2 be continuous random variables in a probability space. Suppose we define May 14

$$Y_1 = g_1(X_1, X_2), \quad Y_2 = g_2(X_1, X_2),$$

where $g_1, g_2 : \mathbb{R}^2 \to \mathbb{R}$. If for every $(y_1, y_2) \in \mathbb{R}^2$, there is a unique $(x_1, x_2) \in \mathbb{R}^2$ where $x_1 = x_1(y_1, y_2), x_2 = x_2(y_1, y_2)$ s.t.

$$y_1 = g_1(x_1, x_2) \text{ and } y_2 = g_2(x_1, x_2),$$

and

$$J(x_1, x_2) = \left| \begin{pmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} \end{pmatrix} \right| \ne 0,$$

then

$$f_{Y_1,Y_2}(y_1, y_2) = f_{X_1,X_2}\left(x_1(y_1, y_2), x_2(y_1, y_2)\right) \left|J(x_1, x_2)\right|^{-1}.$$

Proof. Compute the joint cumulative distribution function for Y_1, Y_2,

$$\begin{aligned}F_{Y_1,Y_2}(y_1, y_2) &= \mathbb{P}\left(Y_1 \le y_1, Y_2 \le y_2\right) = \mathbb{P}\left(\{(x_1, x_2) : g_1(x_1, x_2) \le y_1, g_2(x_1, x_2) \le y_2\}\right)\\ &= \iint_{\{(x_1,x_2): g_1 \le y_1, g_2 \le y_2\}} f_{X_1,X_2}(x_1, x_2)\,\mathrm{d}x_1\,\mathrm{d}x_2.\end{aligned}$$

Now let $u_1 = g_1(x_1, x_2), u_2 = g_2(x_1, x_2)$, we have

$$\begin{aligned}&\iint_{\{(x_1,x_2):g_1\le y_1, g_2\le y_2\}} f_{X_1,X_2}(x_1,x_2)\,\mathrm{d}x_1\,\mathrm{d}x_2 \\ &= \iint_{\{u_1\le y_1, u_2\le y_2\}} f_{X_1,X_2}\left(x_1(u_1,u_2), x_2(u_1,u_2)\right)\left|J(x_1(u_1,u_2), x_2(u_1,u_2))\right|^{-1}\,\mathrm{d}u_1\,\mathrm{d}u_2.\end{aligned}$$

The joint density function is obtained by taking the partial derivatives with respect to y_1 and y_2

$$f_{Y_1,Y_2}(y_1,y_2) = \frac{\partial^2 F_{Y_1,Y_2}}{\partial y_1 \partial y_2} = f_{X_1,X_2}(x_1,x_2)|J(x_1,x_2)|^{-1},$$

where $y_1 = g_1(x_1, x_2)$ and $y_2 = g_2(x_1, x_2)$. ■

Example 8.3.2. $Y_1 = X_1 + X_2, Y_2 = X_1 - X_2$, then

$$X_1 = \frac{1}{2}(Y_1 + Y_2), \quad X_2 = \frac{1}{2}(Y_1 - Y_2),$$

and

$$|\det J(x_1,x_2)| = \left|\det\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\right| = |-2| \ne 0,$$

so

$$f_{Y_1,Y_2}(y_1,y_2) = f_{X_1,X_2}\left(\frac{y_1+y_2}{2}, \frac{y_1-y_2}{2}\right)\cdot\frac{1}{2}.$$

Example 8.3.3. Let $X_1, X_2 \sim N(0,1)$, independent. Let $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$. What can be said about the joint distribution of Y_1, Y_2?

Proof. Recall that if $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$, independent, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$, so $Y_1 \sim N(0,2)$ and $Y_2 \sim N(0,2)$. Thus,

$$f_{X_1,X_2}(x_1,x_2) = f_{X_1}(x_1)f_{X_2}(x_2) = \left(\frac{1}{\sqrt{2\pi}}e^{-\frac{x_1^2}{2}}\right)\left(\frac{1}{\sqrt{2\pi}}e^{-\frac{x_2^2}{2}}\right) = \frac{1}{2\pi}e^{-\frac{x_1^2+x_2^2}{2}}.$$

Thus,

$$\begin{aligned}f_{Y_1,Y_2}(y_1,y_2) &= \frac{1}{2}f_{X_1,X_2}\left(\frac{y_1+y_2}{2}, \frac{y_1-y_2}{2}\right) = \frac{1}{4\pi}e^{-\frac{1}{2}\left[\left(\frac{y_1+y_2}{2}\right)^2 + \left(\frac{y_1-y_2}{2}\right)^2\right]} \\ &= \frac{1}{4\pi}e^{-\frac{1}{2}\left[\frac{y_1^2}{2}+\frac{y_2^2}{2}\right]} = \left(\frac{1}{\sqrt{2\pi\cdot 2}}e^{-\frac{y_1^2}{2\cdot 2}}\right)\left(\frac{1}{\sqrt{2\pi\cdot 2e^{-\frac{y_2^2}{2\cdot 2}}}}\right) = f_{Y_1}(y_1)f_{Y_2}(y_2),\end{aligned}$$

so Y_1, Y_2 are independent $N(0,2)$. ⊛

Fact: If X_1, X_2 are independent and identically distributed continuous random variables, and $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$ are also independent, then X_1, X_2 are normally distributed.

More generally, given $X_1, X_2, \ldots, X_n$, if we define $Y_i = g_i(X_1, X_2, \ldots, X_n)$ s.t.

$$(x_1, x_2, \ldots, x_n) \mapsto (y_1, y_2, \ldots, y_n)$$

is a bijection and

$$J(x_1,\ldots,x_n) = \det\left(\left(\frac{\partial g_i}{\partial x_j}\right)_{i,j=1}^n\right) \ne 0,$$

then

$$f_{Y_1,Y_2,\ldots,Y_n}(y_1,\ldots,y_n) = f_{X_1,X_2,\ldots,X_n}(x_1,x_2,\ldots,x_n)\left|J(x_1,x_2,\ldots,x_n)\right|^{-1}.$$

8.4 Expectation of functions of random variables

Recall that if X is a discrete random variable taking non-negative integer values, then

$$\mathbb{E}[X] = \sum_{n=1}^{\infty} \mathbb{P}(X \geq n),$$

which is proved in the homework.

Lemma 8.4.1. Let X be a continuous random variable.

$$\mathbb{E}[X] = \int_0^{\infty} \mathbb{P}(X \geq t)\,\mathrm{d}t - \int_0^{\infty} \mathbb{P}(X \leq -t)\,\mathrm{d}t.$$

Proof.

$$\begin{aligned}\int_0^{\infty} \mathbb{P}(X \geq t)\,\mathrm{d}t &= \int_0^{\infty}\int_t^{\infty} f_X(x)\,\mathrm{d}x\,\mathrm{d}t = \int_0^{\infty}\int_0^{x} f_X(x)\,\mathrm{d}t\,\mathrm{d}x \\ &= \int_0^{\infty} f_X(x)\int_0^{x} 1\,\mathrm{d}t\,\mathrm{d}x = \int_0^{\infty} x f_X(x)\,\mathrm{d}x.\end{aligned}$$

$$\begin{aligned}\int_0^{\infty} \mathbb{P}(X \leq -t)\,\mathrm{d}t &= \int_0^{\infty}\int_{-\infty}^{-t} f_X(x)\,\mathrm{d}x\,\mathrm{d}t = \int_{-\infty}^{0}\int_0^{-x} f_X(x)\,\mathrm{d}t\,\mathrm{d}x \\ &= -\int_{-\infty}^{0} x f_X(x)\,\mathrm{d}x.\end{aligned}$$

Hence,

$$\int_0^{\infty} \mathbb{P}(X \geq t)\,\mathrm{d}t - \int_0^{\infty} \mathbb{P}(X \leq -t)\,\mathrm{d}t = \int_{-\infty}^{\infty} x f_X(x)\,\mathrm{d}x = \mathbb{E}[X].$$

■

Proposition 8.4.1. Let X_1, X_2 be random variables on a probability space, and let $g : \mathbb{R}^2 \to \mathbb{R}$, then

- (discrete) $\mathbb{E}[g(X_1, X_2)] = \sum_{x_1,x_2} g(x_1, x_2) p_{X_1,X_2}(x_1, x_2)$.
- (continuous) $\mathbb{E}[g(X_1, X_2)] = \iint_{\mathbb{R}^2} g(x_1, x_2) f_{X_1,X_2}(x_1, x_2)\,\mathrm{d}x_1\,\mathrm{d}x_2$.

Proof. Here we only prove the continuous setting. First, we have

$$\mathbb{E}[g(X_1, X_2)] = \int_0^{\infty} \mathbb{P}(g(X_1, X_2) \geq t)\,\mathrm{d}t - \int_0^{\infty} \mathbb{P}(g(X_1, X_2) \leq -t)\,\mathrm{d}t.$$

Now since

$$\mathbb{P}(g(X_1, X_2) \geq t) = \iint_{\{(x_1,x_2):g(x_1,x_2)\geq t\}} f_{X_1,X_2}(x_1, x_2)\,\mathrm{d}x_1\,\mathrm{d}x_2,$$

we have

$$\begin{aligned}\int_0^{\infty} \mathbb{P}(g(X_1, X_2) \geq t)\,\mathrm{d}t &= \int_0^{\infty}\iint_{\{(x_1,x_2):g\geq t\}} f_{X_1,X_2}(x_1, x_2)\,\mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}t \\ &= \iint_{\{(x_1,x_2):g\geq 0\}}\int_0^{g(x_1,x_2)} f_{X_1,X_2}(x_1, x_2)\,\mathrm{d}t\,\mathrm{d}x_1\,\mathrm{d}x_2 \\ &= \iint_{\{(x_1,x_2):g\geq 0\}} g(x_1, x_2) f_{X_1,X_2}(x_1, x_2)\,\mathrm{d}x_1\,\mathrm{d}x_2.\end{aligned}$$

Similarly,

$$\begin{aligned}
\int_0^\infty \mathbb{P}(g(X_1, X_2) \le -t)\,\mathrm{d}t &= \int_0^\infty \iint_{\{(x_1,x_2):g\le -t\}} f_{X_1,X_2}(x_1,x_2)\,\mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}t \\
&= \iint_{\{(x_1,x_2):g\le 0\}} \int_0^{-g(x_1,x_2)} f_{X_1,X_2}(x_1,x_2)\,\mathrm{d}t\,\mathrm{d}x_1\,\mathrm{d}x_2 \\
&= -\iint_{\{(x_1,x_2):g\le 0\}} g(x_1,x_2) f_{X_1,X_2}(x_1,x_2)\,\mathrm{d}x_1\,\mathrm{d}x_2.
\end{aligned}$$

Hence,

$$\begin{aligned}
\mathbb{E}[g(X_1,X_2)] &= \int_0^\infty \mathbb{P}(g(X_1,X_2) \ge t)\,\mathrm{d}t - \int_0^\infty \mathbb{P}(g(X_1,X_2) \le t)\,\mathrm{d}t \\
&= \iint_{\{(x_1,x_2):g\ge 0\}} g(x_1,x_2) f_{X_1,X_2}(x_1,x_2)\,\mathrm{d}x_1\mathrm{d}x_2 - \left(-\iint_{\{(x_1,x_2):g\le 0\}} g(x_1,x_2) f_{X_1,X_2}(x_1,x_2)\,\mathrm{d}x_1\,\mathrm{d}x_2\right) \\
&= \iint_{\mathbb{R}^2} g(x_1,x_2) f_{X_1,X_2}(x_1,x_2)\,\mathrm{d}x_1\,\mathrm{d}x_2.
\end{aligned}$$

■

Example 8.4.1. A road of length L. An accident occurs at portion X, uniformly along the road. An ambulance is at portion Y, uniformly along the road, independent of X. How far must the ambulance travel to get to the accident? What is the expected distance?

Proof. We know $X \sim \mathrm{Unif}([0,L]), Y \sim \mathrm{Unif}([0,L])$ are independent, so

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{L^2}, & \text{if } 0 \le x, y \le L; \\ 0, & \text{otherwise.} \end{cases}$$

Now define $\mathrm{dis}(x,y) = |x-y|$, then

$$\begin{aligned}
\mathbb{E}[\mathrm{dis}(X,Y)] = \iint_{\mathbb{R}^2} \mathrm{dis}(x,y) f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y &= \frac{1}{L^2}\int_0^L \int_0^L |x-y|\,\mathrm{d}x\,\mathrm{d}y \\
&= \frac{1}{L^2}\int_0^L \left(\int_0^y (y-x)\,\mathrm{d}x + \int_y^L (x-y)\,\mathrm{d}x\right)\mathrm{d}y \\
&= \frac{1}{L^2}\int_0^L \left(\left[xy - \frac{1}{2}x^2\right]_0^y + \left[\frac{1}{2}x^2 - xy\right]_y^L\right)\mathrm{d}y \\
&= \frac{1}{L^2}\int_0^L \frac{1}{2}y^2 + \frac{1}{2}L^2 - Ly + \frac{1}{2}y^2\,\mathrm{d}y \\
&= \frac{1}{L^2}\int_0^L y^2 + \frac{1}{2}L^2 - Ly\,\mathrm{d}y \\
&= \frac{1}{L^2}\left[\frac{1}{3}y^3 + \frac{1}{2}L^2 y - \frac{1}{2}Ly^2\right]_0^L \\
&= \frac{1}{L^2}\left[\frac{1}{3}L^3 + \frac{1}{2}L^3 - \frac{1}{2}L^3\right] = \frac{L}{3}.
\end{aligned}$$

⊛

Application (Linearity) If $g(X,Y) = aX + bY$, then

$$\begin{aligned}\mathbb{E}[aX+bY] &= \iint_{\mathbb{R}^2} g(x,y) f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y = \iint_{\mathbb{R}^2} (ax+by) f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y \\ &= a\iint_{\mathbb{R}^2} x f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y + b\iint_{\mathbb{R}^2} y f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y \\ &= a\int_{\mathbb{R}} x \int_{\mathbb{R}} f_{X,Y}(x,y)\,\mathrm{d}y\,\mathrm{d}x + b\int_{\mathbb{R}} y \int_{\mathbb{R}} f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y \\ &= a\int_{\mathbb{R}} x f_X(x)\,\mathrm{d}x + b\int_{\mathbb{R}} y f_Y(y)\,\mathrm{d}y = a\mathbb{E}[X] + b\mathbb{E}[Y].\end{aligned}$$

Application (Monotonicity) Let X, Y be random variables on a probability space. If $X \geq Y$ always, then $\mathbb{E}[X] \geq \mathbb{E}[Y]$.

This is because

$$X \geq Y \Leftrightarrow X - Y \geq 0 \Rightarrow \mathbb{E}[X-Y] \geq 0 \Rightarrow \mathbb{E}[X] - \mathbb{E}[Y] \geq 0 \Rightarrow \mathbb{E}[X] \geq \mathbb{E}[Y].$$

Application (Union Bound/Boole's inequality) If $A_1, A_2, \ldots, A_n$ are events, then

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \mathbb{P}(A_i).$$

Proof. Let X_i be the indicator for A_i:

$$X_i = \begin{cases} 1, & \text{if } A_i \text{ occurs;} \\ 0, & \text{if } A_i \text{ doesn't occurs.} \end{cases}$$

Then,

$$\mathbb{E}[X_i] = 1 \cdot \mathbb{P}(A_i) + 0 \cdot \mathbb{P}(A_i^C) = \mathbb{P}(A_i).$$

Let $X = \sum_{i=1}^n X_i = \#$ of events occur. Let Y be the indicator for $\bigcup_{i=1}^n A_i$:

$$Y = \begin{cases} 1, & \text{if } X \geq 1; \\ 0, & \text{if } X = 0. \end{cases}$$

Hence, $Y \leq X$ always. This shows

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \mathbb{E}[Y] \leq \mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i] = \sum_{i=1}^n \mathbb{P}(A_i).$$

■

Example 8.4.2. We toss a coin n times. Tosses are independent: Heads with probability p, Tails with probability $1-p$. A streak is a maximal subinterval of identical outcomes. What is the expected number of streaks?

Proof. Suppose $X = \#$ of streaks in n tosses. Let

$$X_i = \begin{cases} 1, & \text{if a new streak starting with } i\text{-th toss;} \\ 0, & \text{if not.} \end{cases}$$

Then, $X = \sum_{i=1}^n X_i$ and thus

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i].$$

Suppose
$$\mathbb{E}[X_i] = \mathbb{P}(\text{a new streak starting with } i\text{-th toss}) = p_i,$$
then $p_1 = 1$, and for $i \geq 2$,
$$\begin{aligned}
p_i &= \mathbb{P}(\text{toss } \# \ i \neq \ \text{toss } \# \ i-1) \\
&= \mathbb{P}\left(\{\text{toss } \# \ i = H, \ \text{toss } \# i-1 = T\} \cup \{\text{toss } \# \ i = T, \ \text{toss } \# i-1 = H\}\right) \\
&= \mathbb{P}(\text{toss } \# \ i = H)\mathbb{P}(\text{toss } \# \ i-1 = T) + \mathbb{P}(\text{toss } \# \ i = T)\mathbb{P}(\text{toss } \# \ i-1 = H) \\
&= p(1-p) + (1-p)p = 2p(1-p).
\end{aligned}$$
Thus,
$$\mathbb{E}[X] = \sum_{i=1}^{n} p_i = 1 + 2p(1-p)(n-1).$$

⊛

Lecture 23

May 19.

Example 8.4.3 (Quick Sort).

Algorithm 8.1: Quick Sort

Pick a number uniformly at random.
Compute the other numbers to this pivot, and put them in "smaller" or "bigger" lists accordingly.
Sort sublists recurrsively.
Return "smaller", pivot, "bigger".

How many times do we need to compare two numbers?

Proof. Let $X = \#$ of comparisons. Our goal is to determine $\mathbb{E}[X]$, and suppose after sorted we have
$$x_1 < x_2 < \cdots < x_n.$$
For $1 \leq i < j \leq n$, let
$$X_{i,j} = \begin{cases} 1, & \text{if } x_i \text{ and } x_j \text{ are compared;} \\ 0, & \text{if not.} \end{cases}$$
Then, $X = \sum_{1 \leq i \leq j \leq n} X_{i,j}$. Hence, by linearity of expectation,
$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{1 \leq i < j \leq n} X_{i,j}\right] = \sum_{1 \leq i < j \leq n} \mathbb{E}[X_{i,j}] = \sum_{1 \leq i < j \leq n} \mathbb{P}(x_i, x_j \text{ are compared}).$$
Consider the round in which we choose a pivot in $\{x_i, x_{i+1}, \ldots, x_j\}$, x_i, x_j get compared means one of them is the pivot. Then
$$\mathbb{P}(x_i, x_j \text{ are compared}) = \frac{2}{j+i-1},$$
so
$$\begin{aligned}
\mathbb{E}[X] &= \sum_{i<j} \mathbb{P}(x_i, x_j \text{ are compared}) = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \frac{2}{j-i+1} \\
&= \sum_{i=1}^{n-1} \sum_{j'=1}^{n-i} \frac{2}{j'+1} \approx 2\sum_{i=1}^{n-1} \ln(n-i) \approx 2n \ln n.
\end{aligned}$$

⊛

8.5 Moments of Number of events

Let $A_1, A_2, \ldots, A_n$ be events in a probability space. Let X to be the number of these events that occur, then

$$X = \sum_{i=1}^{n} I_i, \text{ where } I_i = \begin{cases} 1, & \text{if } A_i \text{ occurs;} \\ 0, & \text{if not.} \end{cases}$$

Thus,

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[I_i] = \sum_{i=1}^{n} \mathbb{P}(A_i).$$

Question. What about the variance?

Note that

$$\binom{X}{2} = \#\text{ of pairs of events that both occur} = \sum_{i<j} I_i I_j.$$

Hence,

$$\mathbb{E}\left[\binom{X}{2}\right] = \mathbb{E}\left[\sum_{i<j} I_i I_j\right] = \sum_{i<j} \mathbb{E}[I_i I_j] = \sum_{i<j} \mathbb{P}(A_i \cap A_j),$$

where

$$\binom{X}{2} = \frac{X(X-1)}{2},$$

so

$$\mathbb{E}\left[\binom{X}{2}\right] = \mathbb{E}\left[\frac{1}{2}(X^2 - X)\right] = \frac{1}{2}\mathbb{E}[X^2] - \frac{1}{2}\mathbb{E}[X].$$

Thus,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = 2\mathbb{E}\left[\binom{X}{2}\right] + \mathbb{E}[X] - \mathbb{E}[X]^2.$$

Example 8.5.1. $X \sim \mathrm{Bin}(n,p)$. Let $A_i = \{i\text{-th trial successful}\}$. Then,

$$\begin{aligned} \mathbb{E}[X] &= \sum_{i=1}^{n} \mathbb{P}(A_i) = np. \\ \mathbb{E}\left[\binom{X}{2}\right] &= \sum_{i<j} \mathbb{P}(A_i \cap A_j) = \sum_{i<j} \mathbb{P}(A_i)\mathbb{P}(A_j) = \binom{n}{2} p^2. \\ \mathrm{Var}(X) &= 2\mathbb{E}\left[\binom{X}{2}\right] + \mathbb{E}[X] - \mathbb{E}[X]^2 = 2\binom{n}{2} p^2 + np - n^2p^2 = np(1-p). \end{aligned}$$

Example 8.5.2 (Hat return problem). n people, n hats, hats returned uniformly at random. $X = \#$ of people getting own hats. $A_i = \{i\text{-th person gets own hat}\}$. $I_i = 1_{A_i}$, $p_i = \mathbb{P}(A_i) = \frac{1}{n}$. Then,

$$\mathbb{E}[X] = \sum_{i=1}^{n} p_i = n \cdot \frac{1}{n} = 1.$$

Also,

$$\mathbb{E}\left[\binom{X}{2}\right] = \sum_{i<j} \mathbb{P}(A_i \cap A_j) = \sum_{i<j} \mathbb{P}(A_i)\mathbb{P}(A_j \mid A_i) = \sum_{i<j} \frac{1}{n} \cdot \frac{1}{n-1} = \binom{n}{2} \cdot \frac{1}{n(n-1)} = \frac{1}{2}.$$

Hence,

$$\mathrm{Var}(X) = 2\mathbb{E}\left[\binom{X}{2}\right] + \mathbb{E}[X] - \mathbb{E}[X]^2 = 2 \cdot \frac{1}{2} + 1 - 1^2 = 1.$$

Example 8.5.3 (Coupon collector problem). n types of coupons. Each time, get uniformly random coupon, independent of previous coupons. The goal is to collect n types of coupons. Suppose X is the number of coupons collected before we get every type. X_i is the number of coupons needed to go from having $i-1$ types to having i types. Hence, $X_i \sim \text{Geom}\left(\frac{n-i+1}{n}\right)$. Hence, $X = \sum_{i=1}^{n} X_i$, and

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} \frac{n}{n-i+1} = n \sum_{i'=1}^{n} \frac{1}{i'} \approx n \ln n.$$

Now let Y to be the number of types of coupons that we have a unique copy of when we complete our first set. We can relabel coupons in order they appear, i.e. type i means the i-th coupon we see. Now let $A_i = \{\text{coupon } i \text{ appears uniquely}\}$, and $I_i = 1_{A_i}$, then

$$Y = \sum_{i=1}^{n} I_i \Rightarrow \mathbb{E}[Y] = \sum_{i=1}^{n} \mathbb{P}(A_i).$$

Note that

$$\begin{aligned} A_i &= \{\text{after the first copy of } i, \text{ we get } i+1, \ldots, n \text{ before we get } i \text{ (again)}\} \\ &= \{\text{among } \{i, i+1, \ldots, n\}\, i \text{ is the last coupon we get}\}. \end{aligned}$$

By symmetry, any of those coupons are equally likely to be the last one, so

$$\mathbb{P}(A_i) = \frac{1}{n-i+1}.$$

Hence,

$$\mathbb{E}[Y] = \sum_{i=1}^{n} \mathbb{P}(A_i) = \sum_{i=1}^{n} \frac{1}{n-i+1} \approx \ln n.$$

Also,

$$\mathbb{E}\left[\binom{Y}{2}\right] = \sum_{i<j} \mathbb{P}(A_i \cap A_j)$$

Suppose

$$S_{i,j} = \{\text{In } \{i, i+1, \ldots, j\}, \text{ coupon } i \text{ appears last after seeing coupon } i \text{ first time}\}.$$

Then

$$\mathbb{P}(S_{i,j}) = \frac{1}{j-i+1}.$$

Suppose

$$T_{i,j} = \{\text{among } \{i, j, j+1 \ldots, n\}, i \text{ and } j \text{ are the last 2 after coupon } j \text{ appears}\},$$

then

$$\mathbb{P}(T_{i,j}) = \frac{2}{\binom{n-j+2}{2} \cdot 2!} = \frac{1}{\binom{n-j+2}{2}}.$$

And $S_{i,j}$ and $T_{i,j}$ depend on disjoint sets of coupons, so they are independent. Thus,

$$\mathbb{P}(A_i \cap A_j) = \mathbb{P}(S_{i,j} \cap T_{i,j}) = \mathbb{P}(S_{i,j})\mathbb{P}(T_{i,j}) = \frac{1}{j-i+1} \cdot \frac{2}{\binom{n-j+2}{2}}.$$

Thus,

$$\mathbb{E}\left[\binom{Y}{2}\right] = \sum_{i<j} \frac{1}{j-i+1} \cdot \frac{1}{\binom{n-j+2}{2}}.$$

Remark 8.5.1. When computing $\mathbb{P}(A_i \cap A_j)$, we can observe $A_i \cap A_j = S_{i,j} \cap T_{i,j}$ since after first getting coupon i, coupon $i+1, \dots, j$ should appear first than coupon i again, and after getting coupon j, coupon i, j should be the last 2 appearing among $i, j, j+1, \dots, n$.

Generalization $X = \sum_{i=1}^{n} I_i$ where $I_i = 1_{A_i}$, then

$$\mathbb{E}\left[\binom{X}{k}\right] = \mathbb{E}\left[\#\text{ of } k\text{-sets of events all occuring}\right] = \sum_{i_1 < i_2 < \cdots < i_k} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}).$$

Lecture 24

8.6 Covariance, Variance of Sums, Correlation

May 21.

Question. Given random variables X and Y on a probability space, what is the variance of $X+Y$?

Answer.

$$\begin{aligned}
\operatorname{Var}(X+Y) &= \mathbb{E}\left[(X+Y)^2\right] - \mathbb{E}[X+Y]^2 \\
&= \mathbb{E}\left[X^2 + 2XY + Y^2\right] - (\mathbb{E}[X] + \mathbb{E}[Y])^2 \\
&= \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] - \mathbb{E}[X]^2 - 2\mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[Y]^2 \\
&= \mathbb{E}[X^2] - \mathbb{E}[X]^2 + \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 + 2\left(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]\right) \\
&= \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\left(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]\right).
\end{aligned}$$

⊛

Proposition 8.6.1. Let X, Y be independent and let $g, h : \mathbb{R} \to \mathbb{R}$ be arbitrary. Then

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)].$$

Proof.

$$\begin{aligned}
\mathbb{E}[g(X)h(Y)] &= \iint_{\mathbb{R}^2} g(x)h(y) f_{X,Y}(x,y) \,\mathrm{d}x \,\mathrm{d}y \\
&= \iint_{\mathbb{R}^2} g(x)h(y) f_X(x) f_Y(y) \,\mathrm{d}x \,\mathrm{d}y \\
&= \left(\int_{\mathbb{R}} g(x) f_X(x) \,\mathrm{d}x\right) \left(\int_{\mathbb{R}} h(y) f_Y(y) \,\mathrm{d}y\right) \\
&= \mathbb{E}[g(X)]\mathbb{E}[h(Y)].
\end{aligned}$$

■

Corollary 8.6.1. If X, Y are independent, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

Corollary 8.6.2. If X, Y are independent, then

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y).$$

Proof. By independence,

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

Thus,

$$\begin{aligned}\operatorname{Var}(X+Y) &= \operatorname{Var}(X)+\operatorname{Var}(Y)+2\left(\mathbb{E}[XY]-\mathbb{E}[X]\mathbb{E}[Y]\right)\\ &= \operatorname{Var}(X)+\operatorname{Var}(Y).\end{aligned}$$

■

Example 8.6.1. $X \sim \operatorname{Bin}(n,p)$, and $X = X_1+\cdots+X_n$, where $X_i \sim \operatorname{Ber}(p)$ are independent. Then since

$$\operatorname{Var}(X_i) = p(1-p),$$

we have

$$\operatorname{Var}(X) = \operatorname{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \operatorname{Var}(X_i) = np(1-p).$$

Example 8.6.2. $X \sim N(\mu,\sigma_1^2)$ and $Y \sim N(\mu_2,\sigma_2^2)$ are independent, then we know $X+Y \sim N(\mu_1+\mu_2,\sigma_1^2+\sigma_2^2)$, and we can check

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X)+\operatorname{Var}(Y) = \sigma_1^2+\sigma_2^2.$$

Definition 8.6.1. Given random variables X and Y in a probability space, the covariance of X and Y is defined as

$$\operatorname{Cov}(X,Y) = \mathbb{E}\left[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])\right].$$

Observation.

$$\begin{aligned}\operatorname{Cov}(X,Y) &= \mathbb{E}\left[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])\right]\\ &= \mathbb{E}\left[XY - X\mathbb{E}[Y]-\mathbb{E}[X]Y+\mathbb{E}[X]\mathbb{E}[Y]\right]\\ &= \mathbb{E}[XY]-\mathbb{E}[X]\mathbb{E}[Y]-\mathbb{E}[X]\mathbb{E}[Y]+\mathbb{E}[X]\mathbb{E}[Y]\\ &= \mathbb{E}[XY]-\mathbb{E}[X]\mathbb{E}[Y].\end{aligned}$$

In particular,

(i) $\operatorname{Var}(X+Y) = \operatorname{Var}(X)+\operatorname{Var}(Y)+2\operatorname{Cov}(X,Y)$.

(ii) If X,Y are independent, $\operatorname{Cov}(X,Y)=0$.

Remark 8.6.1. $\operatorname{Cov}(X,Y)=0$ does not imply X,Y are independent.

For example, let $X=-1,0,1$ with probability $\frac{1}{3}$, respectively. Let

$$Y = \begin{cases} 1, & \text{if } X=0;\\ 0, & \text{if } X\neq 0.\end{cases}$$

Then

$$\operatorname{Cov}(X,Y) = \mathbb{E}[XY]-\mathbb{E}[X]\mathbb{E}[Y].$$

Note that $\mathbb{E}[XY]=0$, and $\mathbb{E}[X]=0$, so $\operatorname{Cov}(X,Y)=0$. However, we can show X,Y are not independent.

Note that $\mathbb{P}(X=1,Y=1)=0$, and $\mathbb{P}(X=1)=\frac{1}{3}$, and $\mathbb{P}(Y=1)=\mathbb{P}(X=0)=\frac{1}{3}$, so

$$\mathbb{P}(X=1,Y=1)\neq \mathbb{P}(X=1)\mathbb{P}(Y=1).$$

Proposition 8.6.2. Let $X,Y,X_1,\dots,X_n,Y_1,\dots,Y_m$ be random variables, then

(i) $\operatorname{Cov}(X,Y)=\operatorname{Cov}(Y,X)$.

(ii) $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$.

(iii) $\mathrm{Cov}(aX, Y) = a\,\mathrm{Cov}(X, Y)$ for every $a \in \mathbb{R}$.

(iv) $\mathrm{Cov}\left(\sum_{i=1}^n X_i, \sum_{j=1}^m Y_j\right) = \sum_{i=1}^n \sum_{j=1}^m \mathrm{Cov}(X_i, Y_j)$.

Proof. (i) to (iii) are immediate from the definition. Now we prove (iv).

$$
\begin{aligned}
\mathrm{Cov}\left(\sum_{i=1}^n X_i, \sum_{j=1}^m Y_j\right) &= \mathbb{E}\left[\left(\sum_{i=1}^n X_i\right)\left(\sum_{j=1}^m Y_j\right)\right] - \mathbb{E}\left[\sum_{i=1}^n X_i\right]\mathbb{E}\left[\sum_{j=1}^m Y_j\right] \\
&= \mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^m X_i Y_j\right] - \mathbb{E}\left[\sum_{i=1}^n X_i\right]\mathbb{E}\left[\sum_{j=1}^m Y_j\right] \\
&= \sum_{i=1}^n \sum_{j=1}^m \mathbb{E}[X_i Y_j] - \left(\sum_{i=1}^n \mathbb{E}[X_i]\right)\left(\sum_{j=1}^m \mathbb{E}[Y_j]\right) \\
&= \sum_{i=1}^n \sum_{j=1}^m \left(\mathbb{E}[X_i Y_j] - \mathbb{E}[X_i]\mathbb{E}[Y_j]\right) = \sum_{i=1}^n \sum_{j=1}^m \mathrm{Cov}(X_i, Y_j).
\end{aligned}
$$

■

Corollary 8.6.3. If $X = \sum_{i=1}^n X_i$, then

$$
\mathrm{Var}(X) = \sum_{i=1}^n \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j).
$$

Proof.

$$
\begin{aligned}
\mathrm{Var}(X) = \mathrm{Cov}(X, X) = \mathrm{Cov}\left(\sum_{i=1}^n X_i, \sum_{j=1}^n X_j\right) &= \sum_{i=1}^n \sum_{j=1}^n \mathrm{Cov}(X_i, X_j) \\
&= \sum_{i=1}^n \mathrm{Cov}(X_i, X_i) + \sum_{i \neq j} \mathrm{Cov}(X_i, X_j) \\
&= \sum_{i=1}^n \mathrm{Var}(X_i) + 2\sum_{1 \le i < j \le n} \mathrm{Cov}(X_i, X_j).
\end{aligned}
$$

■

Corollary 8.6.4. If $X_1, X_2, \ldots, X_n$ are pairwise independent,

$$
\mathrm{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \mathrm{Var}(X_i).
$$

Example 8.6.3. We collect independent samples $X_1, X_2, \ldots, X_n$ from some distribution X with mean μ and variance σ^2. We define

$$
\overline{X} = \frac{1}{n}\sum_{i=1}^n X_i
$$

to be the sample mean, and

$$
S^2 = \frac{1}{n-1}\sum_{i=1}^n \left(X_i - \overline{X}\right)^2
$$

to be the sample variance, then

$$\mathbb{E}\left[\overline{X}\right] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \frac{1}{n}\sum_{i=1}^{n}\mu = \mu.$$

We say $\overline{X}$ is an unbiased estimator of μ because it has the right value in expectation. Also,

$$\begin{aligned}\operatorname{Var}(\overline{X}) &= \operatorname{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right)\\ &= \frac{1}{n^2}\sum_{i=1}^{n}\operatorname{Var}(X_i) = \frac{1}{n}\sum_{i=1}^{n}\sigma^2 = \frac{\sigma^2}{n}.\end{aligned}$$

$$\begin{aligned}\mathbb{E}[(n-1)S^2] &= \mathbb{E}\left[\sum_{i=1}^{n}\left(X_i-\overline{X}\right)^2\right] = \mathbb{E}\left[\sum_{i=1}^{n}\left(X_i^2 - 2X_i\overline{X} + \left(\overline{X}\right)^2\right)\right]\\ &= \mathbb{E}\left[\sum_{i=1}^{n} X_i^2 - 2\overline{X}\sum_{i=1}^{n} X_i + n\left(\overline{X}\right)^2\right] = \mathbb{E}\left[\sum_{i=1}^{n} X_i^2 - n\left(\overline{X}^2\right)\right]\\ &= \sum_{i=1}^{n}\mathbb{E}[X_i^2] - n\mathbb{E}\left[\left(\overline{X}\right)^2\right] = \sum_{i=1}^{n}\left(\operatorname{Var}(X_i) + \mathbb{E}[X_i]^2\right) - n\left(\operatorname{Var}(\overline{X}) + \mathbb{E}[\overline{X}]^2\right)\\ &= n\left(\sigma^2+\mu^2\right) - n\left(\frac{\sigma^2}{n}+\mu^2\right) = (n-1)\sigma^2.\end{aligned}$$

Thus, $\mathbb{E}[S^2] = \sigma^2$, i.e. S^2 is an unbiased estimator of σ^2.

Definition 8.6.2. Given random variables X, Y on a probability space, with $\sigma_X = \sqrt{\operatorname{Var}(X)}$ and $\sigma_Y = \sqrt{\operatorname{Var}(Y)}$, we define the correlation of X and Y as

$$\rho(X,Y) = \frac{\operatorname{Cov}(X,Y)}{\sigma_X\sigma_Y} = \frac{\operatorname{Cov}(X,Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}}.$$

Proposition 8.6.3. For any random variables X, Y,

$$-1 \le \rho(X,Y) \le 1.$$

Proof. Consider

$$\begin{aligned}\operatorname{Var}\left(\frac{X}{\sigma_X}+\frac{Y}{\sigma_Y}\right) &= \operatorname{Var}\left(\frac{X}{\sigma_X}\right) + \operatorname{Var}\left(\frac{Y}{\sigma_Y}\right) + 2\operatorname{Cov}\left(\frac{X}{\sigma_X},\frac{Y}{\sigma_Y}\right)\\ &= \frac{\operatorname{Var}(X)}{\sigma_X^2} + \frac{\operatorname{Var}(Y)}{\sigma_Y^2} + \frac{2\operatorname{Cov}(X,Y)}{\sigma_X\sigma_Y} = 2 + 2\rho(X,Y).\end{aligned}$$

Hence,

$$2 + 2\rho(X,Y) \ge 0 \Rightarrow \rho(X,Y) \ge -1.$$

Similarly,

$$0 \le \operatorname{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right) = 2 - 2\rho(X,Y) \Rightarrow \rho(X,Y) \le 1.$$

Remark 8.6.2. We discuss when the equality holds: Suppose $\rho(X,Y)=1$, then

$$\operatorname{Var}\left(\frac{X}{\sigma_X}-\frac{Y}{\sigma_Y}\right)=0.$$

Hence,

$$\frac{X}{\sigma_X}-\frac{Y}{\sigma_Y}=C \quad \text{for some constant } C.$$

This shows

$$Y=\frac{\sigma_Y}{\sigma_X}\cdot X-C\sigma_Y,$$

i.e. $Y=aX+b$ for some $a>0$. Similarly, if $\rho(X,Y)=-1$, then

$$\operatorname{Var}\left(\frac{X}{\sigma_X}+\frac{Y}{\sigma_Y}\right)=0,$$

so $Y=aX+b$ for some $a<0$.

■

Definition 8.6.3. If $\rho(X,Y)=0$, we say X and Y are uncorrelated.

Example 8.6.4. Let A,B be events in a probability space, and let I_A,I_B be their indicator random variables, then

$$\begin{aligned}\operatorname{Cov}(I_A,I_B)&=\mathbb{E}[I_AI_B]-\mathbb{E}[I_A]\mathbb{E}[I_B]=\mathbb{E}[I_{A\cap B}]-\mathbb{E}[I_A]\mathbb{E}[I_B]\\&=\mathbb{P}(A\cap B)-\mathbb{P}(A)\mathbb{P}(B)=\mathbb{P}(B)\left[\mathbb{P}(A\mid B)-\mathbb{P}(A)\right]\end{aligned}$$

Lecture 25

Recall that the conditional distribution of X given Y has pmf/pdf May 26.

$$\begin{aligned}p_{X|Y}(x\mid y)&=\frac{p_{X,Y}(x,y)}{p_Y(y)}\\ f_{X|Y}(x\mid y)&=\frac{f_{X,Y}(x,y)}{f_Y(y)}.\end{aligned}$$

Given the value of Y, we get a new distribution – the conditional distribution of X given Y. Thus, we can computeits expectation or variance.

Definition 8.6.4 (Discrete Case). The conditional expectation of X given Y is defined as

$$\mathbb{E}\left[X\mid Y=y\right]=\sum_x xp_{X|Y}(x\mid y).$$

Definition 8.6.5 (Continuous Case). If X,Y are continuous, the conditional expectation of X given $Y=y$ is

$$\mathbb{E}[X\mid Y=y]=\int_{\mathbb{R}} xf_{X|Y}(x\mid y)\,\mathrm{d}x.$$

Remark 8.6.3. If X,Y are independent, then

$$\mathbb{E}[X\mid Y=y]=\mathbb{E}[X],\quad \text{for all } y\in\operatorname{Im}(Y),$$

i.e. knowing the value of Y doesn't affect the distribution of X (since $p_{X\mid Y}(x \mid y) = p_X(x)$, and $f_{X\mid Y}(x \mid y) = f_X(x)$ if X, Y are independent).

Example 8.6.5. Let X, Y be independent, and $X, Y \sim \mathrm{Bin}(n, p)$. What is $\mathbb{E}[X \mid X + Y = m]$?

Proof. We have to know $p_{X\mid X+Y}(k \mid m)$ to solve this problem.

Note that

$$\begin{aligned} p_{X\mid X+Y}(k \mid m) = \mathbb{P}(X = k \mid X + Y = m) &= \frac{\mathbb{P}(\{X = k\} \cap \{X + Y = m\})}{\mathbb{P}(X + Y = m)} \\ &= \frac{\mathbb{P}(\{X = k\} \cap \{Y = m - k\})}{\mathbb{P}(X + Y = m)}. \end{aligned}$$

Now since $X, Y \sim \mathrm{Bin}(n, p)$, independet, we know $X + Y \sim \mathrm{Bin}(2n, p)$. Hence,

$$\mathbb{P}(X + Y = m) = \binom{2n}{m} p^m (1 - p)^{2n-m}.$$

Also,

$$\begin{aligned} \mathbb{P}(\{X = k\} \cap \{Y = m - k\}) &= \mathbb{P}(X = k)\mathbb{P}(Y = m - k) \\ &= \binom{n}{k} p^k (1 - p)^{n-k} \binom{n}{m-k} p^{m-k} (1 - p)^{n-(m-k)}. \end{aligned}$$

Thus,

$$p_{X\mid X+Y}(k \mid m) = \frac{\binom{n}{k}\binom{n}{m-k} p^m (1 - p)^{2n-m}}{\binom{2n}{m} p^m (1 - p)^{2n-m}} = \frac{\binom{n}{k}\binom{n}{m-k}}{\binom{2n}{m}}.$$

This is a hyper geometric distribution, i.e.

$$X \mid X + Y = m \sim \mathrm{Hypergeometric}(2n, n, m).$$

Hence,

$$\mathbb{E}[X \mid X + Y = m] = \frac{n}{2n} \cdot m = \frac{m}{2}.$$

⊛

Example 8.6.6. Let

$$f_{X,Y}(x, y) = \begin{cases} \dfrac{e^{-\frac{x}{y}} e^{-y}}{y}, & \text{if } x, y > 0; \\ 0, & \text{otherwise.} \end{cases}$$

What is $\mathbb{E}[X \mid Y = y]$?

Proof. To compute

$$f_{X\mid Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)},$$

we need to compute (for $y > 0$)

$$\begin{aligned} f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y)\,\mathrm{d}x &= \int_0^\infty \frac{e^{-\frac{x}{y}} e^{-y}}{y}\,\mathrm{d}x = e^{-y} \int_0^\infty \frac{e^{-\frac{x}{y}}}{y}\,\mathrm{d}x \\ &= e^{-y} \left[-e^{-\frac{x}{y}} \right]_0^\infty = e^{-y}. \end{aligned}$$

Thus,

$$f_Y(y) = \begin{cases} e^{-y}, & \text{if } y > 0; \\ 0, & \text{if } y \leq 0. \end{cases}$$

This shows

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \begin{cases} \frac{\left(\frac{e^{-\frac{x}{y}}e^{-y}}{y}\right)}{e^{-y}} = \frac{e^{-\frac{x}{y}}}{y}, & \text{if } x, y > 0; \\ 0, & \text{if } x \leq 0 \text{ or } y \leq 0. \end{cases}$$

That is, $X \mid Y = y \sim \text{Exp}\left(\frac{1}{y}\right)$. This shows $\mathbb{E}[X \mid Y = y] = y$. ⊛

Remark 8.6.4. Observe that $\mathbb{E}[X \mid Y = y]$ is a function of y:

$$\mathbb{E}[X \mid Y = \cdot] : \text{Im}(Y) \to \mathbb{R}.$$

Thus, we can substitute the random variable Y itself to get a new random variable $\mathbb{E}[X \mid Y]$.

Proposition 8.6.4. Given two random variables X, Y, we have

$$\mathbb{E}\left[\mathbb{E}[X \mid Y]\right] = \mathbb{E}[X].$$

proof of discrete case. First,

$$\mathbb{E}[X \mid Y = y] = \sum_x x p_{X|Y}(x, y).$$

Thus,

$$\begin{aligned} \mathbb{E}[\mathbb{E}[X \mid Y]] &= \sum_y \mathbb{E}[X \mid Y = y] p_Y(y) = \sum_y \sum_x x p_{X|Y}(x,y) p_Y(y) \\ &= \sum_y \sum_x x p_{X,Y}(x,y) = \sum_x x \left(\sum_y p_{X,Y}(x,y)\right) = \sum_x x p_X(x) = \mathbb{E}[X]. \end{aligned}$$

■

Example 8.6.7. A shop receives a random number N of customers each day, $N \sim \text{Poi}(500)$. Each customer spends a uniform amount of money between \$0 and \$1000, independently of N and all other customers. What is the expected daily income?

Proof. Let X_i be the amount spent by the i-th customer, then $X_i \sim \text{Unif}([0, 1000])$. Let X be the total daily income, then $X = \sum_{i=1}^{N} X_i$. Hence, by previous proposition,

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid N]],$$

and

$$\mathbb{E}[X \mid N = n] = \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i] = 500n.$$

Hence,

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid N]] = \mathbb{E}[500N] = 500\mathbb{E}[N] = 500 \cdot 500 = 250000.$$

⊛

Example 8.6.8. We have r big tech companies. The i-th company has n_i billion dollars. Two companies go to court for copyright infringement, with each company equally likely to win. The loser pays the winner \$1 billion. This repeats until all but one company is bankrupt. What is the expected number of court cases? (If we choose the suing companies uniformly at random)

Proof. We first regard the case $r = 2$, and we let $n_1 + n_2 = n$. Let X_i be the number of cases of company 1 starting with $n_1 = i$. Let

$$Y = \begin{cases} 1, & \text{if first case is won by company 1;} \\ 0, & \text{if first case is lost by company 1.} \end{cases}$$

Let A_i be the number of additional rounds after the first if $n_1 = i$. Then

$$X_i = \begin{cases} 0, & \text{if } i = 0, n \text{ (a company is already bankrupt);} \\ 1 + A_i, & \text{if } 1 \le i \le n-1. \end{cases}$$

Let $m_i = \mathbb{E}[X_i]$ then by the proposition

$$\mathbb{E}[X_i] = \mathbb{E}\left[\mathbb{E}[X_i \mid Y]\right] = \frac{1}{2}\mathbb{E}[X_i \mid Y = 1] + \frac{1}{2}\mathbb{E}[X_i \mid Y = 0]$$

by the definition of expectation.

Now note that for $i \ne 0, n$ we have

$$\mathbb{E}[X_i \mid Y = 1] = \mathbb{E}[1 + A_i \mid Y = 1] = 1 + \mathbb{E}[A_i \mid Y = 1].$$

But $A_i \mid Y = 1 \sim X_{i+1}$ (start with i billions and win first round). Thus,

$$1 + \mathbb{E}[A_i \mid Y = 1] = 1 + \mathbb{E}[X_{i+1}] = 1 + m_{i+1}.$$

Similarly,

$$\mathbb{E}[X_i \mid Y = 0] = 1 + m_{i-1}.$$

Thus,

$$m_i = \mathbb{E}[X_i] = \frac{1}{2}\left(1 + m_{i+1}\right) + \frac{1}{2}\left(1 + m_{i-1}\right).$$

Hence,

$$m_i = \begin{cases} 1 + \frac{1}{2}m_{i+1} + \frac{1}{2}m_{i-1}, & \text{if } 1 \le i \le n-1; \\ 0, & \text{if } i = 0, n. \end{cases}$$

Solve the recurrence relation, we have $m_i = i(n-i)$.

Now we consider the general cases: r companies. $n = \sum_{i=1}^{r} n_i$. Let $X_i = \#$ of court cases involving company i, and let X be the number of total cases. Observe that

$$X = \frac{1}{2}\sum_{i=1}^{r} X_i.$$

Thus,

$$\mathbb{E}[X] = \frac{1}{2}\sum_{i=1}^{r} \mathbb{E}[X_i].$$

Notice that X_i follows the two company situation since compny i doesn't really care about which company is suing it. Hence,

$$\mathbb{E}[X_i] = n_i(n - n_i).$$

This shows

$$\mathbb{E}[X] = \frac{1}{2}\sum_{i=1}^{r} n_i(n - n_i) = \frac{1}{2}\left(n^2 - \sum_{i=1}^{r} n_i^2\right).$$

⊛

Lecture 26

March 28.

Example 8.6.9 (Grading oral exams). Professor sees n students one-by-one, in uniformly random order. Wants to give one A+, to the best student.

If the strategy of the professor is to give an A+ to the very first student,

$$\mathbb{P}(\text{success}) = \mathbb{P}\left(\text{best student is the first one}\right) = \frac{1}{n}.$$

We can give a better strategy: See the first k students, and give no A+'s. Give an A+ to a student if they are better than everyone before them.

Define

$$S = \begin{cases} 1, & \text{if we give A+ to the best student;} \\ 0, & \text{if not.} \end{cases}$$

Then $\mathbb{P}(\text{success}) = \mathbb{E}[S]$. Let X be the position of the best student. Then

$$\begin{aligned} \mathbb{P}(\text{success}) = \mathbb{E}[S] = \mathbb{E}[\mathbb{E}[S \mid X]] &= \sum_{i=1}^{n} \mathbb{E}[S \mid X = i]\mathbb{P}(X = i) \\ &= \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[S \mid X = i] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{P}(\text{best student gets A+} \mid \text{they are } i\text{-th}) \\ &= \frac{1}{n}\sum_{i=k+1}^{n} \mathbb{P}(\text{best student gets A+} \mid \text{they are } i\text{-th}) \end{aligned}$$

since first k students are impossible to get A+. Note that best strudent gets A+ | they are the i-th if and only if the best of the first $i-1$ students is in the first k positions, otherwise this student must be better than anyone before them, and they will be given A+ since their position is smaller than i. Hence,

$$\mathbb{P}(\text{best student gets A+} \mid \text{they are } i\text{-th}) = \frac{k}{i-1}.$$

Thus,

$$\begin{aligned} \mathbb{P}(\text{success}) &= \frac{1}{n}\sum_{i=k+1}^{n} \frac{k}{i-1} = \frac{k}{n}\left(\sum_{i=1}^{n-1}\frac{1}{i} - \sum_{i=1}^{k-1}\frac{1}{i}\right) \\ &\approx \frac{k}{n}(\log n - \log k) = \frac{k}{n}\log\left(\frac{n}{k}\right) \quad \text{as } n, k \to \infty. \end{aligned}$$

To find the optimize choice of k, we have

$$\frac{\mathrm{d}}{\mathrm{d}k}\mathbb{P}(\text{success}) = \frac{1}{n}\log\frac{n}{k} + \frac{k}{n}\cdot\frac{1}{\left(\frac{n}{k}\right)}\cdot\left(\frac{-n}{k^2}\right) = \frac{1}{n}\log\frac{n}{k} - \frac{1}{n} = 0$$

which occurs when

$$\log\frac{n}{k} = 1 \Rightarrow \frac{n}{k} = e \Rightarrow k \approx \frac{n}{e}.$$

8.7 Moment Generating Functions

Definition 8.7.1. Given a random variable X, its moment generating function (mgf) is

$$M_X(t) = \mathbb{E}\left[e^{tX}\right], \quad t \in \mathbb{R}.$$

Proposition 8.7.1. Given a random variable X with moment generating function M_X, and some

$n \in \mathbb{N}$, the n-th moment of X is

$$\mathbb{E}[X^n] = M_X^{(n)}(0) = \frac{\mathrm{d}^n}{\mathrm{d}t^n} M_X(t)\bigg|_{t=0}.$$

Proof.

$$\frac{\mathrm{d}^n}{\mathrm{d}t^n} M_X(t)\bigg|_{t=0} = \frac{\mathrm{d}^n}{\mathrm{d}t^n}\mathbb{E}\left[e^{tX}\right]\bigg|_{t=0} = \mathbb{E}\left[\frac{\mathrm{d}^n}{\mathrm{d}t^n} e^{tX}\right]\bigg|_{t=0} = \mathbb{E}[X^n].$$

Remark 8.7.1. For all distributions we care about, we can exchange the expectation and differentiation.

■

Fact: If two random vairables have the same moment generating function, then they have the same distribution.

Example 8.7.1. Let $X \sim \mathrm{Bin}(n, p)$. Then

$$\begin{aligned} M_X(t) = \mathbb{E}\left[e^{tX}\right] &= \sum_{k=0}^{n} e^{tk}\binom{n}{k} p^k (1-p)^{n-k} \\ &= \sum_{k=0}^{n}\binom{n}{k}\left(pe^t\right)^k (1-p)^{n-k} = (1-p+pe^t)^n. \end{aligned}$$

Thus,

$$M_X'(t) = n(1-p+pe^t)^{n-1} pe^t.$$

Hence,

$$\mathbb{E}[X] = M_X'(0) = n(1-p+p)^{n-1} p = np.$$

Also,

$$M_X''(t) = n(n-1)(1-p+pe^t)^{n-2} p^2 e^{2t} + n(1-p+pe^t)^{n-1} pe^t.$$

Thus,

$$\mathbb{E}[X^2] = M_X''(0) = n(n-1)p^2 + np,$$

so

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = np - np^2.$$

Example 8.7.2. $X \sim \mathrm{Poi}(\lambda)$. Then

$$\mathbb{E}[e^{tX}] = \sum_{k=0}^{\infty} e^{tk}\frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda e^t)^k}{k!} = e^{-\lambda}\cdot e^{\lambda e^t} = e^{\lambda(e^t-1)}.$$

Hence,

$$M_X'(t) = e^{\lambda(e^t-1)}\cdot \lambda e^t \Rightarrow \mathbb{E}[X] = M_X'(0) = \lambda.$$

Also,

$$M_X''(t) = \lambda^2 e^{2t} e^{\lambda(e^t-1)} + \lambda e^t e^{\lambda(e^t-1)}.$$

Hence,

$$\mathbb{E}[X^2] = M_X''(0) = \lambda^2 + \lambda \Rightarrow \mathrm{Var}(X) = \mathbb{E}[X^2] - \left(\mathbb{E}[X]\right)^2 = \lambda.$$

Proposition 8.7.2. If X and Y are independent, then

$$M_{X+Y}(t) = M_X(t) M_Y(t).$$

Proof.

$$M_{X+Y}(t) = \mathbb{E}[e^{t(X+Y)}]\mathbb{E}[e^{tX} \cdot e^{tY}] = \mathbb{E}[e^{tX}]\mathbb{E}[e^{tY}] = M_X(t)M_Y(t).$$

∎

Corollary 8.7.1. If $X \sim \mathrm{Poi}(\lambda_1)$ and $Y \sim \mathrm{Poi}(\lambda_2)$ are independent, then $X + Y \sim \mathrm{Poi}(\lambda_1 + \lambda_2)$.

Proof. Note that

$$M_X(t) = e^{\lambda_1(e^t-1)}, \quad M_Y(t) = e^{\lambda_2(e^t-1)}.$$

Since X, Y are independent,

$$M_{X+Y}(t) = M_X(t)M_Y(t) = e^{(\lambda_1+\lambda_2)(e^t-1)}.$$

This si the moment generating function of Poisson random variable with rate $\lambda_1 + \lambda_2$, and since the MGF is unique to the distribution, $X + Y \sim \mathrm{Poi}(\lambda_1 + \lambda_2)$. ∎

Example 8.7.3. $X \sim \mathrm{Exp}(\lambda)$. Then, for $t < \lambda$, we have

$$\begin{aligned} M_X(t) = \mathbb{E}[e^{tX}] &= \int_{\mathbb{R}} e^{tx} f_X(x)\,\mathrm{d}x = \int_0^\infty e^{tx}\lambda e^{-\lambda x}\,\mathrm{d}x \\ &= \lambda \int_0^\infty e^{-(\lambda-t)x}\,\mathrm{d}x = \lambda\left[\frac{e^{-(\lambda-t)x}}{-(\lambda-t)}\right]_{x=0}^\infty = \frac{\lambda}{\lambda-t}. \end{aligned}$$

Thus,

$$M_X'(t) = \frac{\lambda}{(\lambda-t)^2} \Rightarrow \mathbb{E}[X] = M_X'(0) = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}.$$

Also,

$$M_X''(t) = \frac{2\lambda}{(\lambda-t)^3} \Rightarrow \mathbb{E}[X^2] = M''(0) = \frac{2}{\lambda^2} \Rightarrow \mathrm{Var}(X) = \frac{2}{\lambda^2} = \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

Example 8.7.4. $Z \sim N(0,1)$. Then

$$\begin{aligned} M_Z(t) = \mathbb{E}\left[e^{tZ}\right] &= \int_{-\infty}^\infty e^{tz} \cdot \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{z^2}{2}}\,\mathrm{d}z \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^\infty e^{-\frac{1}{2}(z^2-2tz)}\,\mathrm{d}z = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^\infty e^{-\frac{1}{2}(z-t)^2}e^{\frac{1}{2}t^2}\,\mathrm{d}z \\ &= e^{\frac{1}{2}t^2}\frac{1}{\sqrt{2\pi}}\int_{-\infty}^\infty e^{-\frac{1}{2}x^2}\,\mathrm{d}x = e^{\frac{1}{2}t^2}, \end{aligned}$$

where the last equality holds since $\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2}$ is the density function of $N(0,1)$. To compute $\mathrm{Var}(Z)$, we need

$$M_Z'(t) = te^{\frac{1}{2}t^2} \Rightarrow \mathbb{E}[Z] = M_Z'(0) = 0.$$

Also,

$$M_Z''(t) = e^{\frac{1}{2}t^2} + t^2e^{\frac{1}{2}t^2} \Rightarrow \mathbb{E}[Z^2] = M_Z''(0) = 1 \Rightarrow \mathrm{Var}(Z) = 1 - 0^2 = 1.$$

Example 8.7.5. $X \sim N(\mu, \sigma^2)$, then $\frac{X-\mu}{\sigma} = Z \sim N(0,1)$, and we have $X = \mu + \sigma Z$. Note that

$$\begin{aligned} M_X(t) = \mathbb{E}\left[e^{tX}\right] &= \mathbb{E}\left[e^{t(\mu+\sigma Z)}\right] = e^{t\mu}\mathbb{E}\left[e^{(t\sigma)Z}\right] \\ &= e^{t\mu}M_Z(t\sigma) = e^{t\mu}e^{\frac{1}{2}(t\sigma)^2} = e^{t\mu+\frac{1}{2}t^2\sigma^2}. \end{aligned}$$

Now let $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$, independent. We have

$$M_{X+Y}(t) = M_X(t)M_Y(t) = e^{t\mu_1 + \frac{1}{2}t^2\sigma_1^2} \cdot e^{t\mu_2 + \frac{1}{2}t^2\sigma_2^2} = e^{t(\mu_1+\mu_2)+\frac{1}{2}t^2(\sigma_1^2+\sigma_2^2)}.$$

This is the mgf of $N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Example 8.7.6. Let $X_1, X_2, X_3, \dots$ be independent identically distributed random variables. Let N be an independent random variable taking non-negative integer values. Let $X = \sum_{i=1}^N X_i$. Then

$$\mathbb{E}[X] = \mathbb{E}[N]\mathbb{E}[X_1].$$

Proof. We have

$$\begin{aligned} M_X(t) = \mathbb{E}[e^{tX}] &= \mathbb{E}\left[e^{t\sum_{i=1}^N X_i}\right] = \mathbb{E}\left[\mathbb{E}\left[e^{t\sum_{i=1}^n X_i} \mid N = n\right]\right] \\ &= \mathbb{E}_{N=n}\left[\mathbb{E}\left[e^{t\sum_{i=1}^n X_i}\right]\right] = \mathbb{E}_{N=n}\left[\prod_{i=1}^n \mathbb{E}[e^{tX_i}]\right] \\ &= \mathbb{E}_{N=n}\left[M_{X_1}(t)^n\right] = \mathbb{E}\left[M_{X_1}(t)^N\right] \end{aligned}$$

since $X_1, X_2, \dots$ are i.i.d. and thus have same mgf. Note that

$$M_X'(t) = \mathbb{E}\left[\frac{\mathrm{d}}{\mathrm{d}t} M_{X_1}(t)^N\right] = \mathbb{E}\left[N M_{X_1}(t)^{N-1} M_{X_1}'(t)\right].$$

Hence,

$$\mathbb{E}[X] = M_X'(0) = \mathbb{E}\left[N \cdot M_{X_1}(0)^{N-1} \cdot M_{X_1}'(0)\right].$$

Now since

$$M_{X_1}(0) = \mathbb{E}[e^{0 \cdot X_1}] = 1, \quad M_{X_1}'(0) = \mathbb{E}[X_1],$$

we have

$$\mathbb{E}\left[N \cdot M_{X_1}(0)^{N-1} \cdot M_{X_1}'(0)\right] = \mathbb{E}[N \cdot 1 \cdot \mathbb{E}[X_1]] = \mathbb{E}[N] \cdot \mathbb{E}[X_1].$$

⊛

Chapter 9

Limit Theorems

If $X_1, X_2, X_3, \ldots$ are independent and indentically distributed with distribution X, $\mathbb{E}[X] = \mu$, then we can define the sample means

$$\overline{X_n} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

> **Question.** What can we say about the distribution of $\overline{X_n}$?

Recall that

$$\mathbb{E}\left[\overline{X_n}\right] = \mathbb{E}[X] = \mu,$$

and by independence,

$$\operatorname{Var}(\overline{X}_n) = \frac{1}{n}\operatorname{Var}(X).$$

There are three types of results:

(1) Weak Law of Large Numbers:

$$\forall \varepsilon > 0, \quad \mathbb{P}\left(\left|\overline{X_n} - \mu\right| \geq \varepsilon\right) \to 0 \quad \text{as } n \to \infty$$

(2) Strong Law of Large Numbers:

$$\mathbb{P}\left(\lim_{n\to\infty} \overline{X_n} = \mu\right) = 1.$$

(3) Central Limit Theorem: A $n \to \infty$,

$$\frac{(\overline{X_n} - \mu)\sqrt{n}}{\sigma} \approx N(0,1).$$

Lecture 27

Recall the Markov's inequality: If $X \geq 0$, then Jun 2.

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a},$$

the Chebyshev's inequality:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\operatorname{Var}(X)}{a^2},$$

and the First Borel-Cantelli lemma: If $E_1, E_2, \ldots$ are events with $\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty$, then

$$\mathbb{P}\left(\limsup_{n\to\infty} E_n\right) = 0.$$

9.1 Law of Large Numbers

Let $X_1, X_2, \ldots$ be independent and identically distributed. What can we say about

$$\overline{X_n} = \frac{1}{n}\sum_{i=1}^{n} X_i?$$

Theorem 9.1.1 (Weak Law of Large Numbers). Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$. Then for any $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\left(\left|\overline{X_n} - \mu\right| \geq \varepsilon\right) = 0.$$

Proof. We know $\mathbb{E}[\overline{X_n}] = \mu$ since

$$\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \frac{1}{n}n\mu = \mu,$$

and

$$\mathrm{Var}(\overline{X_n}) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\mathrm{Var}(\sum_{i=1}^{n} X_i) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{1}{n^2}n\sigma^2 = \frac{\sigma^2}{n}.$$

By Chebyshev,

$$\mathbb{P}\left(|\overline{X_n} - \mu| \geq \varepsilon\right) \leq \frac{\mathrm{Var}(\overline{X_n})}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \to 0.$$

■

Remark 9.1.1. We need the independence to say

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n}\mathrm{Var}(X_i),$$

but this is true with only pairwise independence. We can still prove the Weak Law of Large Number if the variance is infinite, but harder.

Theorem 9.1.2 (Strong Law of Large Numbers). Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathbb{E}[X_i^4] = K < \infty$, then

$$\mathbb{P}\left(\lim_{n\to\infty}\overline{X_n} = \mu\right) = 1.$$

Proof. We consider the following cases:

Case 1: $\mu = 0$. Let $Y_n = \sum_{i=1}^{n} X_i$, then

$$\begin{aligned} Y_n^4 = \left(\sum_{i=1}^{n} X_i\right)^4 &= \sum_{i,j,k,\ell} X_i X_j X_k X_\ell \\ &= \sum_{i=1}^{r} X_i^4 + 4\sum_{i\neq j} X_i X_j^3 + 6\sum_{i<j} X_i^2 X_j^2 + 12\sum_{i<j\neq k} X_i X_j X_k^2 + 24\sum_{i<j<k<\ell} X_i X_j X_k X_\ell. \end{aligned}$$

Hence,

$$\begin{aligned} \mathbb{E}[Y_n^4] &= \sum_{i=1}^{n}\mathbb{E}[X_i^4] + 4\sum_{i\neq j}\mathbb{E}[X_i X_j^3] + 6\mathbb{E}\sum_{i<j}\mathbb{E}[X_i^2 X_j^2] + 12\sum_{i<j\neq k}\mathbb{E}[X_i X_j X_k^2] + 24\sum_{i<j<k<\ell}\mathbb{E}[X_i X_j X_k X_\ell] \\ &= \sum_{i=1}^{n}\mathbb{E}[X_i^4] + 6\sum_{i<j}\mathbb{E}[X_i^2]\mathbb{E}[X_j^2] \end{aligned}$$

since by independence we have $\mathbb{E}[X_iX_j] = \mathbb{E}[X_i]\mathbb{E}[X_j]$ and $\mathbb{E}[X_i] = 0$.

Thus,

$$\mathbb{E}[Y_n^4] = n\mathbb{E}[X_1^4] + 6\binom{n}{2}\mathbb{E}[X_i^2]^2.$$

Note that

$$\mathbb{E}[X_1^4] = \mathbb{E}\left[\left(X_1^2\right)^2\right] = \mathrm{Var}(X_1^2) + \mathbb{E}[X_1^2]^2 \geq \mathbb{E}[X_1^2]^2$$

since $\mathrm{Var}(X_1^2) \geq 0$. Hence,

$$\mathbb{E}[Y_n^4] \leq \left(n + 6\binom{n}{2}\right)\mathbb{E}[X_1^4] = 3n^2K.$$

Let

$$\begin{aligned} E_n &= \left\{|\overline{X_n}| > \frac{1}{\log n}\right\} = \left\{\left|\frac{1}{n}\sum_{i=1}^n X_i\right| > \frac{1}{\log n}\right\} \\ &= \left\{|Y_n| > \frac{n}{\log n}\right\} = \left\{Y_n^4 > \frac{n^4}{\log^4 n}\right\}. \end{aligned}$$

By Markov,

$$\mathbb{P}(E_n) = \mathbb{P}\left(Y_n^4 > \frac{n^4}{\log^4 n}\right) \leq \frac{\mathbb{E}[Y_n^4]}{\left(\frac{n^4}{\log^4 n}\right)} \leq \frac{3n^2K}{\left(\frac{n^4}{\log^4 n}\right)} = \frac{3K\log^4 n}{n^2}.$$

Therefore

$$\sum_{n=1}^\infty \mathbb{P}(E_n) \leq 3K\sum_{n=1}^\infty \frac{\log^4 n}{n^2} < \infty.$$

By the Borel-Cantelli lemma,

$$\mathbb{P}\left(\limsup_{n\to\infty} E_n\right) = 0.$$

Thus, with probability 1, only finitely many E_n occur. With probability 1, $\exists N \in \mathbb{N}$ s.t. $\forall n \geq N$, E_n doesn't occur, i.e.

$$|\overline{X_n}| \leq \frac{1}{\log n}.$$

Thus, with probability 1, $\lim_{n\to\infty} \overline{X_n} = 0 = \mu$.

Case 2: $\mu \neq 0$, then let $X_n' = X_n - \mu$, then $\mathbb{E}[X_n'] = 0$. Hence,

$$\mathbb{P}\left(\lim_{n\to\infty} \overline{X_n'} = 0\right) = 1 \Leftrightarrow \mathbb{P}\left(\lim_{n\to\infty} \overline{X_n} - \mu = 0\right) = 1 \Leftrightarrow \mathbb{P}\left(\lim_{n\to\infty} \overline{X_n} = \mu\right) = 1.$$

■

Remark 9.1.2. We need independence to say

$$\mathbb{E}[X_iX_jX_kX_\ell] = \mathbb{E}[X_i]\mathbb{E}[X_j]\mathbb{E}[X_k]\mathbb{E}[X_\ell],$$

so it is enough to have 4-independence.

9.2 Central Limit Theorem

Theorem 9.2.1 (Central Limit Theorem)**.** Let $X_1, X_2, \ldots$ be i.i.d. random variables with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$. Then

$$\frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}}$$

converges to $N(0,1)$ as $n \to \infty$, i.e. for any $a \in \mathbb{R}$, let $Z \sim N(0,1)$, then

$$\mathbb{P}\left(\frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \le a\right) \to \mathbb{P}(Z \le a) = \Phi(a).$$

Proof. We first give an observation:

$$\frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \le a \Leftrightarrow \sum_{i=1}^n X_i - n\mu \le a\sigma\sqrt{n} \Leftrightarrow \overline{X_n} \le \frac{a\sigma}{\sqrt{n}}.$$

For our proof, we assume

1. Moment generating function $M_{X_i}(t)$ exists.
2. Fact: If we have random variables $Z_1, Z_2, \ldots$ with mgf M_{Z_n} and cdf F_{Z_n}, and random variables Z with M_Z, F_Z, then if
$$\lim_{n\to\infty} M_{Z_n}(t) = M_Z(t) \quad \forall t,$$
then
$$\lim_{n\to\infty} F_{Z_n}(a) = F_Z(a).$$
This is just the precise way of saying that the momoent generating function determines the distribution.

Now we start the proof of central limit theorem.
Case 1: $\mu = 0, \sigma^2 = 1$. We want to show

$$\frac{\sum_{i=1}^n X_i}{\sqrt{n}} \to N(0,1) \qquad \text{as } n \to \infty.$$

By fact, it is enough to show

$$M_{\frac{\sum_{i=1}^n X_i}{\sqrt{n}}}(t) \to M_{N(0,1)}(t) \quad \forall t.$$

If $Z \sim N(0,1)$, then

$$M_Z(t) = e^{\frac{t^2}{2}}.$$

Let

$$Z_n = \frac{\sum_{i=1}^n X_i}{\sqrt{n}} = \sum_{i=1}^n \frac{X_i}{\sqrt{n}}.$$

By independence,

$$M_{Z_n}(t) = \prod_{i=1}^n M_{\frac{X_i}{\sqrt{n}}}(t) = \left(M_{\frac{X_1}{\sqrt{n}}}\right)^n.$$

Note that

$$M_{\frac{X_1}{\sqrt{n}}}(t) = \mathbb{E}\left[e^{t\frac{X_1}{\sqrt{n}}}\right] = \mathbb{E}\left[e^{\frac{t}{\sqrt{n}}X_1}\right] = M_{X_1}\left(\frac{t}{\sqrt{n}}\right).$$

Want to show

$$\lim_{n\to\infty} M_{X_1}\left(\frac{t}{\sqrt{n}}\right)^n = e^{\frac{t^2}{2}}.$$

Taking logarithm, it is equivalent to show

$$\lim_{n\to\infty} n \log M_{X_1}\left(\frac{t}{\sqrt{n}}\right) = \frac{t^2}{2}.$$

Let $L(t) = \log M_{X_1}(t)$, we want to show

$$\lim_{n\to\infty} nL\left(\frac{t}{\sqrt{n}}\right) = \frac{t^2}{2}.$$

Note that

$$L(0) = \log M_{X_1}(0) = \log \mathbb{E}\left[e^{0\cdot X_1}\right] = \log 1 = 0.$$

Also,

$$L'(t) = \frac{M'_{X_1}(t)}{M_{X_1}(t)} \Rightarrow L'(0) = \frac{M'_{X_1}(0)}{M_{X_1}(0)} = \frac{\mathbb{E}[X_1]}{1} = \mu = 0.$$

Thus, we can compute

$$L''(t) = \frac{M'_{X_1}(t)M_{X_1}(t) - M'_{X_1}(t)^2}{M_{X_1}(t)^2}.$$

This gives

$$L''(0) = \frac{\mathbb{E}[X_1^2]\cdot 1 - \mathbb{E}[X_1]^2}{1^2} = (\sigma^2 + \mu^2) - \mu^2 = 1.$$

Hence,

$$\begin{aligned}
\lim_{n\to\infty} nL\left(\frac{t}{\sqrt{n}}\right) &= \lim_{n\to\infty} \frac{L\left(\frac{t}{\sqrt{n}}\right)}{\left(\frac{1}{n}\right)} = \lim_{n\to\infty} \frac{L'\left(\frac{t}{\sqrt{n}}\right)\cdot\left(-\frac{1}{2}\cdot\frac{t}{n^{\frac{3}{2}}}\right)}{\left(-\frac{1}{n^2}\right)} \\
&= \lim_{n\to\infty} \frac{L'\left(\frac{t}{\sqrt{n}}t\right)}{\left(\frac{2}{\sqrt{n}}\right)} \overset{\text{L'H}}{=} \frac{t}{2}\lim_{n\to\infty} \frac{L''\left(\frac{t}{\sqrt{n}}\right)\cdot\left(-\frac{1}{2}\frac{t}{n^{\frac{3}{2}}}\right)}{\left(-\frac{1}{2n^{\frac{3}{2}}}\right)} \\
&= \frac{t^2}{2}\lim_{n\to\infty} L''\left(\frac{t}{\sqrt{n}}\right) = \frac{t^2}{2}L''(0) = \frac{t^2}{2}.
\end{aligned}$$

Thus, we're done.

General Case: Given X_i with $\mathbb{E}[X_i] = \mu$ and $\delta(X_i) = \sigma^2$. Let $Y_i = \frac{X_i - \mu}{\sigma}$, then $\mathbb{E}[Y_i] = 0$, and $\mathrm{Var}(Y_i) = 1$. Thus, by the previous case, $\forall a$,

$$\mathbb{P}\left(\frac{\sum_{i=1}^n Y_i}{\sqrt{n}} \leq a\right) \to \Phi(a),$$

while

$$\frac{\sum_{i=1}^n Y_i}{\sqrt{n}} = \frac{\sum_{i=1}^n (X_i - \mu)}{\sigma\sqrt{n}} = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}}.$$

■

Remark 9.2.1.

1. Can show convergence is uniform for all a, i.e. $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $\forall a \in \mathbb{R}$ and $n \geq N$,

$$\left|\mathbb{P}\left(\frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \leq a\right) - \Phi(a)\right| \leq \varepsilon.$$

2. Convergence is quite fast

3. CLT holds more generally: If $X_1, X_2, \ldots$ are independent and

   - $\exists M$ s.t. $\mathbb{P}(|X_i| \leq M) = 1$ for all i.
   - $\sum_{n=1}^\infty \mathrm{Var}(X_n) = \infty$.

   Then,

$$\frac{\sum_{i=1}^n (X_i - \mathbb{E}[X_i])}{\sqrt{\sum_{i=1}^n \delta(X_i)}} \to N(0,1).$$

Example 9.2.1. Roll a fair die 10 times, what is $\mathbb{P}(\text{sum} \in [30, 40])$?

Proof. Let X_i be the outcome of the i-th roll. Then $X_i \sim \mathrm{Unif}(\{1,2,3,4,5,6\})$, and $\mathbb{E}[X_i] =$

$3.5, \mathrm{Var}(X_i) = \frac{35}{12}$.

Hence,

$$\mathbb{P}\left(30 \leq \sum_{i=1}^{10} \leq 40\right) = \mathbb{P}\left(29.5 \leq \sum_{i=1}^{10} X_i \leq 40.5\right) = \mathbb{P}\left(-5.5 \leq \sum_{i=1}^{10} X_i - 10 \cdot 3.5 \leq 5.5\right)$$
$$= \mathbb{P}\left(-\frac{5.5}{\sqrt{\frac{350}{120}}} \leq \frac{\sum_{i=1}^{10} X_i - 35}{\sqrt{\frac{350}{12}}} \leq \frac{5.5}{\sqrt{\frac{350}{12}}}\right)$$
$$\approx \Phi\left(\frac{3.5}{\sqrt{\frac{350}{12}}}\right) - \Phi\left(\frac{-5.5}{\sqrt{\frac{350}{12}}}\right) = 2\Phi\left(\frac{5.5}{\sqrt{\frac{350}{12}}}\right) - 1 \approx 0.692.$$

⊛

Chapter 10

Probabilistic Method in Combinatorics

Lecture 28

Combinatorics is the field that study discrete objects. For example, Jun 4.

- graphs
- set systems
- numbers

Extremal Combinatorics

Question. How big can an object be if it satisfies certain properties?

We can decompose this question into two parts:

1. Give a lower bound, give a construction of an object with the property.
2. Give an upper bound, prove that there is nothing better.

10.1 Sum-free sets

Definition 10.1.1. A set $S \subseteq \mathbb{N}$ is sum-free if $\nexists x, y, z \in S$ s.t. $x + y = z$.

Question. How big can a sum-free set $S \subseteq [n]$ be?

We can give some constructions:

- odd numbers $S = \left\{1, 3, 5, \ldots, 2 \cdot \left\lceil \frac{n}{2} \right\rceil - 1\right\}$. $|S| = \left\lceil \frac{n}{2} \right\rceil$.
- Large numbers: $S = \left\{\left\lfloor \frac{n}{2} \right\rfloor + 1, \left\lfloor \frac{n}{2} \right\rfloor + 2, \ldots, n\right\}$. $|S| = \left\lceil \frac{n}{2} \right\rceil$.

Proposition 10.1.1. If $S \subseteq [n]$ is sum-free, then $|S| \leq \left\lceil \frac{n}{2} \right\rceil$.

Proof. Let $m = \max S \leq n$. Then we can partition $1, 2, \ldots, m-1$ into pairs summing to m: $\{1, m-1\}, \{2, m-2\}, \ldots$. Thus S can contain at most one element from each pair. Thus,

$$|S| \leq \# \text{ of partitions} + m \text{ iteself} = \left\lfloor \frac{m-1}{2} \right\rfloor + 1 = \left\lceil \frac{m}{2} \right\rceil \leq \left\lceil \frac{n}{2} \right\rceil.$$

■

Two-player version

- 1st player: Choose a set $A \subseteq \mathbb{N}$ of n natural numbers.
- 2nd player: Find a large sum-free subset $S \subseteq A$.

Definition 10.1.2. Given a finite set $A \subseteq \mathbb{N}$, let $\sigma(A)$ be the size of the largest sum-free subset $S \subseteq A$.

Example 10.1.1. $\sigma([n]) = \left\lceil \frac{n}{2} \right\rceil$.

Theorem 10.1.1 (Erdos, 1965)**.** For any set $A \subseteq \mathbb{N}$ of n natural numbers,

$$\sigma(A) \geq \frac{n+1}{3}.$$

Proof. Instead of constructing an object with the desired property, it is enough to prove the object exists.

Consider a random object and show it has the property with positive probability.

Let p be a prime, $p = 3k+2$ for some $k \in \mathbb{N}$, $p > \max A$. Consider $\mathbb{Z}_p$ and observe $I = \{k+1, k+2, \ldots, 2k+1\} \subseteq \mathbb{Z}_p$ is sum-free. Thus, $A \cap I$ is sum-free (in $\mathbb{N}$). However, $A \cap I$ may be empty.

We can take $x \in \mathbb{Z}_p^\times = \{1, 2, \ldots, p-1\}$ uniformly at random. Then for any $a \in A$, $xa \in \mathbb{Z}_p$ is also uniformly random in $\mathbb{Z}_p^\times$. And

$$A_x = \{a \in A : xa \in I\}$$

is sum-free. This is because if $a_1 + a_2 = a_3$ in $\mathbb{N}$, then $xa_1 + xa_2 = xa_3$ in $\mathbb{N}$, then $xa_1 + xa_2 = xa_3$ in $\mathbb{Z}_p$, but these are in I, which is sum-free.

Now note that

$$\begin{aligned}
\mathbb{E}[|A_x|] &= \mathbb{E}\left[\sum_{a \in A} 1_{a,x}\right], \quad \text{where } 1_{a,x} = \begin{cases} 1, & \text{if } ax \in I; \\ 0, & \text{if } ax \notin I. \end{cases} \\
&= \sum_{a \in A} \mathbb{P}(ax \in I) = \sum_{a \in A} \frac{|I|}{|\mathbb{Z}_p^\times|} = \sum_{a \in A} \frac{k+1}{3k+1} > \frac{1}{3}|A| = \frac{n}{3}.
\end{aligned}$$

Since the average size of A_x is strictly bigger than $\frac{n}{3}$, there must be some x for which $|A_x| \geq \frac{n+1}{3}$. This is sum-free, so

$$\sigma(A) \geq |A_x| \geq \frac{n+1}{3}.$$

■

Remark 10.1.1.

- Bourgan (1990s) used Fourier techniques to improve the bound:

$$\sigma(A) \geq \frac{n+2}{3}.$$

- Eberhard-Green-Manners (2014): There is a set $A \subseteq \mathbb{N}$ of size n for which

$$\sigma(A) \leq \left(\frac{1}{3} + o(1)\right) n.$$

10.2 The Crossing Number Inequality

Definition 10.2.1. A graph is a pair (V, E) where V is a set of vertices and $E \subseteq \binom{V}{2}$ is a set of edges, where every edge is an unordered pair of vertices.

A drawing of the graph is an embedding in the plane:

- vertices $\to$ paths of $\mathbb{R}^2$
- egdes $\to$ closed curves between the endpoints.

Definition 10.2.2. A crossing is an intersection of two edges in a drawing.

Definition 10.2.3. A graph is planar if it can be drawn without crossings.

Question. How many edges can a planar graph on n vertices have?

Remark 10.2.1. A graph of n vertices can have at most $\binom{n}{2}$ vertices.

Theorem 10.2.1 (Euler)**.** A planar graph on $n \geq 3$ vertices has at most $3n - 6$ edges.

Question. What is the minimum number of crossings in a graph with n vertices and m edges?

Notation. We denote this number by $\mathrm{cr}(n, m)$.

Thus, Euler shows if $n \geq 3$, then $\mathrm{cr}(n, m) = 0 \Leftrightarrow m \leq 3n - 6$.

Corollary 10.2.1. For $n \geq 3$, $\mathrm{cr}(n, m) \geq m - 3n + 6$.

Proof. As long as our graph has a crossing, remove an edge in the crossing. Continue until no crossings left. Thus, we will end with $\leq 3n - 6$ edges, i.e. we will removed $\geq m - 3n + 6$ edges, so there are $\geq m - 3n + 6$ crossings. ■

Theorem 10.2.2 (The crossing number inequality)**.** If $m \geq 4n$, then

$$\mathrm{cr}(n, m) \geq \frac{m^3}{64n^2}.$$

Proof. Given a graph on n vertices and m edges, let K be the number of crossings in an optimal drawing of G. Consider a random induced subgraph, where we keep each vertex independently with probability p. Let X be the number of vertices in our random subgraph, Y be the number of edges, and Z be the number of crossings. Then $\mathbb{E}[X] = np$,

$$\mathbb{E}[Y] = \mathbb{E}\left[\sum_{e \in E} 1_{e \text{ surviving}}\right] = \sum_{e \in E} \mathbb{P}\left(e \text{ surviving}\right) = mp^2,$$

and

$$\mathbb{E}[Z] = \sum_{\text{crossings}} \mathbb{P}(\text{crossing survives}) = Kp^4.$$

The random graph has X vertices, Y edges, so number of crossings

$$Z \geq \mathrm{cr}(X, Y) \geq Y - 3X + 6 \geq Y - 3X.$$

Thus, we always have $Z - Y + 3X \geq 0$. This shows

$$\mathbb{E}[Z - Y + 3X] \geq 0.$$

By linearity of expectation,

$$\mathbb{E}[Z] - \mathbb{E}[Y] + \mathbb{E}[X] \geq 0.$$

Thus,

$$Kp^4 - mp^2 + 3np \geq 0,$$

which gives

$$K \geq \frac{mp - 3n}{p^3}.$$

Now we want R.H.S. to be as large as possible, so we can set $p = \frac{4n}{m} \leq 1$, and thus

$$K \geq \frac{n}{\left(\frac{4n}{m}\right)^3} = \frac{m^3}{64n^2}.$$

■

Remark 10.2.2. If $m = \Theta\left(n^2\right)$, then $\operatorname{cr}(n,m) = \Theta\left(n^4\right)$.

Appendix